

MAGNETOM

Replacement of Parts System

Gradient

Avanto
Espree

Document Version

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1.1 General

The following chapter describes the replacement procedures for the "Field Replaceable Units" (FRUs) in the Gradient System K2259.

The information provided also includes references for the required checks and adjustments.



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter between the protective earth connector on the "sub-component" (e.g. front cover) and the protective earth wire connected to the "main component" (e.g. GPA frame).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

Tab. 1 FRUs GPA K2259

GPA K2259					
FRU	Total Time	Instructions	Adjustments	Final Checks	Tools
D10 - DAC - Board (75 62 924)	80 min	(→ D10 - DAC-Board / Page 22)	TuneUp / Phantom Shim, Gradient Delay, Gradient Sensitivity	Quality Assurance / all selected procedures	Standard tool set
D20 - Regulator board (75 62 940)	155 min	(→ D20 - Regulator Board / Page 25)	TuneUp / Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Delay, Gradient Sensitivity, Dynamic Field Map (with VD Software)	Quality Assurance / all selected procedures	Standard tool set
D40 - Modulator board (75 62 965)	155 min	(→ D40 - Modulator Board / Page 28)	TuneUp / Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Delay, Gradient Sensitivity, Dynamic Field Map (with VD Software)	Quality Assurance / all selected procedures	Standard tool set
D60 - CAN/Service (75 62 981)	50 min	(→ D60 - CAN/Service / Page 31)	n.a.	- Regulator Check or Gradient Regulator Check - Quality Assurance Expert / Coil Check	Standard tool set
D70 - I/O-Board (75 63 005)	46 min	(→ D70 - I/O Board / Page 34)	n.a.	- Regulator Check or Gradient Regulator Check - Quality Assurance Expert / Coil Check	Standard tool set
Converter E4A VE-GA650 (30 96 260)	56 min	(→ Converter E4A VE-GA650 / Page 41)	n.a.	- Quality Assurance/Regulator Check or Gradient Regulator Check - Quality Assurance Expert / Coil Check	Standard tool set

GPA K2259					
FRU	Total Time	Instructions	Adjustments	Final Checks	Tools
Current Converter Unit XXL (75 64 037)	295 min	(→ Current Sensor Unit / Page 45)	■ TuneUp / Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Delay, Gradient Sensitivity, Dynamic Field Map (with VD Software)	■ Quality Assurance/ all selected procedures ■ Image Orientation Check	■ Standard tool set ■ SW 19 wrench ■ 10-mm Allen key
Assembly Carrier 6HE 34TE (75 64 060)	56 min	(→ Assembly Carrier / Page 37)	■ n.a.	■ Quality Assurance/ Regulator Check or Gradient Regulator Check ■ Quality Assurance Expert / Coil Check	Standard tool set
24 V - 5.5 W fan (70 64 678)	15 min	n.a.	■ n.a.	■ n.a.	Standard tool set
Cable W520 (75 64 052)	30 min	(→ Cable W520 / Page 51)	■ n.a.	■ n.a.	Standard tool set
Cable W515 (83 62 670)	20 min	n.a.	■ n.a.	■ n.a.	Standard tool set
Cable W305 (83 62 712)	25 min	n.a.	■ n.a.	■ n.a.	Standard tool set
Cable assembly (57 27 297)	20 min	n.a.	■ n.a.	■ n.a.	Standard tool set

GPA K2259					
FRU	Total Time	Instructions	Adjustments	Final Checks	Tools
Power Stage XXL (106 63 309)	160 min	(→ Power Stage XXL / Page 58)	TuneUp / Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Delay, Gradient Sensitivity, Dynamic Field Map (with VD Software)	- Quality Assurance / all selected procedures - Image Orientation Check	- Standard tool set - SW 19 wrench - Manual pump RP50 (material number 81 15 136)
CPC water couplings (6 pcs. water quick couplings for Power Stage) (107 43 825)	60 min	see also SKB: (→ SKB003397, https://skb.healthcare.siemens.com/Pages/SKBAdvancedViewKBDocument.aspx?docid=SKB0033978)	n.a.	- Check the water flows and temperatures in the Service Software under Magnet&Cooling>Cooling Status. - Check water connections for leaks	Standard tool set
Pair of water hoses (83 62 555)	30 min	n.a.	n.a.	n.a.	- Standard tool set - Manual pump RP50 (material number 81 15 136)
Choke Set 13MY 400A 10% for serial no. >1017 (77 51 055) only serial no. 1001-1017 (30 98 977)	175 min	(→ Choke Set / Page 65)	TuneUp / Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Delay, Gradient Sensitivity, Dynamic Field Map (with VD Software)	- Quality Assurance / all selected procedures - Image Orientation Check	Standard tool set, SW 19 wrench

GPA K2259					
FRU	Total Time	Instructions	Adjustments	Final Checks	Tools
Fan box GPA XXL E3 (75 63 971)	66 min	(→ Fan box GPA XXL E3 / Page 70)	n.a.	- Quality Assurance Expert / Coil Check - Image Orientation Check	Standard tool set
D100 - Rectifier Board (75 63 021)	65 min	(→ D100 - Rectifier Board / Page 75)	n.a.	- Quality Assurance / Gradient Rise Time Check - Quality Assurance Expert / Coil Check	- Standard tool set - Heat- conductive paste (CONDU. PASTE Beru-lub, material number 02048650) - Manual pump RP50 (material number 81 15 136)
Circuit breaker 32A (77 48 770)	45 min	(→ Circuit Breaker 32A / Page 82)	n.a.	Quality Assurance Expert / Coil Check	Standard tool set

GPA K2259					
FRU	Total Time	Instructions	Adjustments	Final Checks	Tools
Chilled water distributor (75 64 094)	180 min	(→ Chilled Water Distributor / Page 90)	n.a.	Check water flows and temperatures in the Service Software under Magnet & Cooling> Cooling Status	- Standard tool set - Torx driver bit - Manual pump RP50 (material number 81 15 136) - Filling adapter - Bucket - Water hose
Main Transformer GPAXXL (75 63 989)	8 hrs.	(→ Main Transformer / Page 97)	n.a.	- Quality Assurance - Regulator Check or Gradient Regulator Check - Gradient Rise Time Check - Eddy Current Compensation Check - Gradient Delay Check - Gradient Sensitivity Check - Dynamic Field Map (with Software VD) - Image Orientation Check - Quality Assurance Expert / Coil Check - Run Duty Cycle Sequence	- Standard tool set - Transformer Lifting Device (51 37 711) - A second person is required

GPA K2259					
FRU	Total Time	Instructions	Adjustments	Final Checks	Tools
D110 - Power-Up circuit board (75 63 047)	36 min	(→ D110 - Power-Up Circuit Board / Page 85)	n.a.	- Quality Assurance / Regulator Check or Gradient Regulator Check - Quality Assurance Expert / Coil Check	Standard tool set
100A Contactor 24VDC 3S 100A 400V (30 90 560)	90 min	(→ 100A Contactor 24VDC 3S 100A 400V / Page 88)	n.a.	Quality Assurance Expert / Coil Check	Standard tool set
Line Power Filter Assembly (100 46 634)	65 min	(→ Circuit Breakers, Line Filter, and Transformer / Page 112)	n.a.	Quality Assurance/ Coil Check	- Standard tool set - Duspol - Step Ladder
Small Signal Transformer T11 and Circuit Breaker F2 (100 46 633)	50 min	(→ Circuit Breakers, Line Filter, and Transformer / Page 112)	n.a.	Quality Assurance/ Coil Check	- Standard tool set - Duspol
Current Control LA305S/ 500 1:2000 (30 96 070)	37 min	(→ Current Control LA305S/ 500A / Page 53)	n.a.	- Quality Assurance/ Regulator Check or Gradient Regulator Check - Image Orientation Check - Quality Assurance Expert/ Coil Check	- Standard tool set - SW 19 wrench - Step Ladder

GPA K2259					
FRU	Total Time	Instructions	Adjustments	Final Checks	Tools
Current Transformer IT700 700A Single transformer unit: (86 89 296)	145 min	(→ Current Transformer 700A/IT700 700A/ / Page 55)	TuneUp / Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Delay, Gradient Sensitivity, Dynamic Field Map (with VD Software)	- Quality Assurance/ All selected Procedures - Image Orientation Check	- Standard tool set - SW 19 wrench - Step Ladder
Door GPA_XXL (75 63 765)	10 min	n.a.	n.a.	n.a.	Standard tool set
Door handle ACC (also SEP, GPA-XXL) (81 13 891)	10 min	n.a.	n.a.	n.a.	Standard tool set
Guide roll f. Schroff cab. (lock) (81 13 875)	30 min	n.a.	n.a.	n.a.	Standard tool set

Tab. 2 Gradient Coil Avanto/Espree

Gradient Coil Avanto/Espree					
FRU	Time	Instructions	Adjustments	Final Checks	Tools
Avanto Gradient Coil (81 19 690)	approx. 2 days	(→ Avanto Gradient Coil / Page 127)	- TU/ Phantom Shim - if necessary: Magnet Shim - Complete Tune up	Complete Quality Assurance	Refer to (→ Tools and time required / Page 128).
Espree Gradient Coil (100 48 100)	approx. 2days (5 days in case a straight Body/ Gradient coil was installed)	(→ Upgrade Espree straight bore Gradient/ Body Coil to Slim-line / Page 147)	- TU/ Phantom Shim - if necessary: Magnet Shim - Complete Tune up	Complete Quality Assurance	Refer to (→ Tools and time required / Page 148).

Tab. 3 Components of Gradient Coil

Components of Gradient Coil					
FRU	Time	Instructions	Adjustments	Final Checks	Tools
Water distributor for Avanto (81 19 641) Espree (81 21 290)	120 min	(→ GC Water Distributor (Avanto/ Espree) / Page 179)	n.a.	- Check the water flows and temperatures in the Service Software under Magnet& Cooling > Cooling Status - Check water distributor and hose connections for leaks.	- Non-magnetic tool kit - Non-magnetic scalpel

Components of Gradient Coil					
FRU	Time	Instructions	Adjustments	Final Checks	Tools
NTC sensors in the Gradient Coil	120 min	(→ Defective NTC-Sensors of Gradient Coil / Page 185)	n.a.	n.a.	- Non-magnetic tool kit - Pin ejector tool - DMM
Gradient Coax Cables Avanto (106 78 832) (replaces material number 81 19 625)	3 hr	(→ Gradient Coax Cables Avanto (material number 106 78 832) / Page 188)	n.a.	- QA/ Spike Check - QA/ Image Orientation Check	- Non-magnetic Tool Kit - Non-magnetic Torque Wrench (20-50 Nm) - Non-magnetic torque wrench (5 - 20 Nm) - Jaw wrench SW17 nonmag (material number 105 63 219) - Accessory for torque wrenches (refer to instructions) - Set of NORD-LOCK Washers GC (103 58 905)

Components of Gradient Coil					
FRU	Time	Instructions	Adjustments	Final Checks	Tools
Gradient Coax Cables Avanto (original version) (81 19 625)	3 hr	(→ Gradient Coax Cables Avanto (material number 81 19 625) / Page 215)	n.a.	- QA/ Spike Check - QA/ Image Orientation Check	- Non-magnetic Tool Kit (including 32-mm open-ended wrench) - Non-magnetic Torque Wrench (20-50 Nm) - Accessory for torque wrenches (refer to instructions) 8-s. socket gradient coil, non-magnetic (103 99 445) - Set of NORD-LOCK Washers GC (103 58 905) - Heat gun - Loctite 221 (soft-locking, 20 48 874)

Components of Gradient Coil					
FRU	Time	Instructions	Adjustments	Final Checks	Tools
Gradient Coax Cables Espree (81 23 684) Coax Straight - complete (serial numbers <30099) or Gradient cables Espree (081 21 860) Coax Slim - complete	3 hr	(→ Gradient Cables (Espree) / Page 236)	n.a.	- QA/ Spike Check - QA/ Image Orientation Check	- Non-magnetic Tool Kit - Non-magnetic Torque Wrench (20-50 Nm) - Accessory for torque wrenches (refer to instructions) 8-s. socket gradient coil, non-magnetic (103 99 445) - Set of NORD-LOCK Washers GC (103 58 905) - Heat gun - Loctite 221 (soft-locking, 20 48 874)

Tab. 4 Gradient Supervision (VLM)

Gradient Supervision VLM					
FRU	Total Time	Instructions	Adjustments	Final Checks	Tools
Aspirating_Smoke_Detector_VLM-000-S (106 83 917) or Smoke_Detector_Configur._VLM-001-SIE (with higher alarm threshold for countries with strong environmental pollution) (109 15 504)	90 min	Please refer to: CB-DOC / Installation / Startup / System Start-up / Startup of the Gradient Supervision			Standard tool set

**WARNING**

Troubleshooting and repairs involving current sensors, gradient filters, gradient coil, and gradient cables may lead to problems with the system image orientation.

This could result in a misdiagnosis.

- » Perform the "Image Orientation Check" after performing all repair procedures and disconnecting/opening the gradient coil power circuit, e.g., replacement of:
 - » - Current sensors
 - » - Gradient filters
 - » - Gradient coil
 - » - Gradient cables



Adhere to the general safety guidelines in the safety chapter.



Never unplug the quick connectors of a water-cooled component when the water pump of the SEP is running or the KKT chiller has not been switched to the bypass via the IFP (high impact pressure).



Before returning the power stage, drain the water from the pipes by opening both ends.

**CAUTION**

High voltage across capacitor

- » Voltage decreases to a harmless level 20 minutes after turning off the cabinet.

2.1 D10 - DAC-Board

(Material number 75 62 924)

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment type 8005 from 3M Deutschland GmbH).

2.1.1 Tools and Time Required

2.1.1.1 Time

80 min.

2.1.1.2 Tools

- Standard tool set
- Ground test meter
- ESD equipment (097 02 606)

2.1.2 Preliminary



WARNING

High Voltage !

Contact with voltage-conducting parts leads to death or serious physical injury.

- » Adhere to the described steps for your own safety.



CAUTION

Switching-off circuit breaker F1/GPA does not remove the power from the GSSU Power Supply.

Risk of electrical shock !

- » Also switch-off circuit breaker F2/GPA before you start to work on the GSSU.

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged

3. Remove the front cover (→ Fig. 1 Page 23)

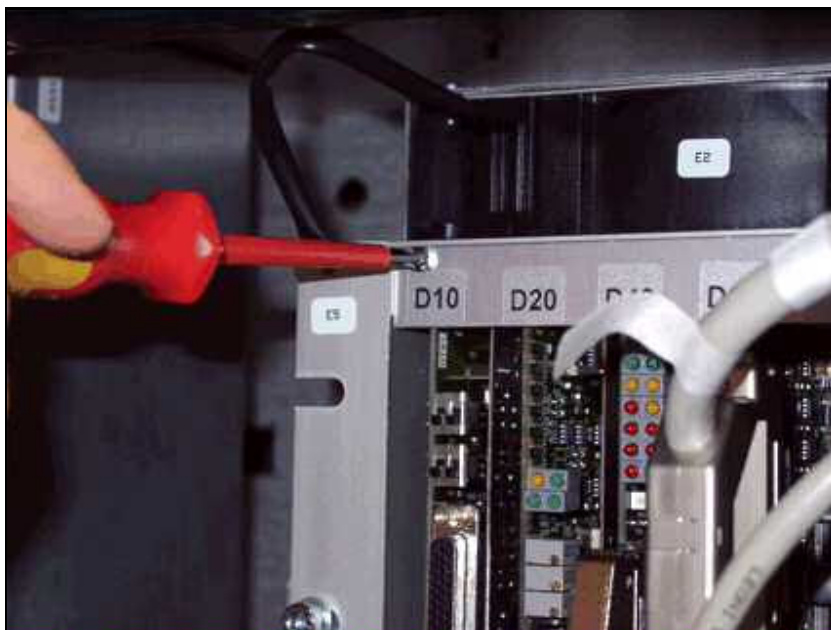
Fig. 1: GSSU-front cover - remove the 4 Allen screws



2.1.3 Disassembly

- Remove connector X3
- Remove upper and lower bracket

Fig. 2: Remove upper and lower bracket of the GSSU



- Carefully pull out D10

2.1.4 Assembly

- To install D10, proceed in reverse order

2.1.5 Concluding Steps

- Install the front cover
- Measure the protective earth connection resistance ≤ 100 mOhm with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.



Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

- Switch on GPA circuit breakers F1 and F2

2.1.6 Adjustments

Perform Tune-up

- Phantom Shim
- Gradient Delay
- Gradient Sensitivity
- Time: 5 min.

2.1.7 Final Checks

Perform Quality Assurance

- All selected procedures

2.2 D20 - Regulator Board

(Material number 75 62 940)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

2.2.1 Tools and Time Required

2.2.1.1 Time

155 min.

2.2.1.2 Tools

Standard tool set

2.2.2 Preliminary



WARNING

High Voltage !

Contact with voltage conducting parts leads to death or serious physical injury

» Adhere to the described steps for your own safety.



CAUTION

Switching-off circuit breaker F1/GPA does not remove the power from the GSSU Power Supply.

Risk of electrical shock !

» Also switch-off circuit breaker F2/GPA before you start to work on the GSSU.

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

3. Remove the front cover (→ Fig. 3 Page 26)

Fig. 3: GSSU-front cover - remove the 4 Allen screws



2.2.3 Disassembly

- Remove connector X10
- Remove upper and lower bracket (→ Fig. 2 Page 23)
- Carefully pull out D20

2.2.4 Assembly

- To install D20, proceed in reverse order
- Check jumper and switch settings if applicable

2.2.5 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2

2.2.6 Adjustments

Perform Tune-Up

- Regulator Adjust
- Phantom Shim
- CTC

- ECC
- Gradient Sensitivity
- Gradient Delay
- Dynamic Field Map (VD Software)
- Time: 45 min.

2.2.7 Final Checks

Perform Quality Assurance

- All selected procedures
- Time: 70 min.

2.3 D40 - Modulator Board

(Material number 75 62 965)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)



The jumper settings of the D40 (modulator assembly), p.n. 75 62 965, in the gradient cabinet is different for the various system types.

To prevent malfunctions, the D40 jumper setting ex factory has been set so it will not work in any system.

In case the D40 (Modulator assembly) needs to be replaced, check and adapt the D40 jumper settings according to the label attached to the D40.



Ignore this notice for system serial numbers >25096. For system serial numbers <25097 adhere to the following: if exchanging D40, p.n. 75 62 965 on system serial numbers <25097, check the revision level of the defective D40. If rev. level is 01 or higher, no additional action is necessary. For D40 with rev. level 00, also check the rev. level of the final stages, p.n. 75 63 955. The revision level of the final stages has to be >00 (check all three final stages); otherwise, also exchange the final stage(s) for rev.level 01, or higher.

2.3.1 Tools and Time Required

2.3.1.1 Time

155 min.

2.3.1.2 Tools

Standard tool set

2.3.2 Preliminary



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» Adhere to the described steps for your own safety

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

3. Remove the front cover (→ Fig. 4 Page 29)

Fig. 4: GSSU-front cover - remove the 4 Allen screws



2.3.3 Disassembly

- Remove connectors X3, 4, 5 from the top down
- Remove upper and lower bracket (→ Fig. 2 Page 23)
- Carefully pull out D40

2.3.4 Assembly

- To install D40, proceed in reverse order
- Check jumper and switch settings if applicable

2.3.5 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2

2.3.6 Adjustments

Perform Tune-Up

- Regulator Adjust
- Phantom Shim
- CTC

- ECC
- Gradient Sensitivity
- Gradient Delay
- Dynamic Field Map (only VD Software)
- Time: 45 min.

2.3.7 Final Checks

Perform Quality Assurance

- All selected procedures
- Time: 70 min.

2.4 D60 - CAN/Service

(Material number 75 62 981)

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment type 8005 from 3M Deutschland GmbH).

2.4.1 Tools and Time Required

2.4.1.1 Time

50 min.

2.4.1.2 Tools

- Standard tool set
- ESD equipment (097 02 606)

2.4.2 Preliminary



WARNING

High Voltage !

Contact with voltage conducting parts leads to death or serious physical injury.

- » Adhere to the described steps for your own safety



CAUTION

Switching-off circuit breaker F1/GPA does not remove the power from the GSSU Power Supply.

Risk of electrical shock !

- » Also switch-off circuit breaker F2/GPA before you start to work on the GSSU.

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

3. Remove the front cover (→ Fig. 5 Page 32)

Fig. 5: GSSU-front cover - remove the 4 Allen screws



2.4.3 Disassembly

- Remove connectors X20, X30
- Remove upper and lower bracket (→ Fig. 2 Page 23)
- Carefully pull out D60

2.4.4 Assembly

- To install D60, proceed in reverse order
- Check jumper and switch settings if applicable

2.4.5 Concluding Steps

- Install the front cover
- Measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.



Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

- Switch on GPA circuit breakers F1 and F2

2.4.6 Final Checks

- Regulator Check or Gradient Regulator Check
- Quality Assurance Expert / Coil Check

2.5 D70 - I/O Board

(Material number 75 63 005)

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment type 8005 from 3M Deutschland GmbH).

2.5.1 Tools and Time Required

2.5.1.1 Time

46 min.

2.5.1.2 Tools

- Standard tool set
- Ground test meter
- ESD equipment (097 02 606)

2.5.2 Preliminary



WARNING

High Voltage !

Contact with voltage conducting parts leads to death or serious physical injury

» Adhere to the described steps for your own safety



CAUTION

Switching-off circuit breaker F1/GPA does not remove the power from the GSSU Power Supply.

Risk of electrical shock !

» Also switch-off circuit breaker F2/GPA before you start to work on the GSSU.

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

3. Remove the front cover (→ Fig. 6 Page 35)

Fig. 6: GSSU-front cover - remove the 4 Allen screws



2.5.3 Disassembly

- Remove connectors X3, X4 and light conductor U1
- Remove upper and lower bracket (→ Fig. 2 Page 23)
- Carefully pull out D70

2.5.4 Assembly

- To install D70, proceed in reverse order
- Check jumper and switch settings if applicable

2.5.5 Concluding Steps

- Install the front cover
- Measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.



Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

- Switch on GPA circuit breakers F1 and F2

2.5.6 Final Checks

Perform Quality Assurance

- Regulator Check

Perform Quality Assurance / Expert

- Coil Check

2.6 Assembly Carrier

(Material number 75 64 060)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

2.6.1 Tools and Time Required

2.6.1.1 Time

56 min.

2.6.1.2 Tools

Standard tool set

2.6.2 Preliminary



WARNING

High voltage

Contact with voltage conducting parts will lead to death or serious physical injury

» Adhere to the described steps for your own safety.

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

3. Remove the front cover (→ Fig. 7 Page 38)

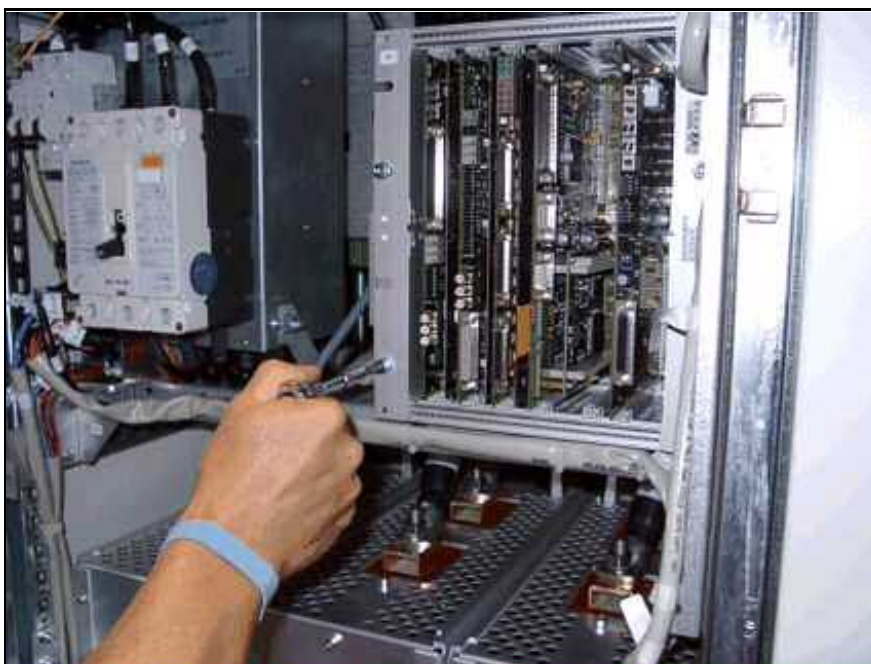
Fig. 7: GSSU-front cover - remove the 4 Allen screws



2.6.3 Disassembly

- Disconnect all cables connected to the GSSU boards
- Loosen the 4 Allen screws M5 securing the GSSU rack

Fig. 8: Remove Allen screws at GSSU rack



- Remove cable tie for power cable of Fan E2 unit

Fig. 9: Remove cable tie at the top of the GSSU



Fig. 10: Remove cable tie at the bottom of the GSSU



- Remove the flat connector from fan assembly E2 (red +17V/blue 0V)
- Pull out the sub-rack
- Pull out the boards

2.6.4 Assembly

- To install the GSSU rack, proceed in reverse order.

- When connecting the cables, ensure polarity is correct.

2.6.5 Final Checks

Perform Quality Assurance

- Regulator Check or Gradient Regulator Check

Perform Quality Assurance Expert

- Coil Check
- Time: 6 min.

2.7 Converter E4A VEGA650

(Material number 30 96 260)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

2.7.1 Tools and Time Required

2.7.1.1 Time

56 min.

2.7.1.2 Tools

Standard tool set

2.7.2 Preliminary



WARNING

High voltage

Contact with voltage conducting parts will lead to death or serious physical injury

» **Adhere to the described steps for your own safety**

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

3. Remove the front cover (→ Fig. 11 Page 42)

Fig. 11: GSSU-front cover - remove the 4 Allen screws



2.7.3 Disassembly

- Cover the final stage with a blanket to prevent the screws from falling in



If a metallic part (e.g. nut, washer) disappears into a power stage, you have to remove it, turn it upside down, and shake it carefully until the missing part falls out.

Each loose metallic particle in the power stage is highly dangerous and may cause a short circuit under power.

- Remove all cables from the front

- Loosen the 4 Allen screws (3 mm Allen key) to remove the converter

Fig. 12: Allen screw front side



- Remove the cable tie
- Move converter E4 to the front
- Remove the line power cable from the back side

Fig. 13: AC-DC back of E4



Cable connections converter E4:

SL1	I	I	I	I
	Gnd	0V	+17V	+17V
		Dark blue	Blue	Red
SL2	I	I	I	I
	Gnd1	Gnd2	V1	V2
SL3	I	I	I	I
	SL3.0VB	SL3.15VB	SL3.0VA	SL3.15VA
SL4	I	I	I	I
	SL4.Gnd1	SL4.Gnd2	SL4.24V1	SL4.24V2

2.7.4 Assembly

- To install converter E4A, proceed in reverse order

2.7.5 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2

2.7.6 Final Checks

Perform Quality Assurance

- Regulator Check or Gradient Regulator Check

Perform Quality Assurance Expert

- Coil Check
- Time: 6 min.

3.1 Current Sensor Unit

(Material number 75 64 037)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

3.1.1 Tools and Time Required

3.1.1.1 Time

295 min.

3.1.1.2 Tools

- Standard tool set
- SW 19 wrench
- 10 mm Allen key

3.1.2 Preliminary



WARNING

High voltage

Contact with voltage conducting parts will lead to death or serious physical injury

» Adhere to the described steps for your own safety.

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged
3. Remove the front cover (→ Fig. 20 Page 52)
4. Remove the GSSU assembly (→ Assembly Carrier / Page 37)

3.1.3 Disassembly

- Remove top cover of current sensor unit (4x5 mm Allen screws)

Fig. 14: Top cover of current sensor unit



- Remove gradient cable strain relief

Fig. 15: Top part of cord grip



- Remove gradient cable from current sensor unit (SW 19 wrench, Allen key 10 mm)

Fig. 16: Disconnect the gradient cable



- Remove the cable tie
- Disconnect cable W520 from D20
- Pull the cable W520 to the top

Fig. 17: W520 current sensor unit top view



- Use a ratchet with extension and a SW 19 wrench to open the screw connections of the gradient cables inside the cabinet

Fig. 18: Remove power cable from the current sensor unit



- Remove the screws for the sensor assembly on top of the GPA cabinet
- Lift the current sensor unit to the top

3.1.4 Assembly

- To install the Current Sensor Unit, proceed in reverse order

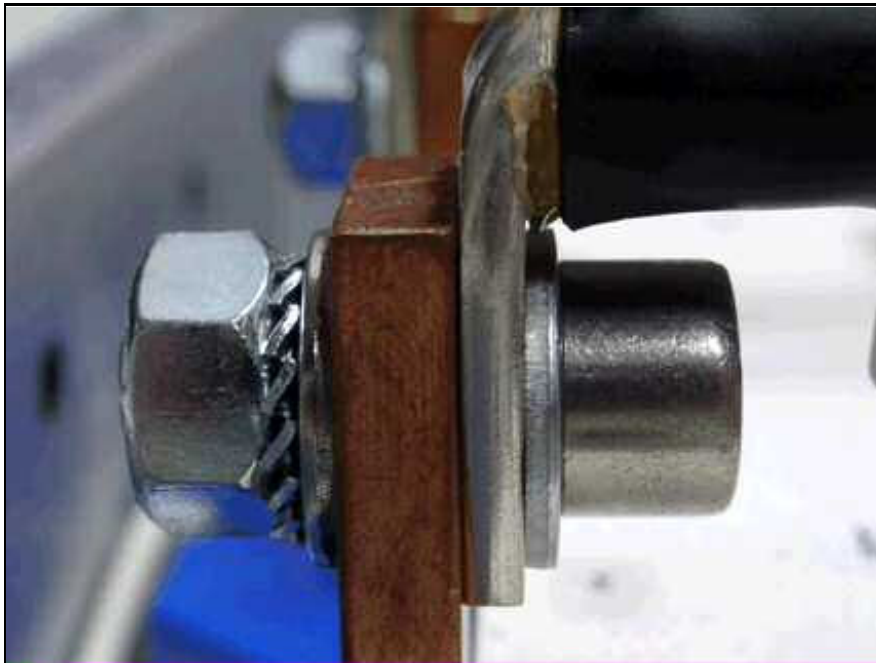
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Ensure that the screws, washers, lock washers, and nuts are positioned in the right order.

Apply a torque of 40 ± 4 Nm.

Fig. 19: Correct order



3.1.5 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2

3.1.6 Adjustments

Perform Tune-Up

- Regulator Adjust
- Phantom Shim
- CTC
- ECC
- Gradient Sensitivity
- Gradient Delay
- Dynamic Field Map (VD Software)
- Time: 45 min.

3.1.7 Final Checks

Perform Quality Assurance

- All selected procedures
- Image Orientation Check

- Time: 70 min.

3.2 Cable W520

(Material number 75 64 052)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

3.2.1 Tools and Time Required

3.2.1.1 Time

30 min.

3.2.1.2 Tools

- Standard tool set

3.2.2 Preliminary



WARNING

High Voltage !

Contact with voltage conducting parts leads to death or serious physical injury

» Adhere to the described steps for your own safety.

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged

3. Remove the front cover (→ Fig. 20 Page 52)

Fig. 20: GSSU-front cover - remove the 4 Allen screws



3.2.3 Disassembly

- Remove top cover of current sensor unit (4x5 mm Allen screws) (→ Fig. 14 Page 46)
- Disconnect cable W520 from D20
- Remove cable tie
- Pull cable W520 to the top
- Disconnect W520 from current sensor units

3.2.4 Assembly

- To install cable W520, proceed in reverse order.

3.2.5 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2

3.3 Current Control LA305S/500A

(Material number 30 96 070)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

3.3.1 Tools and Time Required

3.3.1.1 Time

37 min.

3.3.1.2 Tools

- Standard tool set
- SW 19 wrench

3.3.2 Preliminary



WARNING

High voltage

Contact with voltage conducting parts will lead to death or serious physical injury

» **Adhere to the described steps for your own safety**

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.
3. Remove the front cover (→ Fig. 20 Page 52)

3.3.3 Disassembly

- Remove top cover of current sensor unit (4x5 mm Allen screws) (→ Fig. 14 Page 46)
- Remove the gradient cable strain relief (→ Fig. 15 Page 46)
- Remove gradient cable from current sensor unit (SW 19 wrench, 10 mm Allen key) (→ Fig. 16 Page 47)
- Remove the cable tie
- Unplug connector P4, P5, P6 and P1, P2, P3
- Pull cable W520 out of the current sensor unit to the inside of the cabinet
- Lift up complete carrier with the 6 current sensor controls

- Remove the 2 Torx screws and replace current control

Fig. 21: Replace current control



3.3.4 Assembly

- To install the current control, proceed in reverse order.

3.3.5 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2

3.3.6 Final Checks

Perform Quality Assurance

- Regulator Check or Gradient Regulator Check
- Image Orientation Check

Perform Quality Assurance Expert

- Coil Check
- Time: 7 min.

3.4 Current Transformer 700A/IT700 700A/

Old single transformer unit: 86 75 329

New single transformer unit: 86 89 296

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

3.4.1 Tools and Time Required

3.4.1.1 Time

145 min.

3.4.1.2 Tools

- Standard tool set
- SW 19 Wrench

3.4.2 Preliminary



WARNING

Contact with voltage conducting parts will lead to death or serious physical injury
Adhere to the described steps for your own safety

» High voltage

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.
3. Remove the front cover (→ Fig. 20 Page 52)

3.4.3 Disassembly

- Remove top cover of current sensor unit (4x5 mm Allen screws) (→ Fig. 14 Page 46)
- Remove the gradient cable strain relief (→ Fig. 15 Page 46)
- Remove gradient cable from current sensor unit (SW 19 wrench, 10 mm Allen key) (→ Fig. 16 Page 47)
- Remove the cable tie
- Remove connector P1 (P2, P3) using a small screwdriver

- Remove the stabilization plate

Fig. 22: Loosen Allen screw



- Remove the current transformer

Fig. 23: Current transducer screw



3.4.4 Assembly

- To install the current transformer, proceed in reverse order.

3.4.5 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2

3.4.6 Adjustments

Perform Tune-Up

- Regulator Adjust
- Phantom Shim
- CTC
- ECC
- Gradient Sensitivity
- Gradient Delay
- Dynamic Field Map (VD Software)
- Time: 45 min.

3.4.7 Final Checks

Perform Quality Assurance

- All selected procedures
- Image Orientation Check
- Time: 70 min.

4.1 Power Stage XXL

(material number 106 63 309)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

4.1.1 Tools and Time required

4.1.1.1 Time

160 min.

4.1.1.2 Tools

- Standard tool set
- SW 19 wrench
- Manual pump RP50 (material number 81 15 136)

4.1.2 Preliminary



WARNING

High Voltage !

Contact with voltage conducting parts leads to death or serious physical injury

» **Adhere to the described steps for your own safety.**

1. Switch off GPA circuit breakers F1 and F2
2. **System with SEP**
 - Switch off F3 in the ACS (water pump)
 - Close ball valves OUTLET to ACC and INLET from ACC at roof of SEP
3. **System with IFP**
 - Switch off ACC/F102 (water cooling circuit)
 - Close ball valves ACC supply and ACC return on the roof of IFP
4. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

5. Remove the front cover (→ Fig. 24 Page 59)

Fig. 24: GSSU-front cover - remove the 4 Allen screws



4.1.3 Disassembly

System with IFP/SEP



Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit.

If drained cooling fluid (38% ethylene glycol water mixture) requires refilling, use a hand pump (service tool) or water pump with drill.

- Remove the quick connections of the water hoses from the inlet and outlet sides

Fig. 25: Disconnect water pipes



NOTICE

The water connections shut off automatically.

- » However, ensure that there are no water spills and that splashes of water are cleaned up with a cloth.

- Disconnect X1 from the final stage

Fig. 26: Removing X1 (final stage)



- Loosen the gradient cables on top of the final stage (SW 19 wrench)

Fig. 27: Power stage



- Disconnect the corresponding DCU supply connectors X31, X2, X32

Fig. 28: Disconnect X31, X2, X32 final stage



- Remove the cross beam in front of the three power stages

Fig. 29: Remove ground rail



- Remove the power stage with the help of the handles on top and front side (weight of power stage approximately 30 kg)

Fig. 30: Pull out final stage



To prevent the water connections from being damaged, do not set the final stage on the hoses.

4.1.4 Assembly

- To install final stage XXL, proceed in reverse order

4.1.5 Concluding Steps

- Install the GSSU front cover
- Re-open the ball valves at the SEP or IFP
- To refill the water circuit, refer to Installation Instructions.
- To check the water pressure, refer to Startup description.
- Switch on ACC/F102 or ACS/F3 (chilled water circuit)
- Switch on GPA circuit breakers F1 and F2
- For systems with Chiller/IFP: Bleed the chiller water pump, refer to Startup description.

4.1.6 Adjustments

Perform Tune-Up

- Regulator Adjust

- Phantom Shim
- CTC
- ECC
- Gradient Sensitivity
- Gradient Delay
- Dynamic Field Map (only VD Software)
- Time: 45 min.

4.1.7 Final Checks

Perform Quality Assurance

- All selected procedures
- Image Orientation Check
- Time: 70 min.



Before returning a replaced power stage, drain any residual water out of it. Hold the water pipes in the lowest possible position and open both quick connections with a suitable adapter or tool.



Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit. If drained cooling fluid (38% ethylene glycol water mixture) requires refilling, use a hand pump (service tool) or water pump with drill.

5.1 Choke Set

Choke set 13MY 400 A 10% serial no. 1001-1017 only

(Material number 30 98 977)

Choke set 120X138DRS76 028 from serial no. 1017

(Material number 77 51 055)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

5.1.1 Tools and Time Required

5.1.1.1 Time

175 min,

5.1.1.2 Tools

Standard tool set

SW 19 wrench

5.1.2 Preliminary



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» **Adhere to the described steps for your own safety.**

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

3. Remove the front cover (→ Fig. 31 Page 66)

Fig. 31: GSSU-front cover - remove the 4 Allen screws

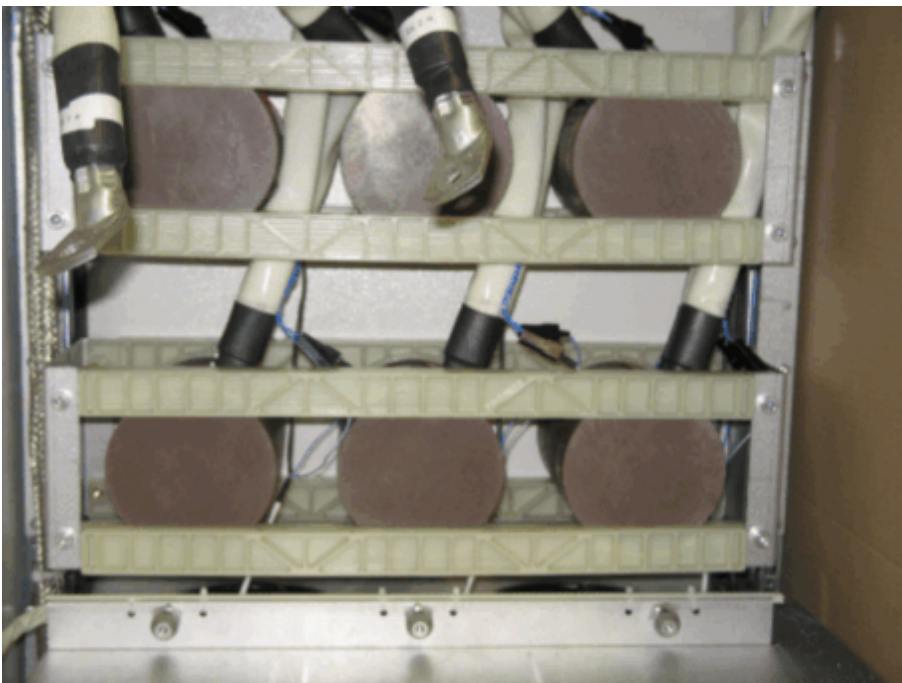


4. Remove all three final stages according to the replacement instructions

5.1.3 Disassembly

- Loosen the gradient cables of the choke set (affected axis) from the current sensor unit

Fig. 32: Access to the chokes after removal of the power stages

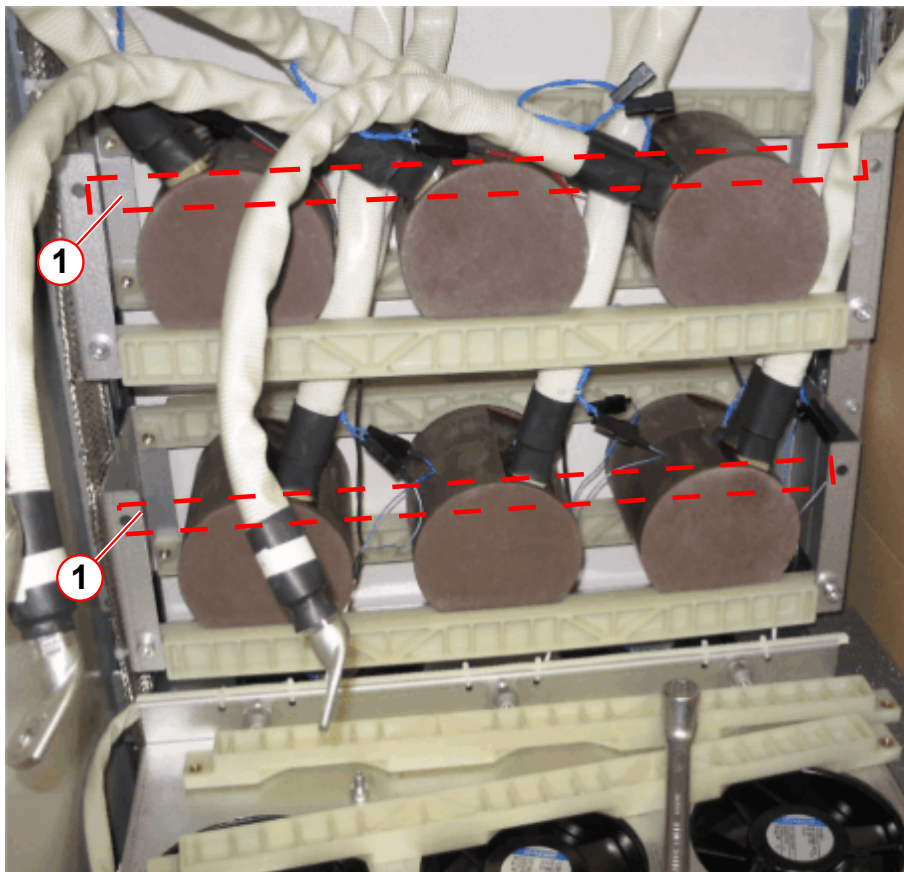


- Remove the upper securing bars of the chokes

Fig. 33: Loosen the screws of the securing bars



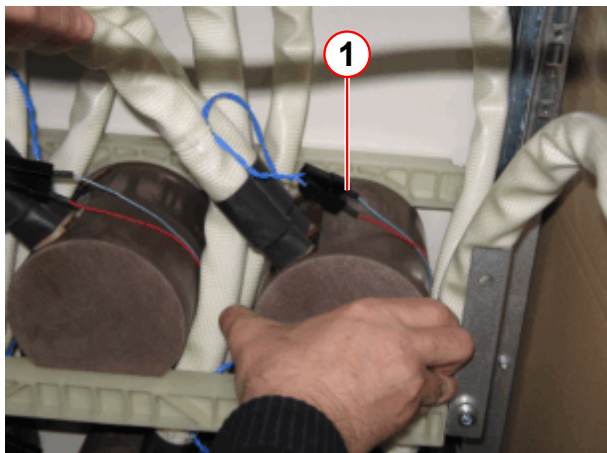
Fig. 34: Removing the Noise Suppression Chokes



(1) Upper securing bars have been removed.

- Disconnect the temperature sensor cable from the chokes

Fig. 35: Remove the choke



(1) Temperature sensor connector

- Remove the choke set

5.1.4 Assembly

- To install the choke set, proceed in reverse order

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Ensure that the screws, washers, and nuts are positioned in the correct sequence.

Apply a torque of 40 ± 4 Nm at the gradient cable connections of the Current Converter Unit box and the Power Stages.

5.1.5 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2

5.1.6 Adjustments

Perform Tune-Up

- Regulator Adjust
- Phantom Shim
- CTC
- ECC
- Gradient Sensitivity
- Gradient Delay

- Dynamic Field Map (VD Software)
- Time: 45 min,

5.1.7 Final Checks

Perform Quality Assurance

- All selected procedures
- Image Orientation Check
- Time: 70 min,

5.2 Fan box GPA XXL E3

(Material number 75 63 971)

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment type 8005 from 3M Deutschland GmbH).

5.2.1 Tools and Time Required

5.2.1.1 Time

70 min

5.2.1.2 Tools

- Standard tool set
- SW 19 open-ended wrench
- Ground test meter

5.2.2 Preliminary



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» Adhere to the described steps for your own safety.

1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

3. Remove the front cover (→ Fig. 36 Page 71)

Fig. 36: GSSU-front cover - remove the 4 Allen screws

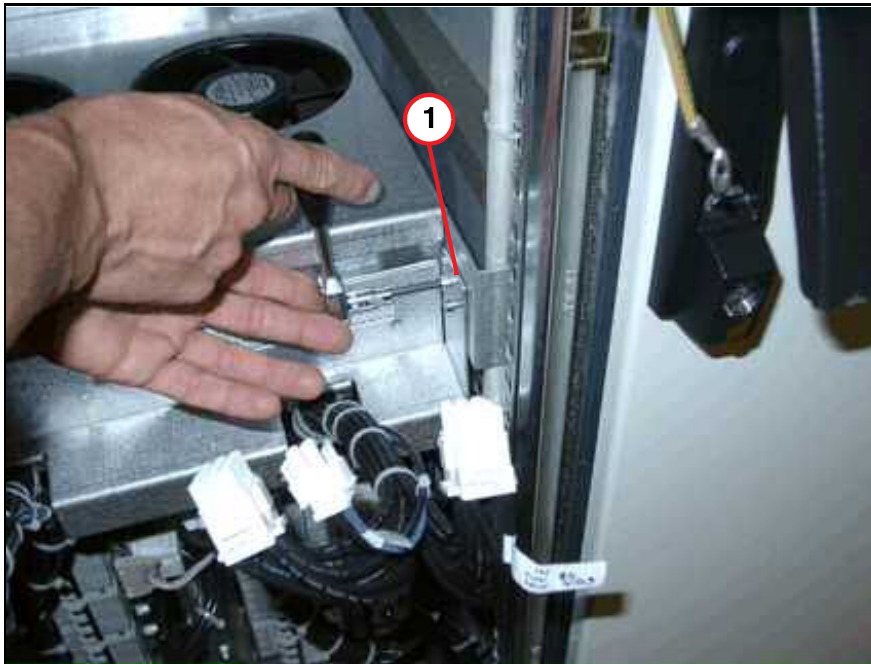


5.2.3 Disassembly

- Remove all 3 final stages (→ Power Stage XXL / Page 58)
- Remove Plexiglas cover from rectifier assembly

- Loosen the 2 screws for the mounting plate

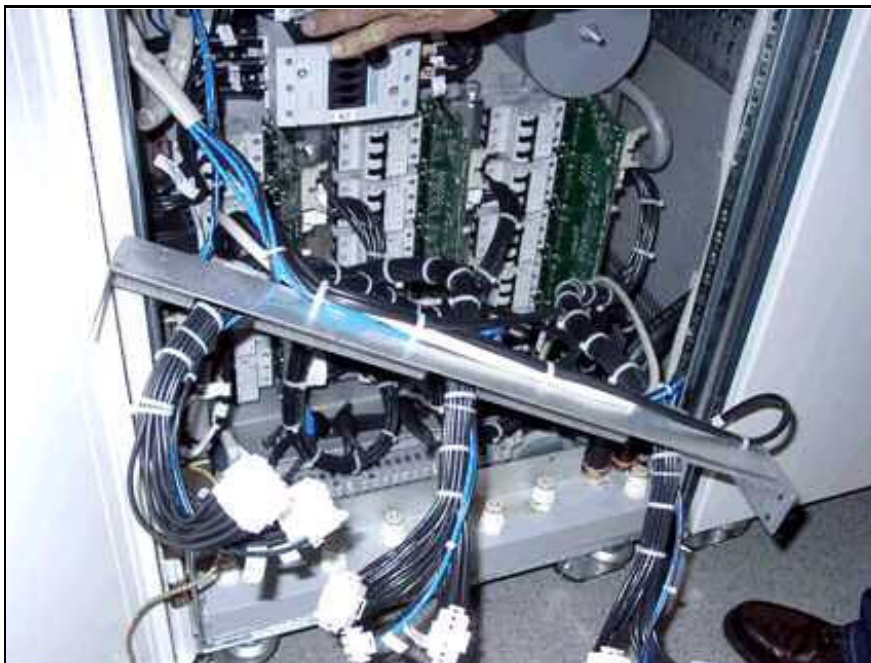
Fig. 37: Loosen the 2 screws at the E3 mounting plate



(1) screws

- Remove the mounting plate

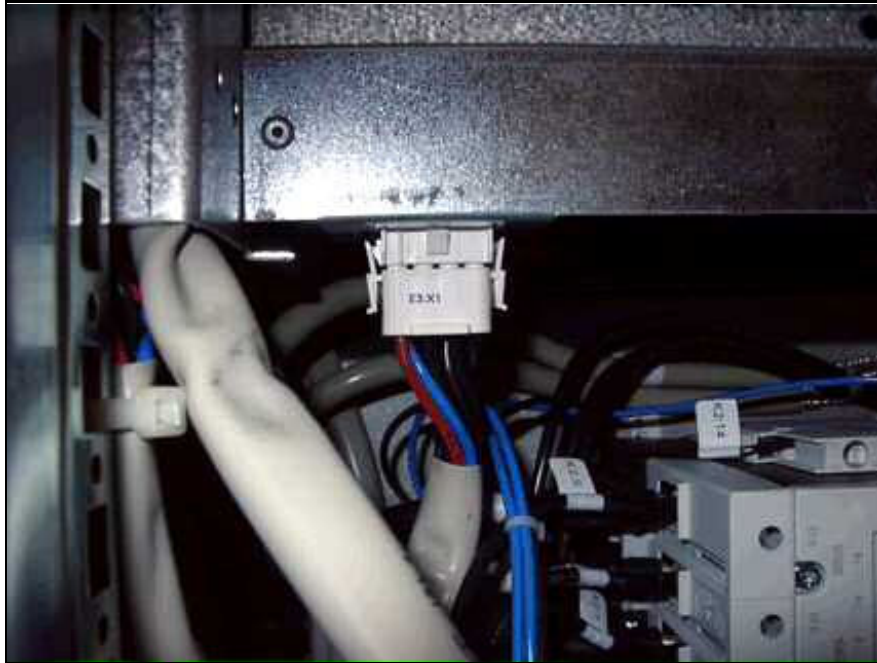
Fig. 38: Mounting plate E3 removed



- Disconnect the cables from rectifier board D100 (3 x X1)

- Disconnect cable E3.X1 from fan box

Fig. 39: Disconnecting power plug E3



- Pull out fan box E3

Fig. 40: Pull out fan box



5.2.4 Assembly

- To install the fan box, proceed in reverse order.

- Install final stage XXL (→ Power Stage XXL / Page 58)

5.2.5 Concluding Steps

- Install the front cover
- Switch on ACC/F102 or ACS/F3 (chilled water circuit)
- Switch on GPA circuit breakers F1 and F2

5.2.6 Final Checks

- Run **SeSo / Magnet & Cooling** and check EPC cooling status.
- Quality Assurance/ Image Orientation Check

6.1 D100 - Rectifier Board

(Material number 75 63 021)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

6.1.1 Tools and Time Required

6.1.1.1 Time

65 min,

6.1.1.2 Tools

Standard tool set

Heat- conductive paste (CONDU.PASTE Berulub, material number 02048650)

Manual pump RP50 (material number 81 15 136)

6.1.2 Preliminary



WARNING

High Voltage !

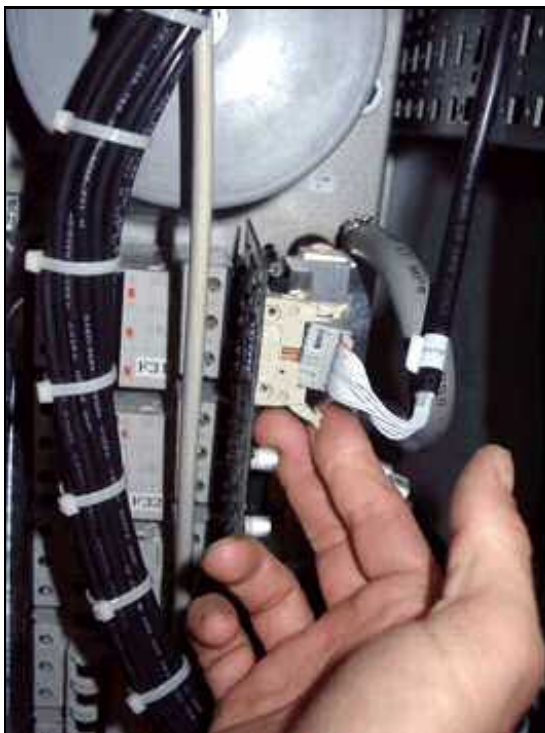
Contact with voltage conducting parts leads to death or serious physical injury

» Adhere to the described steps for your own safety.

1. Switch off GPA circuit breakers F1 and F2.
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.
3. **Systems with SEP:**
 - Close ball valves OUTLET to CCS and INLET from CCS at roof of SEP.
4. **Systems with IFP:**
 - Close ball valves CCS supply and CCS return on the roof of IFP.
5. Remove the quick connections of the water hoses from the inlet and outlet sides of the power stage.
6. Remove Perspex cover from rectifier assembly.

6.1.3 Disassembly

Fig. 41: Remove X1 on rectifier board



- Remove connector X1, X32, X31 from rectifier board D100.

Fig. 42: Remove X32 at the rectifier board

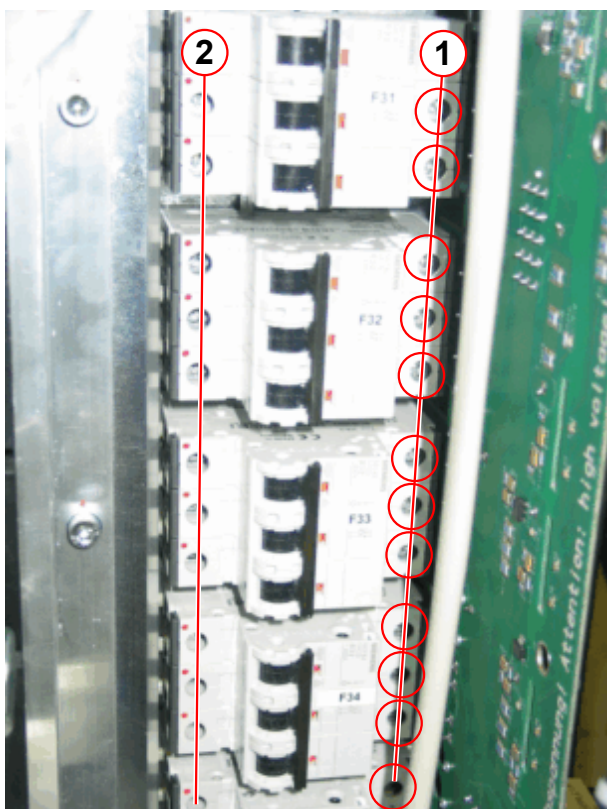


Fig. 43: Remove X31 at rectifier board



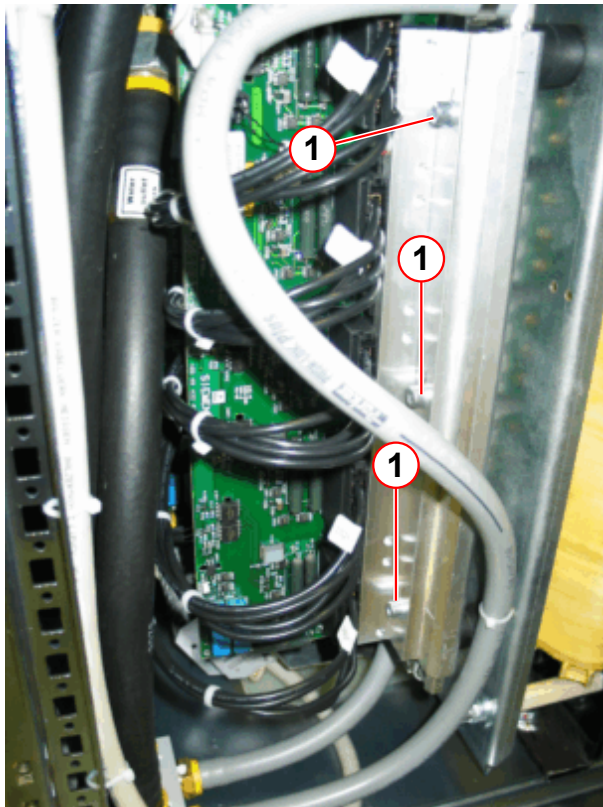
- Loosen all screws on the outlet of the circuit breakers(F11-F15, or F21-F25, or F31-F35) on the right side (Pos. 1).

Fig. 44: Circuit breaker F3x (32A)



- (1) Outlet side of circuit breaker
- (2) Inlet side of circuit breaker

Fig. 45: Heat sink on D100



(1) Screw - secures heat sink to water cooler

- Remove the 3 screws which secure the heat sink of the rectifier board to the water cooler.

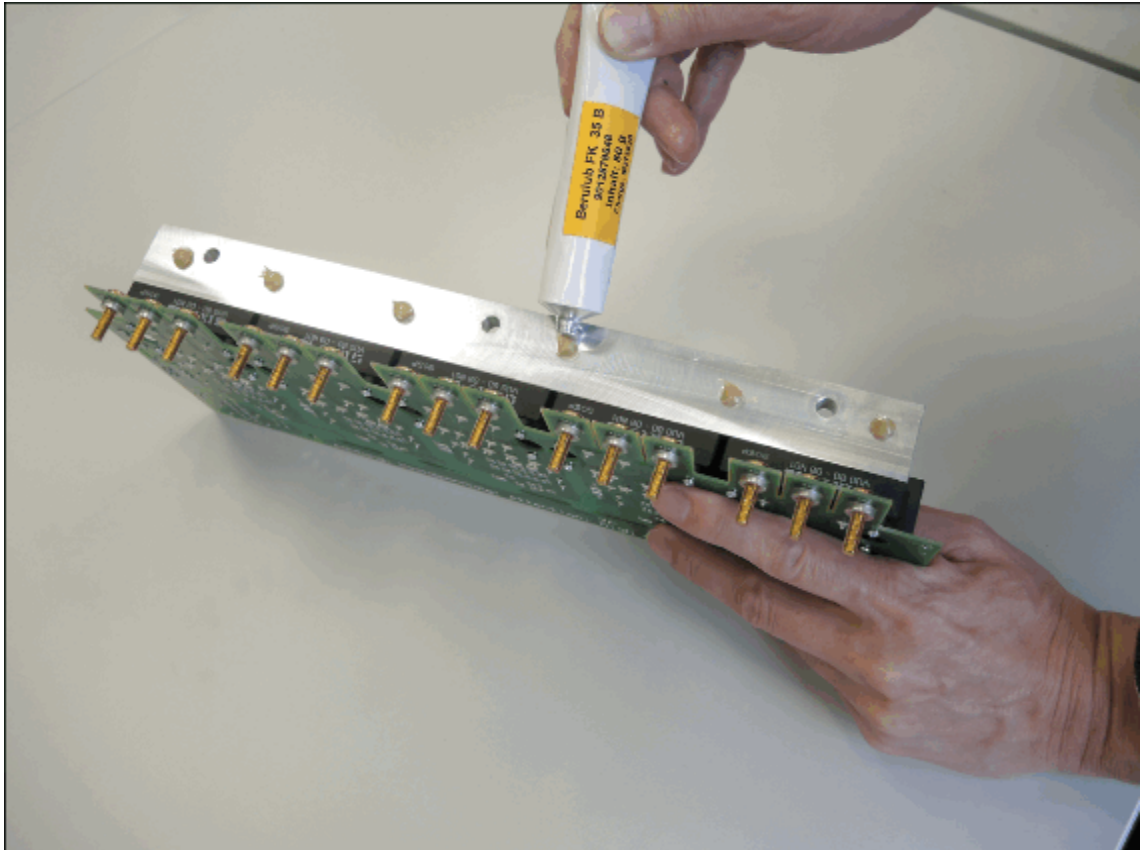
- Remove D100

6.1.4 Assembly

To install the D100, proceed in reverse order.

- Apply condu. paste Berulub to the heat sink, 6 dots like shown on (→ Fig. 46 Page 80).
- Install the D100.

Fig. 46: D100 with heat conductive paste



Make sure that the D100 connection bolts are in the right position as shown below.

Fig. 47: Wrong connected

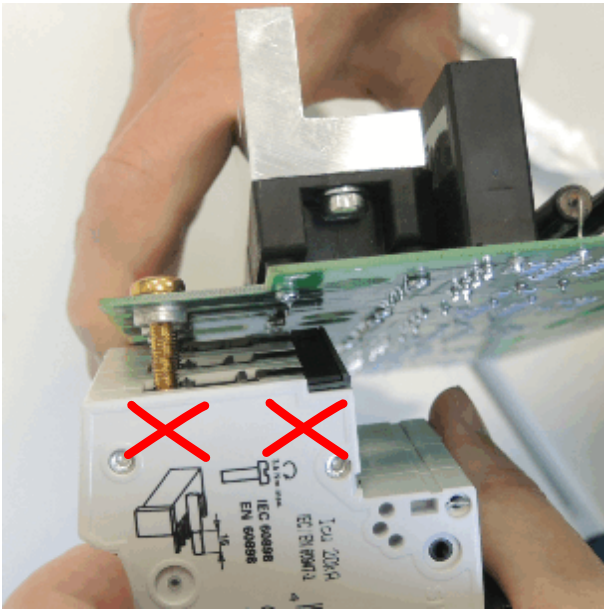
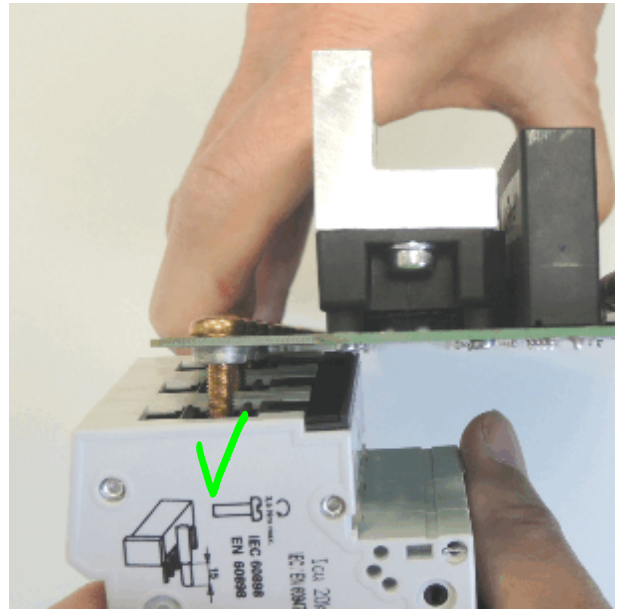


Fig. 48: Right connected



6.1.5 Concluding Steps

- Reconnect the connectors on D100.
- Install the Perspex cover.
- Reconnect water hoses.
- Re-open the ball valves at the SEP or IFP.
- To refill the water circuit, refer to Installation Instructions.
- To check the water pressure, refer to Startup description.
- Switch on ACC/F102 or ACS/F3 (chilled water circuit).
- Switch on GPA circuit breakers F1 and F2.
- For systems with Chiller/IFP: Bleed the chiller water pump, refer to Startup description.

6.1.6 Final Checks

Perform Quality Assurance

- Gradient Rise Time Check

Perform Quality Assurance / Expert

- Coil Check or BC Image Check

6.2 Circuit Breaker 32A

(Material number 77 48 770)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

6.2.1 Tools and Time Required

6.2.1.1 Time

45 min,

6.2.1.2 Tools

Standard tool set

6.2.2 Preliminary



WARNING

High voltage

Contact with voltage conducting parts will lead to death or serious physical injury

» Adhere to the described steps for your own safety.

1. Switch off SEP/MRE circuit breaker F102 on top of ACC cabinet.
2. Switch off GPA circuit breakers F1 and F2
3. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

6.2.3 Disassembly

- Remove the corresponding power stage
- Remove Plexiglas cover
- Unplug X1, X30, X31 D100 (→ Fig. 41 Page 76), (→ Fig. 42 Page 76)
- Loosen output screws of circuit breakers (F1- F15, F21-F25, F31- F35) of the associated D100 board (→ Fig. 44 Page 78)
- Remove the 3 screws for the heat sink
- Loosen the 3 input screws on the left of the defective circuit breaker (→ Fig. 45 Page 79)
- Remove D100 board

- Unlock circuit breaker mounting mechanism and pull it out

Fig. 49: Removing the circuit breaker



6.2.4 Assembly

- if needed, the D100 to the left of the defective circuit breaker should be removed for better access to the transformer tapping

Fig. 50: Installing the circuit breaker



- Install circuit breaker 32A.

- Reassemble the cabinet in the reverse order.

6.2.5 Concluding Steps

- Switch on SEP/MRE circuit breaker F102 on top of the ACC cabinet
- Switch on GPA circuit breakers F1 and F2.

6.2.6 Final Checks

Perform Quality Assurance Expert

- Coil Check
- Time: 5 min,

6.3 D110 - Power-Up Circuit Board

(Material number 75 63 047)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

6.3.1 Tools and Time Required

6.3.1.1 Time

36 min,

6.3.1.2 Tools

Standard tool set

6.3.2 Preliminary



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» Adhere to the described steps for your own safety.

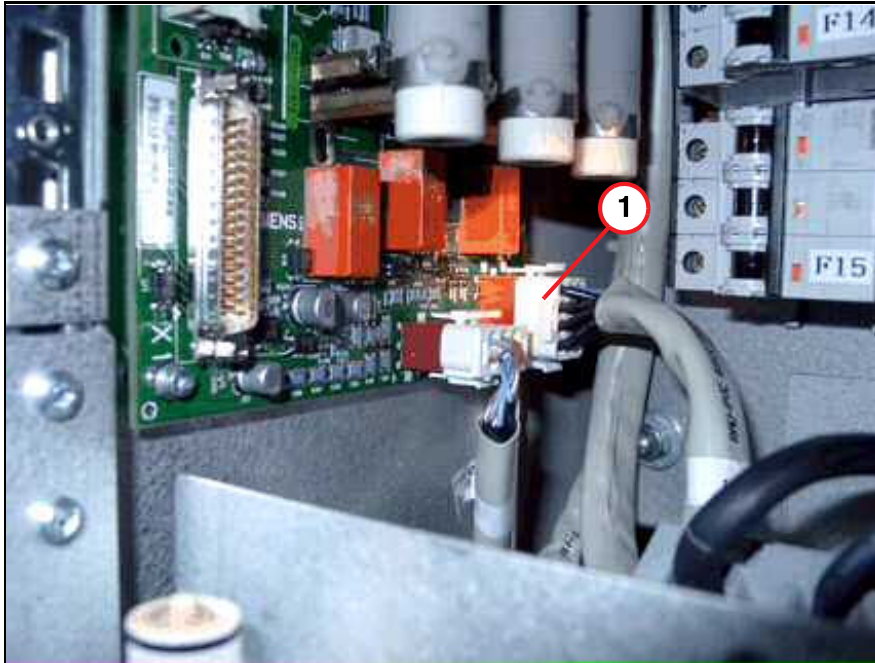
1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

6.3.3 Disassembly

- Remove the final stage water hoses from the water distributor and tie them up to one side using cable ties.
- Remove the Plexiglas cover.

- Unplug X5, X12, X10, X11, X12, X1, X25, and X24 from D110

Fig. 51: D110 - Remove connector



(1) Connector

Fig. 52: D110 - Remove 3 Torx screws



- Remove 3 Torx screws from the cabinet frame

- Remove D110 and carrier

6.3.4 Assembly

- To install the D110, proceed in reverse order.

6.3.5 Concluding Steps

- Install the Plexiglas cover
- Switch on GPA circuit breakers F1 and F2

6.3.6 Final Checks

Perform Quality Assurance

- Regulator Check or Gradient Regulator Check

Perform Quality Assurance Expert

- Coil Check
- Time: 6 min,

6.4 100A Contactor 24VDC 3S 100A 400V

(Material number 30 90 560)

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment type 8005 from 3M Deutschland GmbH).

6.4.1 Tools and Time Required

6.4.1.1 Time

90 min,

6.4.1.2 Tools

- Standard tool set
- Torx 25 bit

6.4.2 Preliminary



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» Adhere to the described steps for your own safety

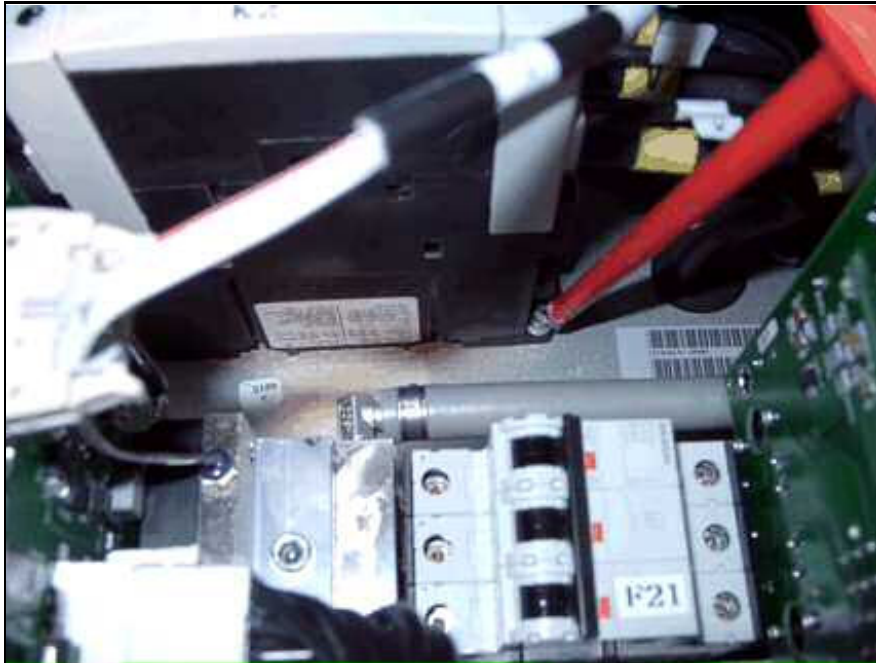
1. Switch off GPA circuit breakers F1 and F2
2. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.
3. Remove Plexiglas cover

6.4.3 Disassembly

- Remove X and Y final stages (→ Power Stage XXL / Page 58)
- Disconnect all leads from the K2 contactor

- Remove the K2 contactor (2 diagonal screws)

Fig. 53: Remove screw at K2



6.4.4 Assembly

- To install contactor K2, proceed in reverse order.

6.4.5 Concluding Steps

- Install Plexiglas cover
- Switch on GPA circuit breakers F1 and F2

6.4.6 Final Checks

Perform Quality Assurance Expert

Quality Assurance /

- Gradient Rise Time Check
- Image Orientation Check

Quality Assurance / Expert

- Coil Check or BC Image Check

7.1 Chilled Water Distributor

(material number 75 64 094, cooling water distributor without rectifier heat sinks and hoses)

 Adhere to all ESD regulations and use the appropriate tool set: Anti-static wristband and grounded ESD Field Service Kit (material number 097 02 606).

7.1.1 Tools and Time required

7.1.1.1 Time

180 min

7.1.1.2 Tools

- Standard tool set
- Torx driver bit
- Manual pump RP50 (material number 81 15 136)
- Filling adapter, delivered with the system.
- Bucket
- Water hose, delivered with the system.

7.1.2 Preliminary Steps



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» Adhere to the described steps for your own safety.

1. Switch the MR system to stand-by under **System>Control**
2. Switch off GPA circuit breakers F1 and F2.1
3. Wait at least 20 minutes to ensure that the final stage capacitors are fully discharged before opening the cover inside the GPA.
4. Switch off the helium compressor (warning at the alarm box can be reset with "acknowledge")
5. System with **SEP**:
 - Switch off SEP/MREF circuit breaker F102 on top of the ACC cabinet.
 - Close ball valves OUTLET to CBB and INLET from CBB on the roof of SEP.
6. System with **IFP**
 - Switch off the main switch on the chiller.

- Close the water supply and return valve on the IFP.

7.1.3 Disassembly

7.1.3.1 Drain Water from the ACS



Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit.

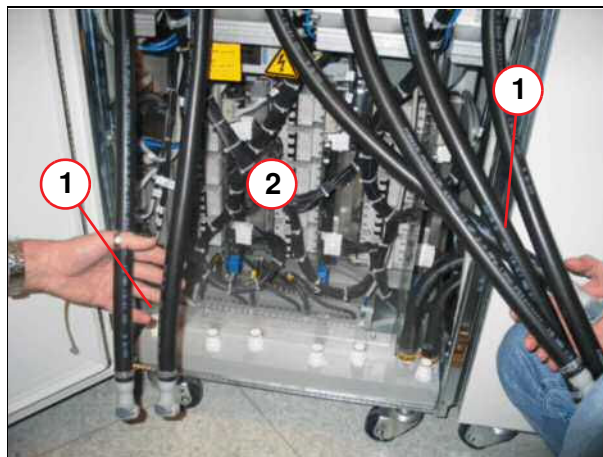
Fig. 54: ACS shutoff valve



(1) Open-end spanner

- Close the valves on top of the ACS.

Fig. 55: GPA cover



(1) Hoses
(2) Protective cover

- Disconnect the water hoses (1) to the power stages from the water distributor.

Fig. 56: ACS water filling



- (1) Return circuit
- (2) Supply circuit
- (3) Filling pipe

- Connect water hose with filling adapter fitted to the return circuit valve.

Fig. 57: Filling port

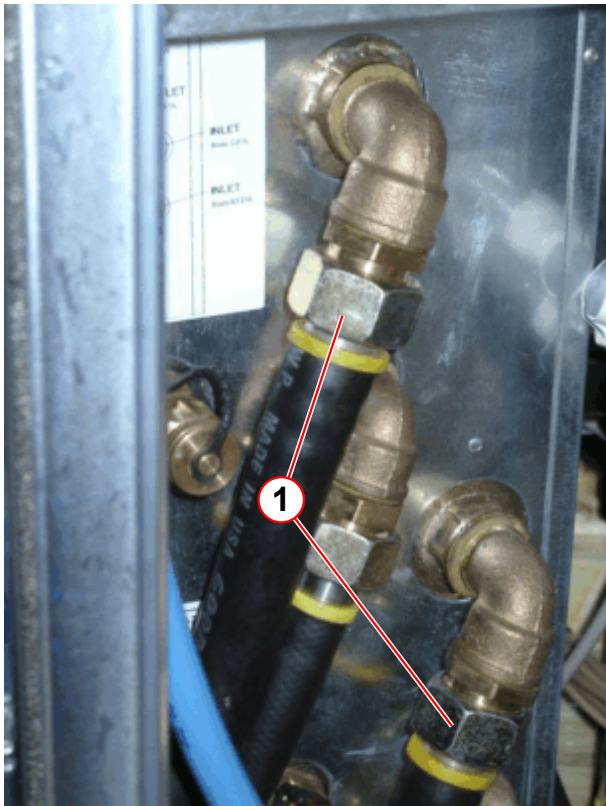


(1) Filling port valve

- Position the bucket in front of the cooler and put the cooler drain hose in it (the ACS holds approximately 8 l of water).
- Open the valve of the ACS return circuit port.
- Connect the water hose with the filling adapter fitted to the supply circuit valve.
- Open the valve of the ACS supply circuit valve.
- Vent ACS. Loosen the screws at the connection on top of the cabinet. Air entering the system will help drain the ACS.

7.1.3.2 Mechanical Removal of Water Distributor

Fig. 58: Water connection in ACS



- Remove the water supply and return hoses at the ACS.
(→ 1/ Fig. 58 Page 94)

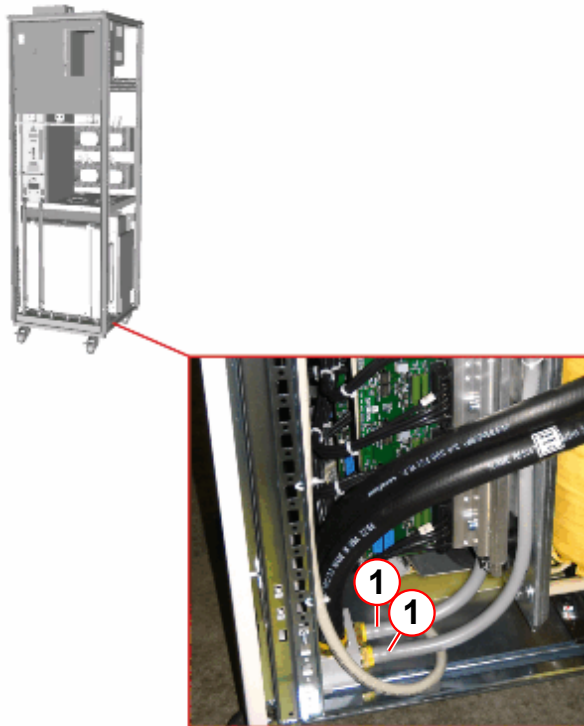
(1) Water connectors to be released

Fig. 59: Remove the water distributor.



- Ensure that the power stage capacitors are discharged and F1 is switched off. **Adhere to the Safety Information.**
- Remove the Perspex cover of the power stage supply in the GPA. (→ 2/Fig. 55 Page 91)
- Remove the lower part of the water distributor by loosening the screws on the right and left side.

Fig. 60: Cut the hoses from the water distributor to the heat sinks of D100



- At the old water distributor, the water hoses to rectifier D100 heat sinks must be cut off. Place absorbent paper under the hoses to catch leaking water. Cut directly behind the water nozzles, otherwise, the residual hose might be too short!
- Remove the plate from the water distributor and mount it at the new one.

(1) Cut the water hoses directly behind the connections

NOTICE

Incorrectly cut off water hoses may lead to insufficient residual hose lengths!
Pay attention and cut directly behind the water nozzles.

- » In case the remaining hose is too short,
- » order p.n. 100 48 115 (hose 10 mm).

7.1.4 Assembly

- To install the chilled water distributor, proceed in reverse order.
- Re-open the ball valves at the SEP or IFP
- To refill the water circuit, refer to Installation Instructions.
- To check the water pressure, refer to Startup description.

7.1.5 Concluding Steps

- System with **SEP**:
 - Switch on SEP/MREF circuit breaker F102 on top of the ACC cabinet
- System with **IFP**
 - Switch on the chiller via the main switch.
 - Bleed the chiller water pump, refer to Startup description.
- Switch on GPA circuit breakers F1 and F2.1.
- Switch on the helium compressor.
- Switch the MR system "ON" at the alarm box.
- Check the water flows and temperatures in the Service Software under **Magnet & Cooling> Cooling Status**.

8.1 Main Transformer GPA XXL

(Material number 75 63 989)

 Adhere to all ESD regulations and use the appropriate tool kit: anti-static wristband and grounded tabletop pad (TDK equipment type 8005 from 3M Deutschland GmbH).



In order to verify that the main transformer is defective, perform the measurements described under (→ Test of Main Transformer GPAXXL / Page 98) before ordering the transformer and the lifting device.

8.1.1 Tools and time required

8.1.1.1 Time

8 hrs

8.1.1.2 Tools

- Standard tool set
- Transformer lifting device (51 37 711)



A second person is required to support the job.

8.1.2 Preliminary Steps

8.1.2.1 Powering down of the cabinet

1. Switch off GPA circuit breakers F1 and F2.1



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

- » Wait at least 20 minutes to ensure that the final stage capacitors are fully discharged.



In order to verify that the main transformer is defective, perform the measurements before ordering the transformer and the lifting device.

8.1.3 Test of Main Transformer GPAXXL

8.1.3.1 Measurement preparation

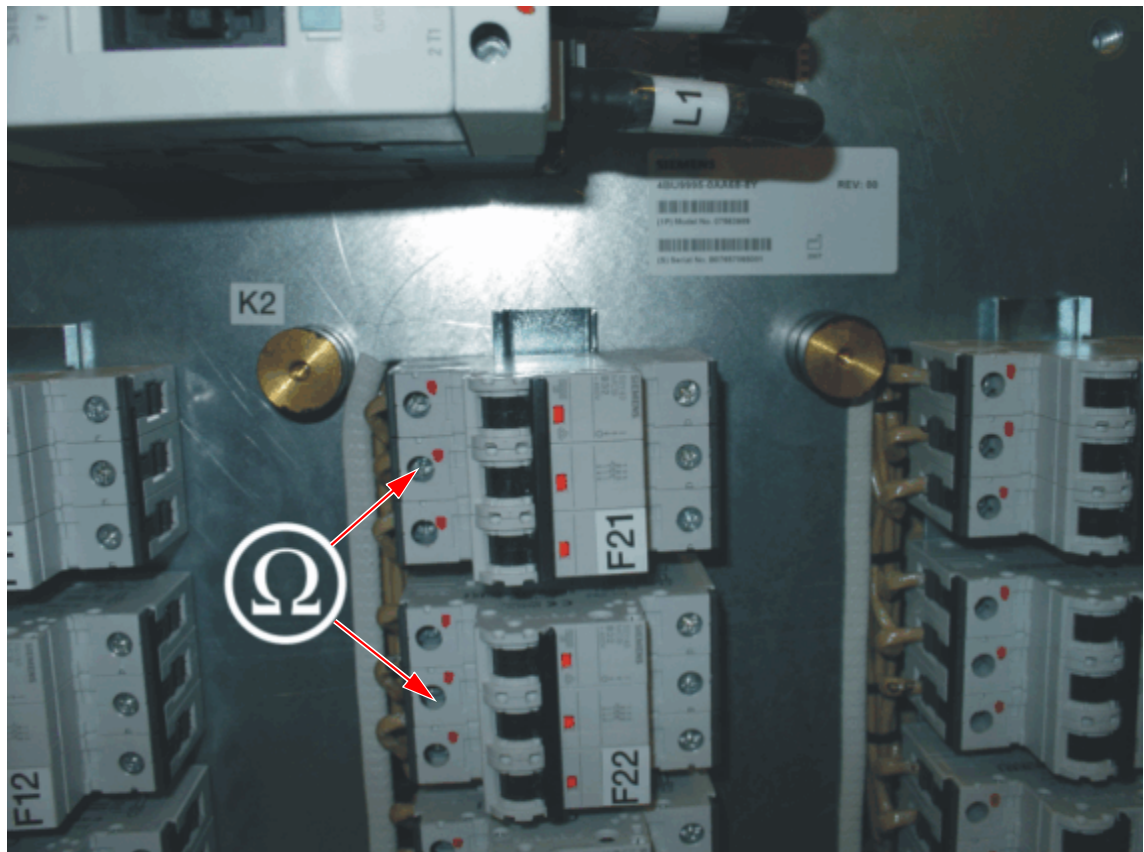
- Make sure that the cabinet has been powered down according to (→ Powering down of the cabinet / Page 97).
- Remove the final stage water hoses from the water distributor and tie them up to one side using cable ties.
- Remove the Plexiglass cover from the front of the transformer.
- Switch off all 15 automatic circuit breakers.

8.1.3.2 Measurement

1. Secondary - secondary

- With a DVM in the Ohm range, measure the resistance between every 2 secondary systems of the transformer.
- To do so, a test probe is to be connected with any of the three automatic circuit breaker terminals on the left (the three left-hand terminals of an automatic circuit breaker are connected together at a low impedance level using the wye-connected secondary winding of the transformer, this means that either terminal 2, 4 or 6 can be used).
- The other test probe is to be connected with a left-hand terminal from a different automatic circuit breaker.
- In this way, all $14+13+12+\dots+2+1 = 105$ combinations are to be tested.
 - » Normally, the measured value must be $>20\text{M}\Omega$, which in the case of a typical DVM is displayed as 'OL' or similar. Should a value $<20\text{M}\Omega$ be output, the transformer is defective.

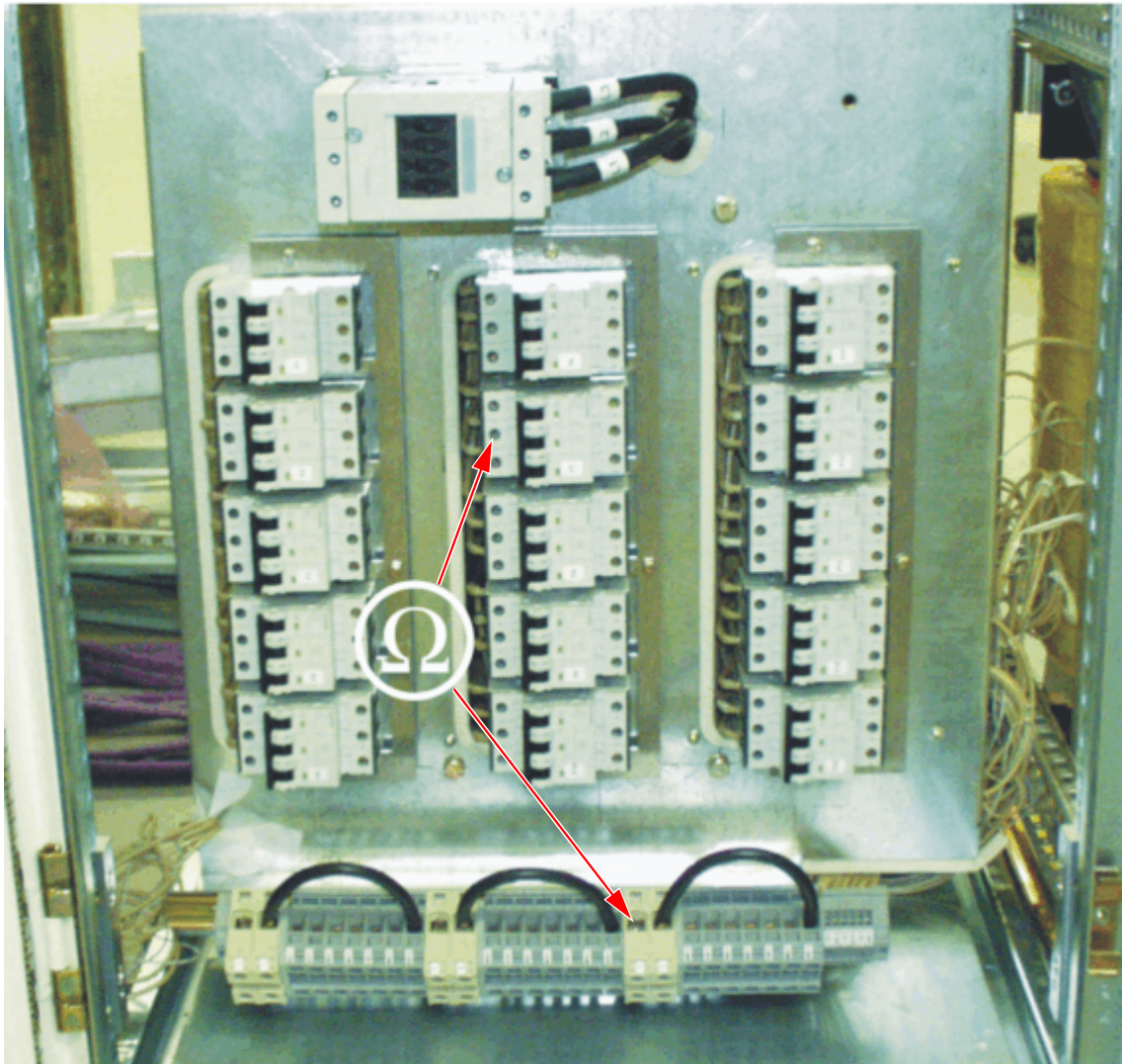
Fig. 61: Example: Measurement between F21 and F22



2. Primary - secondary

- To check the insulation between the primary and secondary side of the transformer, each of the 15 secondary systems must be measured against the primary side with the ohmmeter.
- To do so, a test probe is to be connected with one of the three left-hand terminals of an automatic circuit breaker. (The three left-hand terminals of an automatic circuit breaker are connected together at a low impedance level using the wye-connected secondary winding of the transformer, this means that either terminal 2, 4 or 6 can be used).
- The other test probe is to be connected with a terminal from the primary side. (Provided that the 3 black wire jumpers are inserted for voltage matching, all primary terminals are connected together at a low impedance level due to the delta connection on the primary side.)

Fig. 62: Example: Measurement between F22 and the primary side



3. Results

All measured values are to be entered into the white fields of the table (⇒ Fig. 63 Page 101). This is generally "∞" (infinite, >20MΩ).

- » In case one of the measured values deviates from the expected value, the transformer is defective and must be replaced (alternatively the complete GPA cabinet can be replaced).

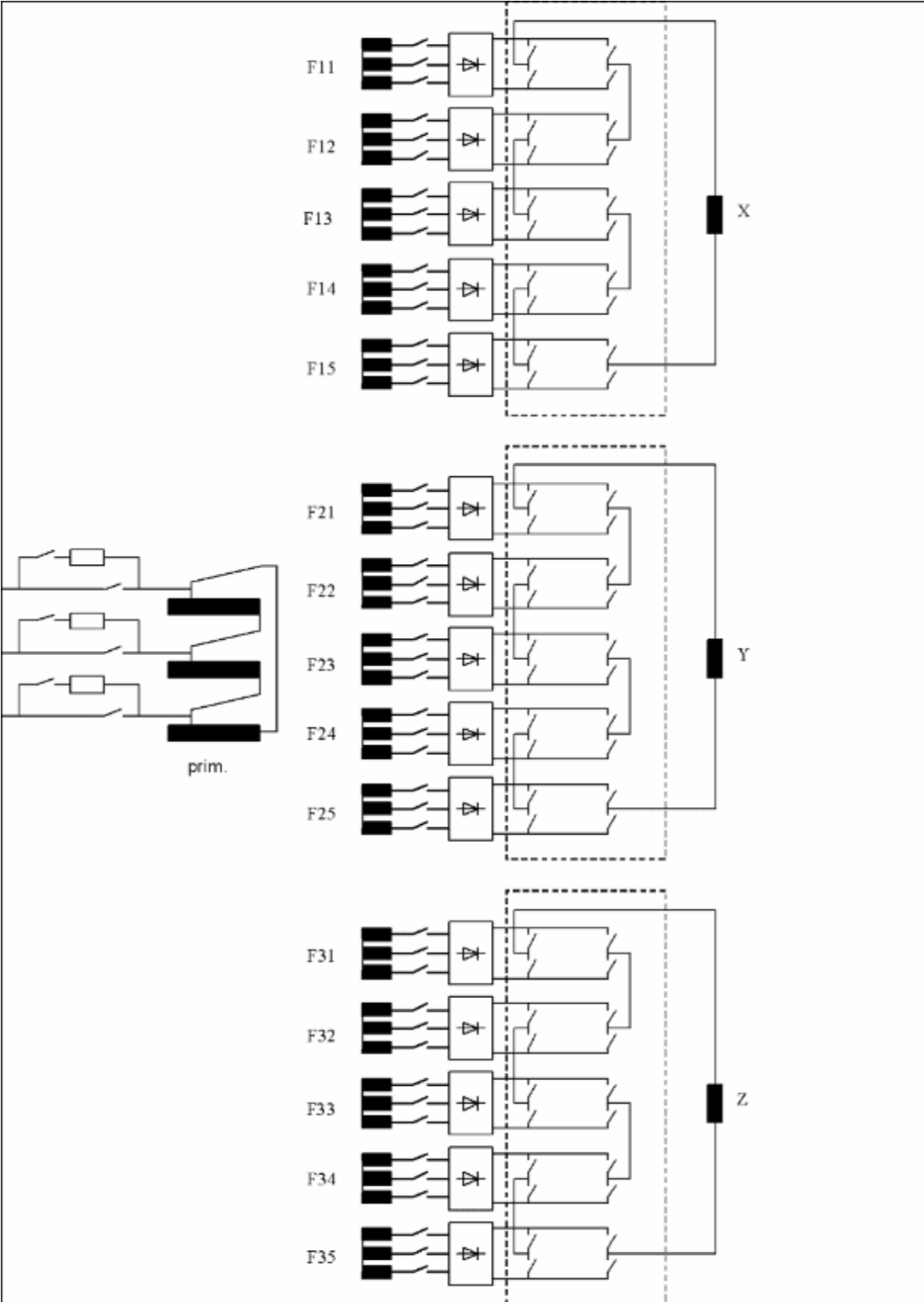
Fig. 63: Table with measured resistance values

[	F11	F12	F13	F14	F15	F21	F22	F23	F24	F25	F31	F32	F33	F34	F35
F11															
F12															
F13															
F14															
F15															
F21															
F22															
F23															
F24															
F25															
F31															
F32															
F33															
F34															
F35															
prim.															

4. Further Information

The transformer has a delta-connected primary winding with tapplings for adjusting to the country specific line voltage. The secondary windings consist of 45 windings which are located on the 3 legs. These are interconnected in wye to 15 secondary 3-phase systems (→ Fig. 64 Page 102). Each secondary system is fuse-protected by a 3-pole 32 A circuit - breaker and feeds a partial output stage of the 3 gradient axes by means of a three-phase bridge rectifier.

Fig. 64: Schematics regarding transformer wiring



8.1.4 Disassembly



The main transformer assembly must be removed from the rear side of the cabinet. It weighs approx. 450 kg and a second person is required to support the job.



WARNING

High voltage is present on the main power terminals on top of the GPA cabinet. Disconnecting the cables on top could lead to death or serious injury.

- » Switch off the primary electrical on-site mains power to the GPA cabinet.
- » Protect it against unintentional power-on.



WARNING

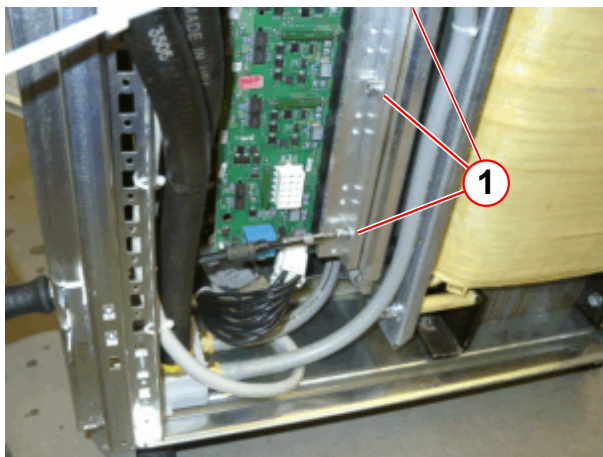
High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

- » Wait at least 20 minutes to ensure that the final stage capacitors are fully discharged.

- Isolate all electrical connections from the top plate of the GPA cabinet.
- Disconnect the GPA from the ACC and isolate the connections.
- Remove the side sections and disassemble the grounding cables.
- Remove the output stages.
- Remove the lower Plexiglas® cover (rectifier module).
- Remove connectors X1, X31, X32 from all 3 rectifier modules (D100).

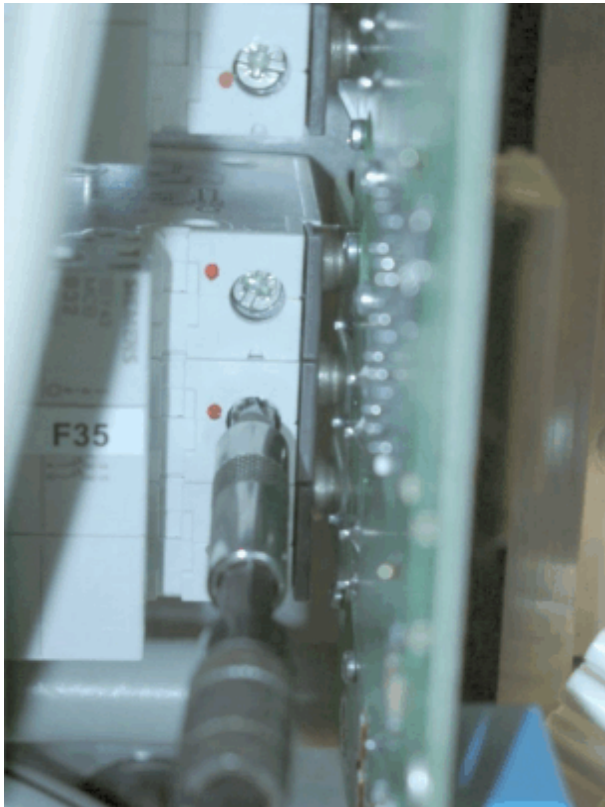
Fig. 65: Removal of aluminum cooling unit (D100 rectifier)



(1) 3 x Allen screws

- Remove 3 x hexagon Allen screws (→ 1/ Fig. 65 Page 103) of the aluminum cooling unit (D100 rectifier module).

Fig. 66: Loosening the 45 screws of the rectifier modules



- For each of the 3 rectifier modules, loosen all 15 screws on the right-hand side of the circuit breakers (→ Fig. 66 Page 104).

- Remove all three D100 rectifier modules.

Fig. 67: Disassembling the extruded profiles



- Disassemble the extruded profiles (→ Fig. 67 Page 104).
 - » The Pikatron transformer shown in the left photograph was replaced by another type on later GPA cabinets.

- Remove the cable ties from the water hoses and the supply line from the ON/OFF assembly.

Fig. 68: Removal of the fan assembly



- Disassemble the front plate of the fan (4 mm Allen screw, 4 x)

Fig. 69: Disconnecting the contactor cables and auxiliary switch cables



- Disconnect the contactor input cables (4 mm Allen screw) and auxiliary switch cables, these are not the wide cables on the right-hand side of the contactor. (depending on the manufacturer, L1, L2, L3 or U, V, W is labeled).



There are 2 different types of transformers used on the GPA cabinet providing the power for the small signal power supply. Please follow the appropriate description in either (→ Fig. 70 Page 106) or (→ Fig. 71 Page 106).

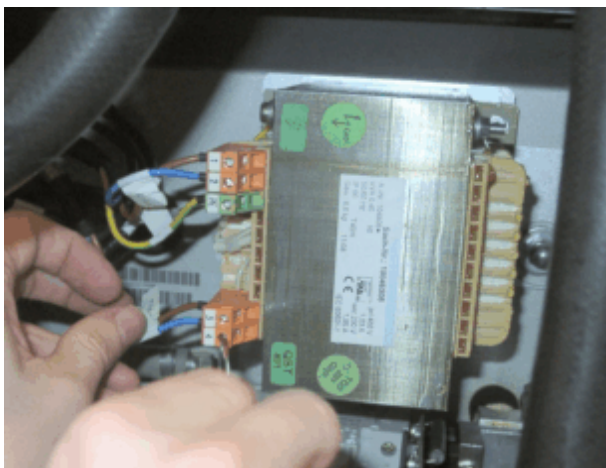
Fig. 70: Disconnecting the Pikatron transformer



- Depending on your site-specific configuration, either remove the Pikatron transformer output connections (or see below).

» The Pikatron transformer shown on the left photograph was replaced by another type from Tramag on later GPA cabinets (→ Fig. 71 Page 106).

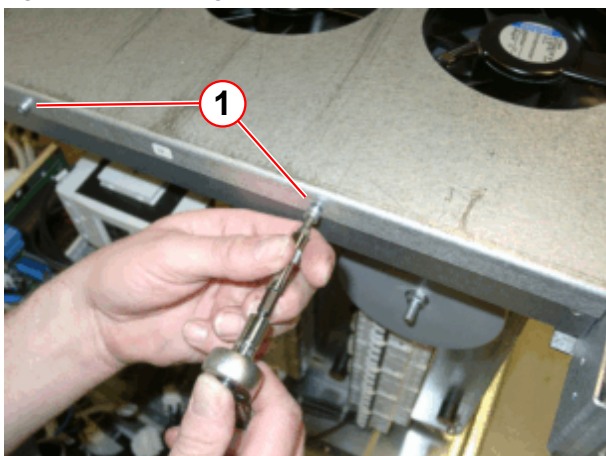
Fig. 71: Disconnecting the Tramag transformer



- Or remove the Tramag transformer output connections.

» The Tramag transformer shown in the left photograph is the replacement for the previously used Pikatron transformer, shown in (→ Fig. 70 Page 106).

Fig. 72: Unscrewing the fan module



- Remove the fan module at the front (4 mm Allen screw, 2 x).

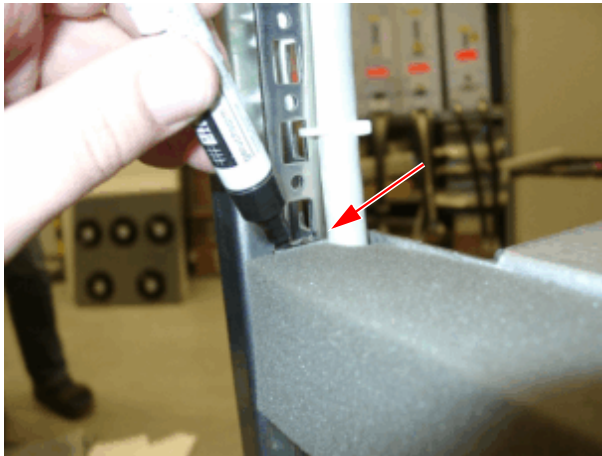
(1) 2 x 4mm Allen screws

Fig. 73: Removing the power connector from the fan module



- Remove the power connector X1 from the E3 fan module and take out the module.

Fig. 74: Marking the position of the fan module fastening plate



- Highlight the position of the fan module fastening plate.

Fig. 75: Removal of the fan module fastening plate



- Remove the fan module fastening plate (8 x Torx, size 25).

- Remove the mounting bracket of the main transformer (19 mm wrench and 4 x Torx, size 25)

Fig. 76: Position of the transformer mounting brackets

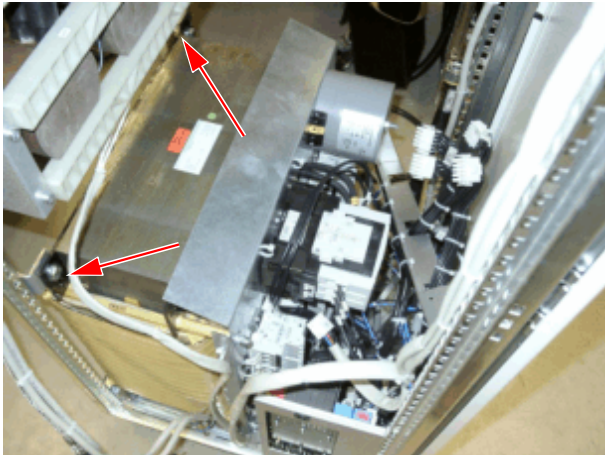


Fig. 77: Removing the main transformer mounting brackets

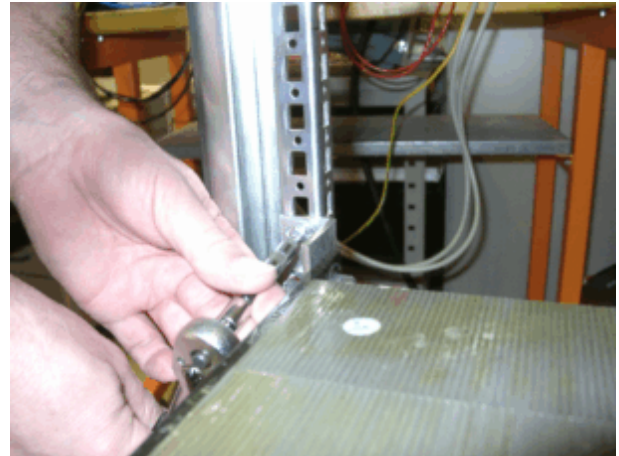


Fig. 78: Transformer base plate fixing points



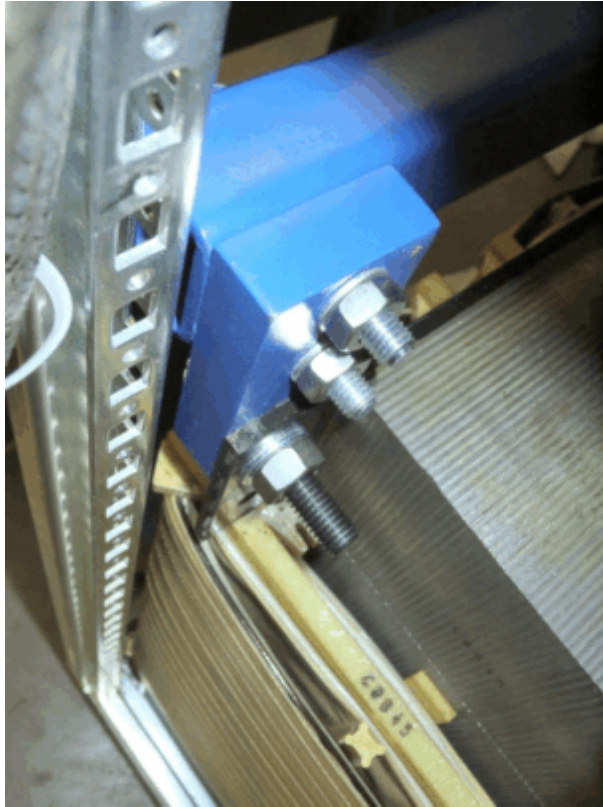
- Unscrew the transformer from the base plate (19 mm, 4 x).

Fig. 79: Grounding connection of the transformer



- Unscrew the grounding cable of the transformer (5 mm Allen screw).

Fig. 80: Lifting lugs of the lifting device attached to the transformer



- Move the lifting lug into position and attach it with screws to the transformer (if required, adjust the height with the hydraulic lifting device)

- Close the valve of the hydraulic lifting device by using the rod provided with the lifting device and by turning it clockwise.

Fig. 81: Closing the hydraulic valve of the lifting device (by turning it clockwise)

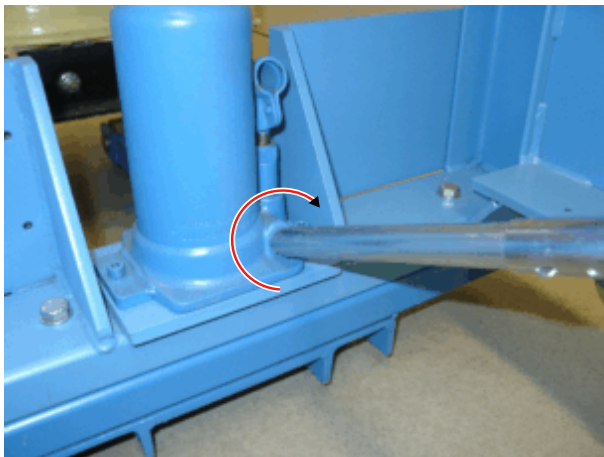
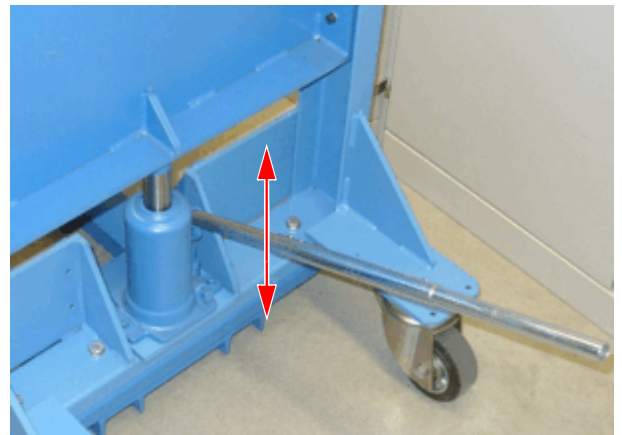


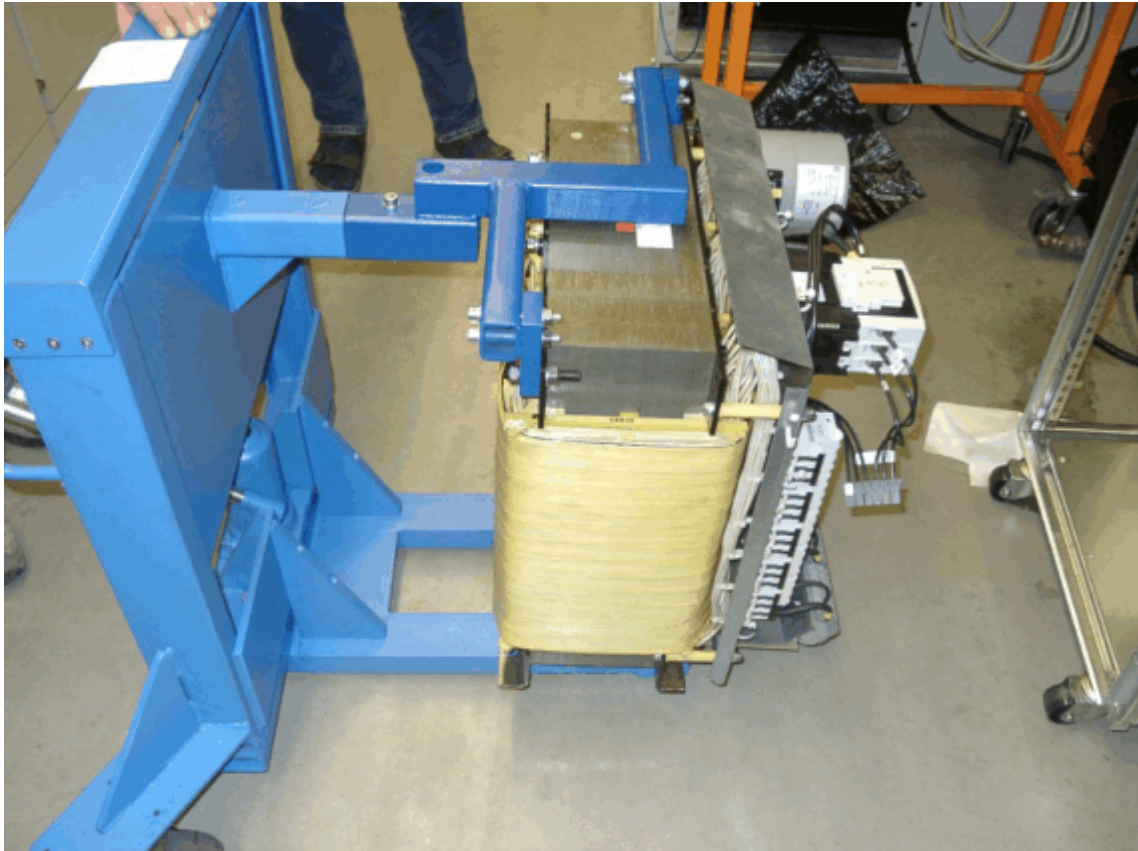
Fig. 82: Lifting the hydraulic piston



- » Lift the transformer until the cabinet can be moved freely.

- Push the cabinet out of the way (or retract the transformer).

Fig. 83: Removing the transformer unit



- Put the transformer down on a pallet or squared timbers for an intermediate period and unscrew the lifting lugs from the transformer.
- Attach the lifting lugs to the lifting eyes of the new transformer and pull all screws tight.

8.1.5 Assembly

- To install the transformer, proceed in reverse order.

8.1.6 Concluding steps

- The transformer is not defined as a repairable part and can be scrapped locally.

8.1.7 Final Checks

Performing Quality Assurance Checks:

- Regulator Check or Gradient Regulator Check
- Gradient Rise Time Check
- Eddy Current Compensation Check

- Gradient Delay Check
- Gradient Sensitivity Check
- Dynamic Field Map (with VD Software)
- Coil Check (QA- Expert mode)
- Run the Duty Cycle sequence
 - » Time: 25 min

9.1 General Information

Some spare parts in the GPAXXL (Avanto, Espree) and GPA90/22 (Verio) have changed.

In order to use the modified spare parts, some components have to be replaced along with the defect part (due to compatibility reasons). Please refer to (→ Tab. 5 Page 112) for information about the parts which must be replaced at the same time as the defective part (last column of the table).

The major change is caused by the introduction of a new small signal transformer which replaces the old style toroidal Pikatron transformer. This new transformer requires a different circuit breaker rated with 1.6 A (rather than the previously used 6A version).



The direct replacement of components (marked with “new” in the left column of the table below) is not explicitly described in this document, as it is a one-to-one replacement.

Tab. 5 FRU table line power distribution

Defective part	Kit to be ordered	Parts to be changed
■ Circuit breaker 6A (77 48 788)	Small signal transformer kit (100 46 633)	1.6A circuit breaker+small signal transformer+mount- ing plate
■ Pikatron transformer (30 98 381)	Small signal transformer kit (100 46 633)	1.6A circuit breaker+small signal transformer+mount- ing plate
■ Circuit breaker 100A (77 48 762)	Line power filter and small signal transformer kit (100 46 634)	Cpl.line power filter (includ- ing 100A and 1.6A circuit breaker)+small signal trans- former+mounting plate
■ Interference suppr. filter 3x80A (75 63 963)	Line power filter and small signal transformer kit (100 46 634)	Cpl.line power filter (includ- ing 100A and 1.6A circuit breaker)+small signal trans- former+mounting plate
■ Line power filter new	Line power filter and small signal transformer kit (100 46 634)	Cpl.line power filter (includ- ing 100A and 1.6A circuit breaker)
■ 100A circuit breaker new	Line power filter and small signal transformer kit (100 46 634)	Cpl.line power filter (includ- ing 100A and 1.6A circuit breaker)
■ 1.6A circuit breaker new	Small signal transformer kit (100 46 633)	1.6A circuit breaker
■ Small signal transformer new	Small signal transformer kit (100 46 633)	Small signal transformer

9.2 Line Power Filter Assembly

(Material number 100 46 634)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

9.2.1 Tools and Time Required

9.2.1.1 Time

65 min.

9.2.1.2 Tools

- Standard tool set
- Duspol or DVM
- Step ladder

9.2.2 Preliminary



In the case either the F2 circuit breaker, 100A circuit breaker or line power filter are defective, the complete line filter assembly has to be replaced.

If an old toroidal small signal transformer is used in the GPA cabinet, the transformer has to be replaced (in order to adapt the transformer to the new fuse rating of 1.6A). Refer to (→ Small Signal Transformer and Circuit Breaker F2 / Page 119) for the transformer replacement procedure.



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

- » Adhere to the described steps for your own safety.

1. Switch off the on-site power supply for the MR system



Ensure that it cannot be switched on again

Fig. 84: Switch off circuit breaker for the MR system



2. Remove the front cover of the small signal unit (→ Fig. 36 Page 71)
3. Check voltage on all contacts of F1 (use Duspolt or equivalent measurement device)

Fig. 85: Duspolt power check

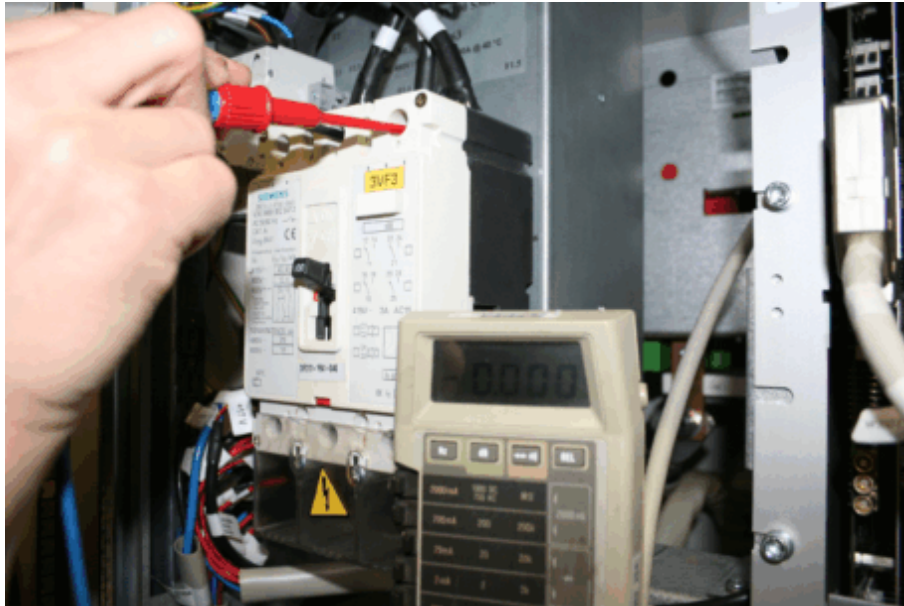


4. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

9.2.3 Disassembly

1. Remove the upper front cover (4 mm Allen key, 4x).
2. Verify that zero voltage is present at the 100A circuit breaker output (while it is switched on) (→ Fig. 86 Page 115).

Fig. 86: Checking the voltage at 100A circuit breaker output



3. Unscrew and remove the 3 power cables (F1.1, F1.3, F1.5) from the output of the 100A circuit breaker (→ Fig. 87 Page 115).
4. Unscrew the 2 leads on top of F2 (Phillips screwdriver / (→ Fig. 88 Page 115)).

Fig. 87: Removing the output cables from F1



Fig. 88: Removing leads from F2



5. Remove the cover from the mains input terminal on top of the GPA cabinet (4 mm Allen key, 4x / (→ Fig. 89 Page 116)).

Fig. 89: Removing the cover for the mains input terminal



6. Unscrew the cable clamp (→ Fig. 90 Page 116), all connections from the mains input terminal (L1, L2, L3, PE) and the 4 Allen screws (5 mm Allen key) holding the line filter in place (→ Fig. 91 Page 116).

Fig. 90: Removing the cable clamp and mains leads

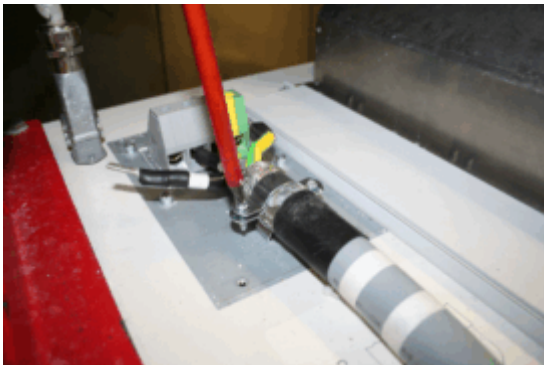
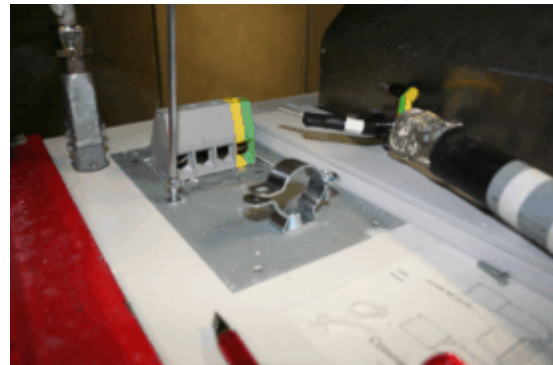


Fig. 91: Removing Allen screws from the line filter



7. Remove the nut from the right side of the line filter housing (13 mm open wrench / (→ Fig. 92 Page 117)).

Fig. 92: Removing the nut from the line filter housing



8. Carefully remove the line filter assembly (if it is secured with a cable tie, cut it / (→ Fig. 93 Page 117)).



Be prepared for the weight of the filter (~12 kg) and hold it firmly while removing it.

Fig. 93: Cutting the cable tie at the line filter assembly



9.2.4 Assembly

1. Insert the new line filter assembly (→ Fig. 94 Page 118) and secure it with the nut (13 mm open wrench) and the 4 Allen screws.

Fig. 94: Inserting the new line filter assembly



2. Reassemble cables and covers in reverse order

9.2.5 Concluding Steps

- Turn both circuit breakers (1.6 A and 100 A) ON and start up the main power.



Make sure the compressor unit restarts after on site power-on.

9.2.6 Final Checks

Perform Quality Assurance Expert

- Coil Check
- Time: 5 min.

9.3 Small Signal Transformer and Circuit Breaker F2

(Material number 100 46 633)

 Adhere to ESD regulations and use the appropriate tool kit: Anti-static wristband and grounded tabletop pad. (TDK equipment Type 8005 from 3M Deutschland GmbH)

9.3.1 Tools and Time Required

9.3.1.1 Time

50 min.

9.3.1.2 Tools

- Standard tool set
- Duspol or DVM

9.3.2 Preliminary



If the old style 6 A circuit breaker and the old style toroidal small signal transformer are defective, both parts must be replaced simultaneously. The new transformer requires a different rated fuse current. For this reason, a different circuit breaker (1.6 A) has to be used.



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

- » Adhere to the described steps for your own safety

1. Switch off the on-site power supply for the MR system.



Ensure that it cannot be switched on again

Fig. 95: Switch off circuit breaker for the MR system



2. Remove the front cover of the small signal unit (→ Fig. 36 Page 71)
3. Check voltage on all contacts of F1 (use Duspolt or equivalent measurement device)

Fig. 96: Duspolt power check



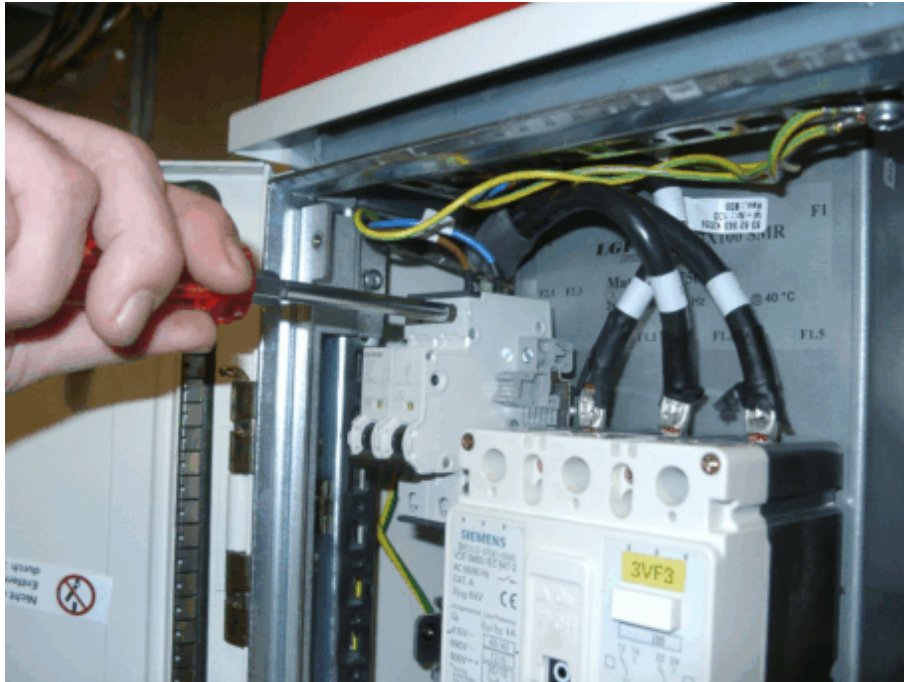
4. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.

9.3.3 Disassembly

Circuit Breaker F2

1. Remove the upper front cover (4 mm Allen key, 4x).
2. Unscrew the 4 leads from F2 (Phillips screwdriver / (→ Fig. 97 Page 121)).

Fig. 97: Removing 4 leads from F2



3. Unlock the old 6 A circuit breaker and pull it out (→ Fig. 98 Page 121).

Fig. 98: Unlocking and removing the old circuit breaker

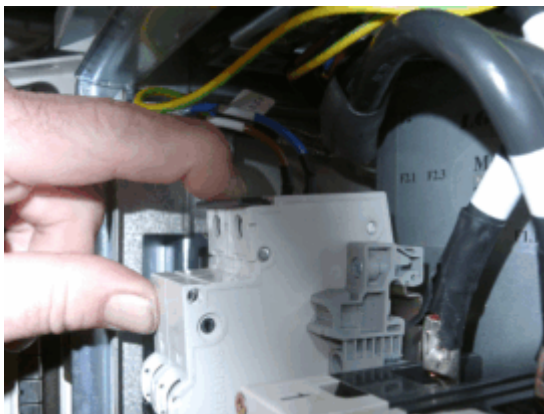


Fig. 99: Inserting and locking the new circuit breaker



9.3.4 Assembly

Circuit Breaker F2

1. Insert and lock the new 1.6A circuit breaker (→ Fig. 99 Page 121).
2. Reinsert the 4 leads and pull them tightly.

9.3.5 Disassembly

Small Signal Transformer T11

1. Uncouple the 6 water hoses from the water distributor (→ Fig. 100 Page 122).
2. Remove the lower perspex cover (4mm Allen key, 2x / (→ Fig. 101 Page 122)).

Fig. 100: Uncoupling the water hoses



Fig. 101: Removing the perspex cover



3. Unplug the lower connectors from the final stages (X31, X32 and X2).
4. Remove the cross bar at the lower end of the final stages (→ Fig. 102 Page 122).

Fig. 102: Removing the cross bar



5. Unscrew and remove the 5 power leads from the old transformer with a small screwdriver (→ Fig. 103 Page 123).

Fig. 103: Removing the 5 power leads

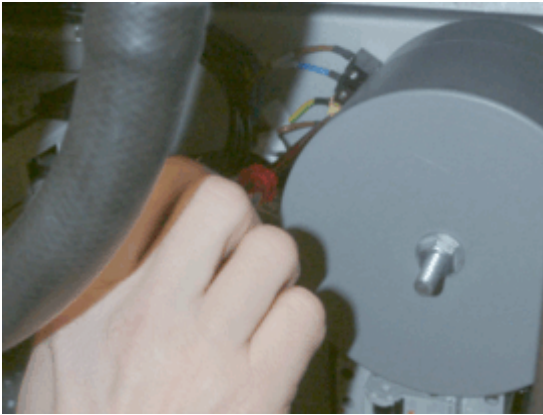
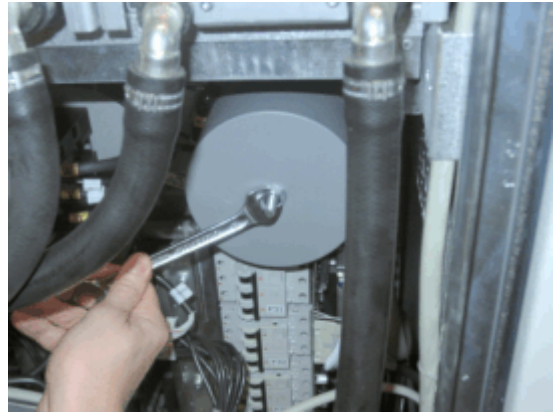


Fig. 104: Unscrewing the old transformer



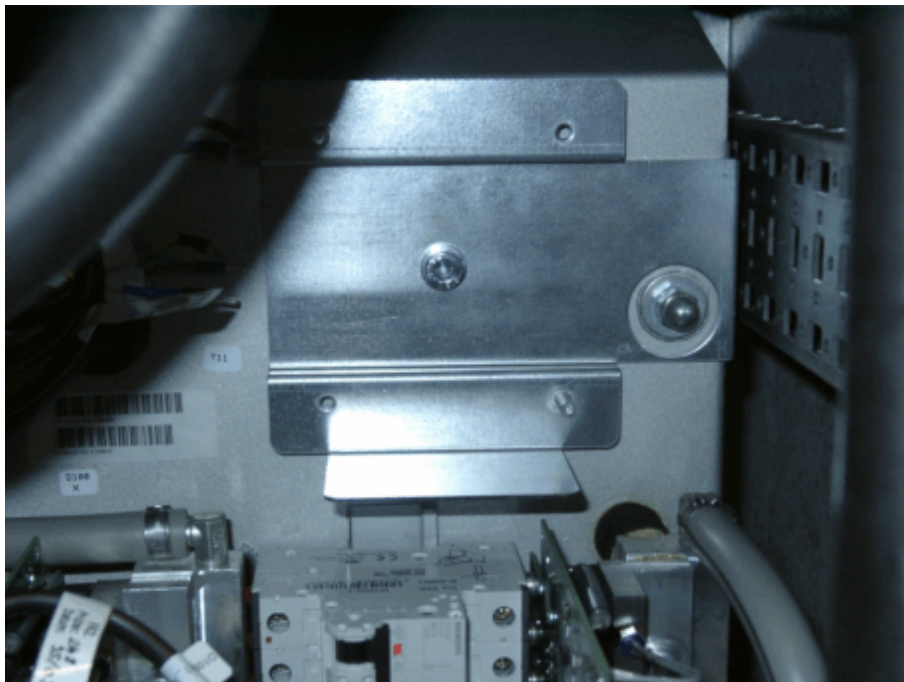
6. Remove the 13 mm nut from the center bolt of the old transformer and pull it out slowly, be prepared for the weight (→ Fig. 104 Page 123).
7. Remove the center bolt completely by turning it counter-clockwise.

9.3.6 Assembly

Small Signal Transformer T11

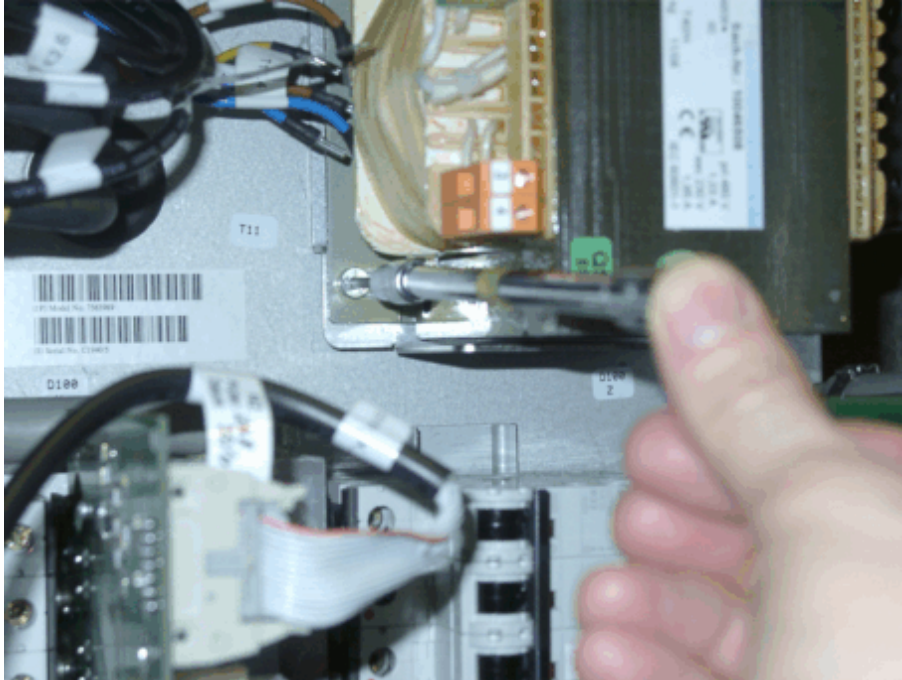
1. Fix the new mounting plate provided with the kit, using a 6 mm Allen screw and washer (→ Fig. 105 Page 123).

Fig. 105: New mounting plate installed



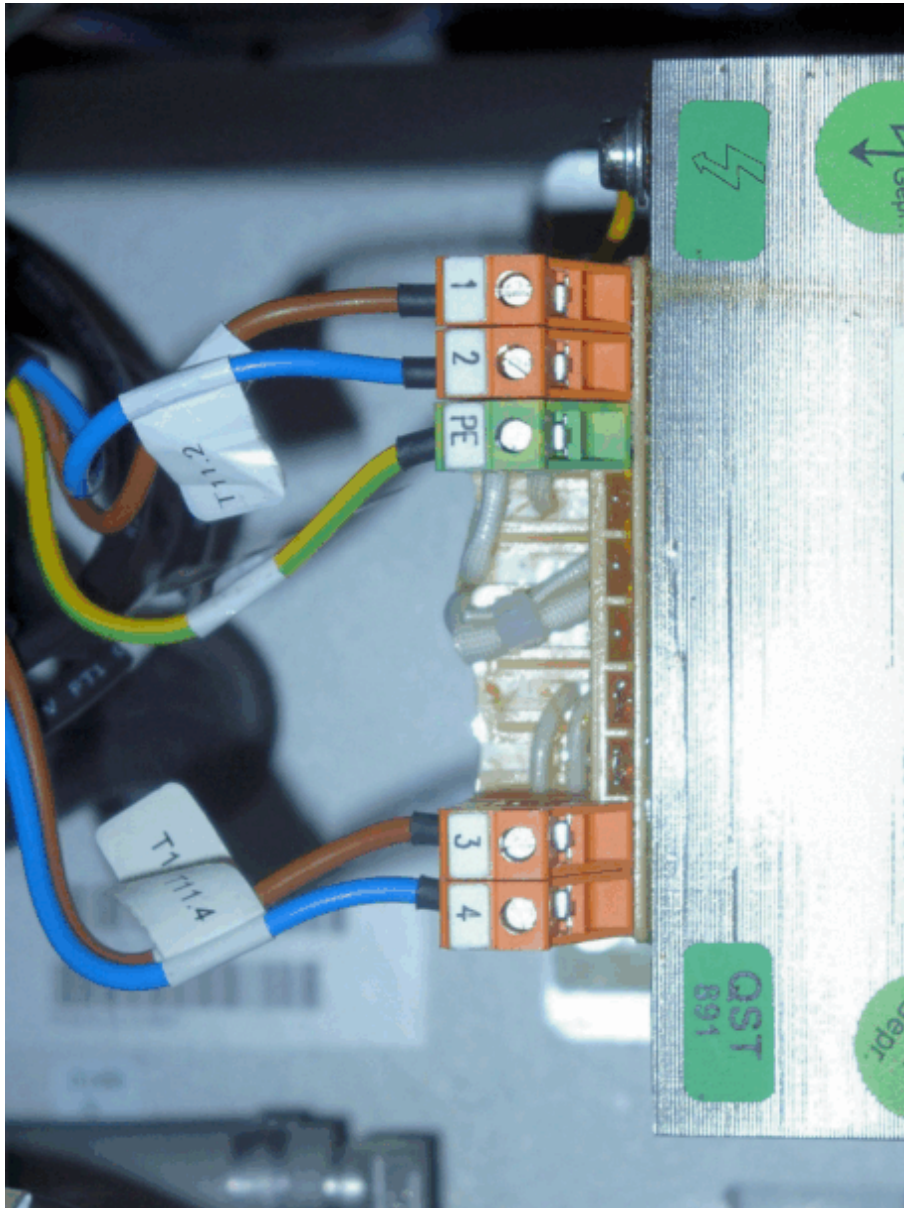
2. Attach the new transformer to the mounting plate using the 2 diagonal threaded bolts as a guide and fasten it with 3 mm Allen screws (x 2). No nuts are used. (→ Fig. 106 Page 124).

Fig. 106: Mounting the new transformer



3. Reconnect the power leads to the new transformer and pull them tight (→ Fig. 107 Page 125).

Fig. 107: Reconnected power leads



4. Reassemble the GPA cabinet in reverse order.

9.3.7 Concluding Steps

- Install the front cover
- Switch on GPA circuit breakers F1 and F2.

9.3.8 Final Checks

Perform Quality Assurance Expert

- Coil Check
- Time: 5 min.

10.1 Avanto Gradient Coil

(Material number 8119690)



CAUTION

During BC/GC disassembly, you have to work rather close to the non-isolated connections of the gradient cables at the GC.

Hazardous voltages may be present at the connections of the gradient cables.

- » Switch off the gradient power amplifier via circuit breaker F2/GSSU and the F1/100A contactor in the GPA. Wait at least for 20 minutes before you start working on the GC.

10.1.1 Overview of GC Replacement Work Steps

Work step	Instructions / References	Special Tools
1. Magnet Ramp Down	Magnet ramp instructions (→ System Start-up / M6-020.815.01)	MPSU 3600 (8395613)
2. Disassembling the Front/Back/Side Covers:	(→ Body Coil (BC Avanto) / M6-020.841.14) (→ Avanto Covers / M6-020.841.10)	
3. Disassembling the BC	(→ Body Coil (BC Avanto) / M6-020.841.14)	
4. Disassembling the old GC	(→ Avanto Gradient Coil / Page 127) for the Espree gradient coil, refer to the Espree upgrade from the straight bore gradient/body coil to Slimline: (→ RF System / M6-020.841.14)	Lifting device (7014678) Coil Support (4765926) Short Beam, 4 m (1141725) Torque wrench 20-50 Nm non-magnetic (8395803) Accessory for torque wrench (08396116) 8-s. socket gradient coil, non-magnetic (10399445) Heat gun
5. Assembling the new GC		
6. Reassembling the BC	(→ Body Coil (BC Avanto) / M6-020.841.14) (→ Avanto Covers / M6-020.841.10)	Torque wrench 5-20 Nm non-magnetic (8395829) Accessory for torque wrench (08396116)

Work step	Instructions / References	Special Tools
7. System restart: - Refilling the cooling system - Refilling LHe	(→ FRU Overview / M6-020.841.18) (→ System Start-up / M6-020.815.01)	for the KKT chiller: 38% ethylene glycol-water mixture Antifreeze tester (8115094) If necessary: Helium fill kit
8. Magnet ramping and shimming (if necessary)	Magnet ramp instructions (→ System Start-up / M6-020.815.01) Shim instructions (Service Software/Shim, Online Help Shim, TU/Phantom Shim)	MPSU 3600 (8395613) Array Shim device (8366978)
9. Reassembling the Front/Back/Side Covers:	(→ Body Coil (BC Avanto) / M6-020.841.14)	Non-magnetic tool set (7757706)
10. General TuneUp	Service Software/Tune up, Online Help for General TuneUp	Adjustment tool for BC decoupling (7366771)
11. Quality Assurance	Service Software QA, Online Help General QA	

10.1.2 Prerequisites

- Measure the pathway from the exterior to examination room; minimum clearance for 1 m (37") for the gradient coil. Allow additional room to maneuver the lifting devices around corners.
- Liquid helium and helium gas are on site if refill is necessary.

10.1.3 Tools and time required

10.1.3.1 Time

Approximately 2 days with two people; removing the BC requires three people.

10.1.3.2 Additional Tools

- Transport pallet for the GC when removed
- Standard tool kit
- Tools for hammering in the wedges (→ Fig. 130 Page 143)
- Measuring tape or gauge; if available, digital depth measure (→ Fig. 129 Page 142)

10.1.4 Preliminary

- The lifting device is available on-site.
- The array shim device is available on-site.
- The magnetic field is ramped down.
- Switch off GPA/GSSU/F2 and 100 A/F1 contactor in the GPA.
- Switch off ACC/F11 (RFPA).
- Switch off ACC/F24 (RFIS).
- Switch off electrical shim F21.
- Disassemble the covers and the BC (→ Body Coil (BC Avanto) / M6-020.841.14).

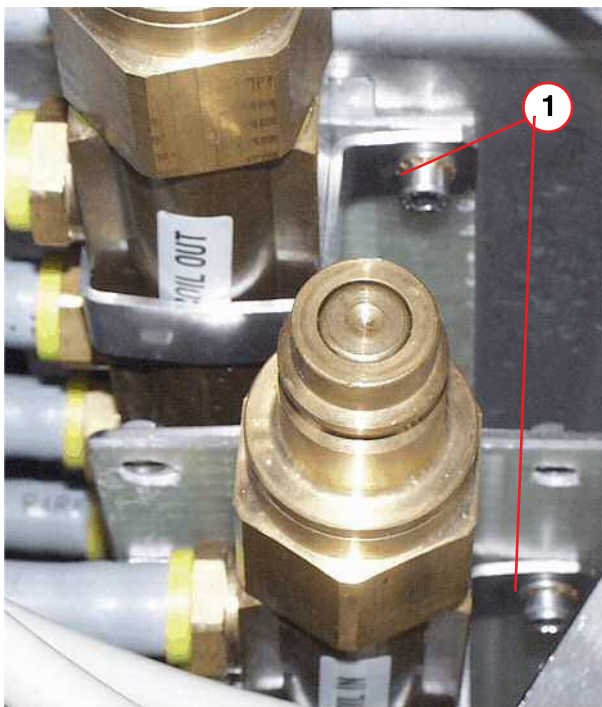
10.1.5 Disassembling the GC

10.1.5.1 Prerequisites

The GC is freely accessible.

10.1.5.2 Disconnecting and disassembling the water hoses

Fig. 108: Water distributors for GC



(1) Mounting clamps

Fig. 109: Water hoses mounted at the left side of the magnet



Fig. 110: Water hoses mounted at the right side of the magnet



Systems with SEP

- Switch off cutter switch F3 in the ACS (water pump)
- Disconnect the water couplings and loosen the water distributors (→ Fig. 108 Page 129).
- Remove the water hoses from the clamps at the magnet (→ Fig. 109 Page 130), (→ Fig. 110 Page 130).
- Place the water hoses with the water distributors in the GC bore.

System with IFP

- Switch off ACC/F1 (water cooling circuit).
- Disconnect the water couplings and loosen the water distributors (→ Fig. 108 Page 129).
- Remove the water hoses from the clamps at the magnet (→ Fig. 109 Page 130), (→ Fig. 110 Page 130).
- Place the water hoses with the water distributors in the GC bore.

10.1.5.3 Disconnecting cables

- Remove the gradient coaxial link cables from the magnet.

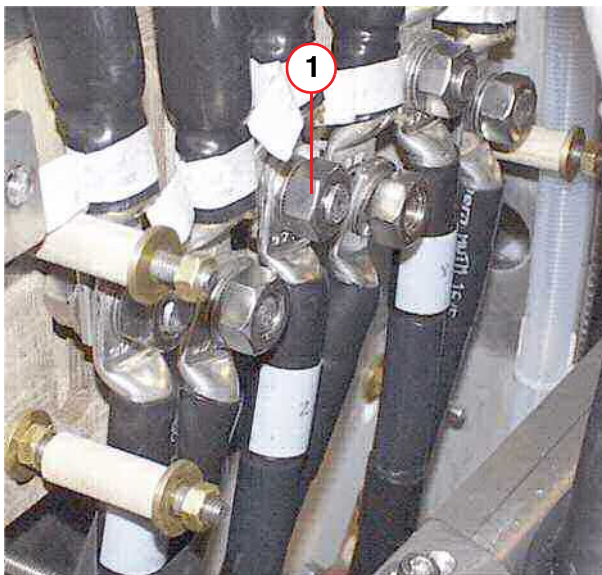
Fig. 111: Coaxial link cable mounted at GC



(1) Screw locking

- Remove the cover of the gradient cable connections at the magnet.
- Loosen the gradient cables; pull the shim plug and the plugs of the coil temperature monitoring.

Fig. 112: Upper GC cable connection at magnet



(1) Connection between GC cable and coaxial link cable (tightened with 40 +/- 4 Nm)

Fig. 113: Connectors at GC



- (1) Shim connector
- (2) Temperature probe connector

Fig. 114: Ground connection - GC to magnet

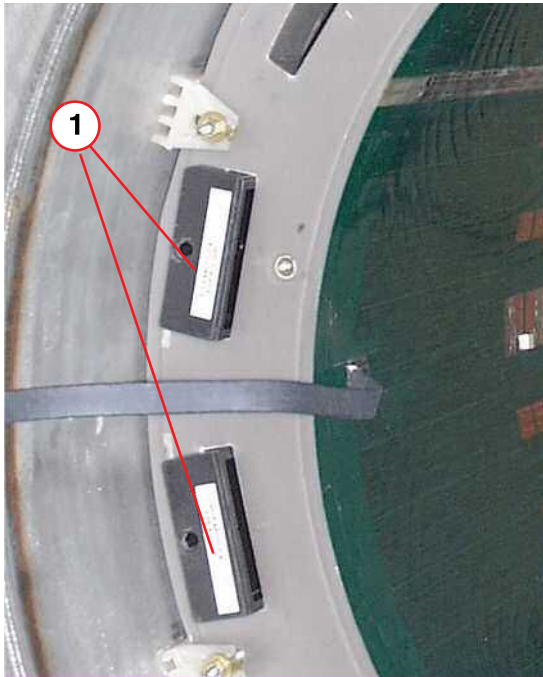


- Loosen the two grounding braids from the GC in back and front of the magnet.

10.1.5.4 Disassembling mechanical components

- Remove the shim trays from the GC. (Attach any missing numbering).

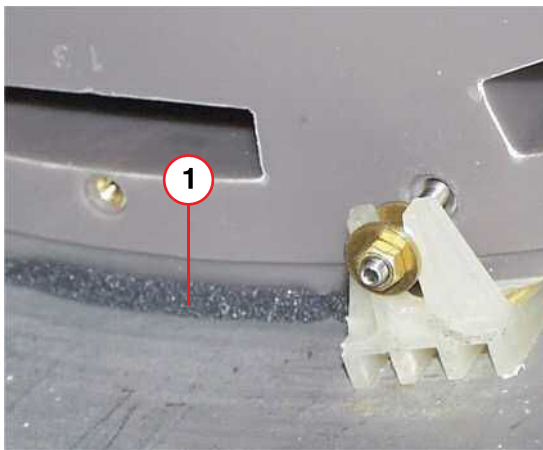
Fig. 115: Assembled shim trays in the GC



(1) Shim trays

- Loosen the front nuts and washers on the threaded bolts of the wedges (using a 3 mm Allen key) securing the GC.
- Carefully remove the foam rubber strips between the GC and the magnet bore (front and back of magnet) (→ Fig. 116 Page 133).

Fig. 116: Sound absorption



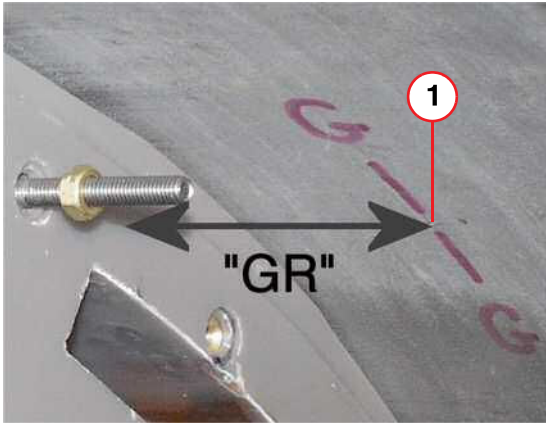
(1) Foam

10.1.5.5 Measurement before removing the GC

- Measure the distance **"GR"** between the **front face** of the **GC** and the **"G-points"** at the patient end of the magnet bore (at the upper right and left sides) (→ Fig. 117 Page 134).

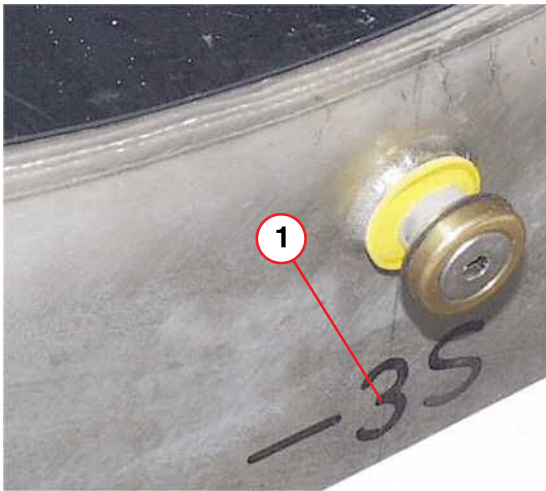
The following table shows the specific distance **GR** depending on **measure M**, printed on the front of the magnet (→ Fig. 118 Page 134).

Fig. 117: Z-alignment - measure "GR"



(1) G-point

Fig. 118: Offset measure "M" on front side of magnet



(1) Example: Measure "M" = -35

Distance "GR" (mm) (+/- 1 mm)	Measure M on magnet
45	- 5S
44	- 4S
43	- 3S
42	- 2S
41	- 1S
40	0

Distance "GR" (mm) (+/- 1 mm)	Measure M on magnet
39	+1P
38	+2P
37	+3P
36	+4P
35	+5P

The new GC has to be adjusted in the Z-direction at the same "GR" distance.

Examples:

If $M = -3S$, the distance "GR" between the front face of the GC to the "G-points" on the magnet bore (patient side) has to be **GR = 43 mm**.

If $M = 0$, the distances between the faces of the GC to the magnet edges (front and back) are equal.

10.1.5.6 Removing the GC using the Lifting Device

Lifting Device Assembly Configuration

- Mobile support 1 in front of the patient table (→ Fig. 119 Page 135)

Fig. 119: Front beam on mobile support 1



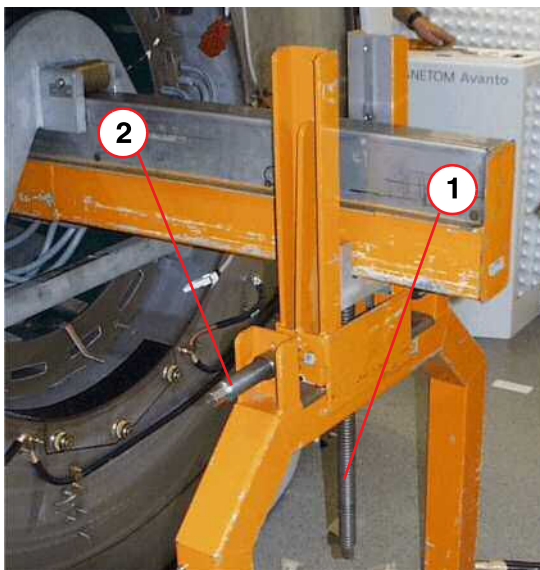
- Mobile support 2 under the patient table in front of magnet (→ Fig. 120 Page 136)

Fig. 120: Beam on mobile support 2 not yet connected to the back beam



- Stationary support behind the magnet (**Caution: Use foot extensions!**)
(→ Fig. 121 Page 136)

Fig. 121: Lifting device - stationary beam support



- (1) Vertical spindle
- (2) Horizontal spindle

- Two connected beams

Fig. 122: Connected beams with level



- Two sliding supports running on the connected beams.

Procedure

- Slide the two beams of the coil support through shim pockets 10 and 15 (→ Fig. 123 Page 137).

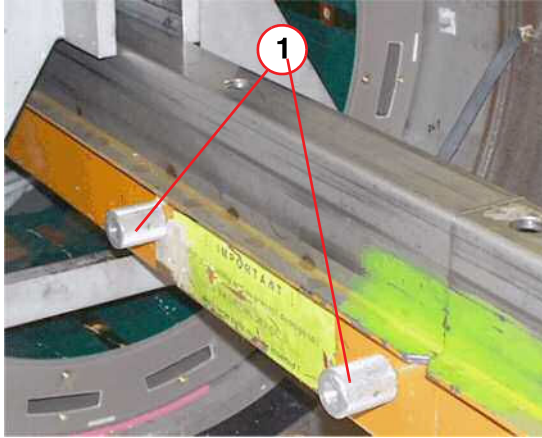
Fig. 123: Sliding carriages of lifting device mounted on the beams



- Mount the back/front sliding carriage on the support beams (on shim pocket positions 10 and 15) at the magnet back/front.
- Position the stationary beam support behind the magnet.
- To protect the RF shield, put a large piece of foam or similar inside the GC/BC.
- Put the rear (short) support beam through the GC/BC and sliding carriages and place its end on the stationary support.

- Position the front support beam using both mobile supports 1, 2.
If necessary adjust the height of the patient table.
- Bring the two beams together and secure the connection with two bolts (→ Fig. 124 Page 138).

Fig. 124: The two securing bolts on the two beams connected together



(1) Securing bolts

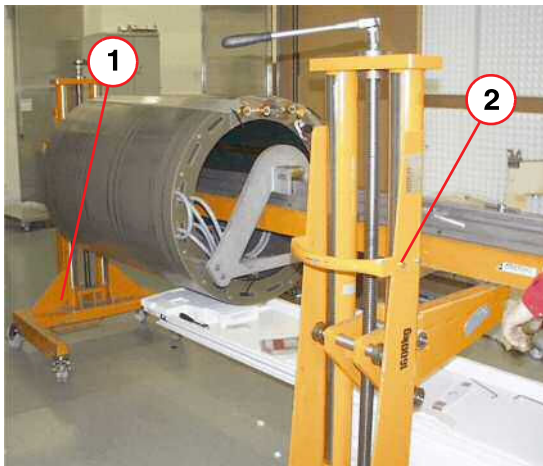
- Adjust the height of the beam to the position of the sliding carriages.
- Remove the mobile support stand 2.
- Deinstall all wedges:
First remove the upper wedges; carefully loosen the wedges (by lightly hammering against both sides of the wedge with a plastic hammer).
For removing the lower wedges, raise the beam to lift the GC/BC in the magnet bore. Then remove the lower wedges.
- Remove the back washers, nuts, and threaded rods from the GC (3 mm hex socket).

- Push out the GC. Move it up to the front of mobile support 1 using the sliding carriages. **Take care not to strike and damage components at the end of the patient table; adjust the vertical position of the patient table.**

Fig. 125: Be careful not to collide with the patient table



Fig. 126: GC on the beam between mobile supports 1 and 2



- (1) Mobile support 1
(2) Mobile support 2

- Reconnect mobile support stand 2 with the foremost support beam; separate the back beam.
- Transport the GC out of the magnet room using mobile support stands 1 and 2 (→ Fig. 126 Page 139).

10.1.5.7 Preparing the GC for Shipment

- Place the GC on the transport pallet, disassemble the sliding carriages and support beams. Remove the support beam of the lifting device.
- Attach the GC to the transport pallet with transport safety items.



WARNING

Prior to shipping the GC, any remaining water must be emptied from the gradient coil.

Otherwise, the internal water channels may freeze.

- » Open the quick connectors using the delivered adapters or by pressing in the quick connects with a piece of hose or a screwdriver. If available, use compressed air for emptying the GC/BC. The fluid should be refilled in the cooling circuit.

10.1.5.8 Installing the new GC

- To protect the RF shield, put a large piece of foam or similar inside the GC.
- Connect the sliding carriages to the two support beams in shim pockets 10 and 15 of the new GC.
- Mount the beam (routed through the GC and sliding carriages) on mobile support stands 1 and 2.
- Transport the GC using mobile transport stands 1 and 2 in the magnet room over the patient table.
- Reattach both support beams, secure with two bolts and remove mobile support stand 2.
- Prior to pushing the GC into the magnet, clean the magnet bore with a vacuum cleaner.

- Center the GC to the magnet bore by adjusting the height of the support beam and repositioning the mobile support. Adjust the support beam position in the Z-direction with a level.

Fig. 127: Connected beams with level



- Push the GC into the magnet until the Z-distance "**GR**" is reached. Measure "**GR**" at the "G-points" between the front face of the GC to the "G-points" (→ Fig. 117 Page 134).

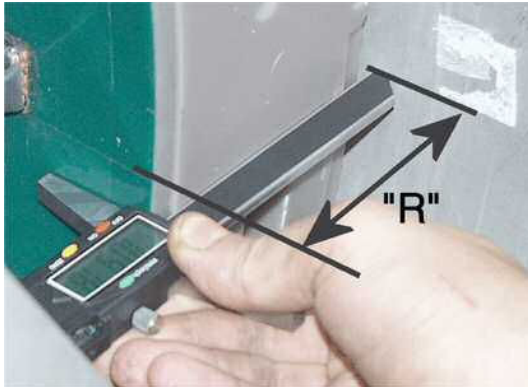
10.1.5.9 Adjusting the GC and assembling the wedges

- Put Loctite 221 on the threaded rods. Screw in the threaded rods (3 mm hex socket) for the wedge attachment and attach the back nuts and washers (up to a minimum distance of approximately 10 mm from the edge of the gradient coil).
- Verify the correct Z-adjustment and radial adjustment of the GC:
The horizontal position of the support beams is required for the correct radial position of the GC.
Check the position of the beam with a level and adjust the height of the supports, if required (→ Fig. 127 Page 141).
- Using the level, which is placed on the sliding carriage, check the correct position of the GC (with respect to the X-axis).
Alternatively, you can mount two screws securing the wedge in threaded holes of equal height as a support base for level measurement.

- Central adjustment:

The distance of the inner surface of the GC to the bore of magnet "R" has to be measured in front and back of the magnet at several locations around the coil (e.g. right and left, top and bottom).

Fig. 128: Radial alignment - Measure "R"



All measurements of "R" must be within a tolerance margin of ± 0.5 mm (i.e. $R \pm 0.5$ mm).

Use a (digital, if available) depth measure.

Fig. 129: Digital depth measure



- Attach all wedges on both sides by hand (lightly press them in) and test for appropriate fit:

Begin with the two lower wedges, then add two upper wedges etc. (Total: on the front side 12 wedges, in back 12 wedges).

The length of the threaded bolts can be adjusted to attach the safety washers and nuts if the wedge cannot be pushed in far enough. (Use a 3mm Allen key).

- Using a plastic hammer and synthetic or delivered brass holders, lightly hammer in all wedges.

Fig. 130: Example of tools: Plastic hammer and synthetic holders



Fig. 131: Hammering in the wedges



- Verify the correct Z-distance "**GR**" repeatedly.
- Recheck the horizontal position (i.e. possible shifting) of the GC by using a level placed on the sliding carriages.
- Recheck the radial distances of the GC "**R**" repeatedly.
- If all distances are correct, hammer in the wedges as follows:
 - Simultaneously hammer in the wedges that are located in the same position in back and front of the magnet.

Start with those in the lower back and front of the magnet, then move on to the upper back and front of the magnet.

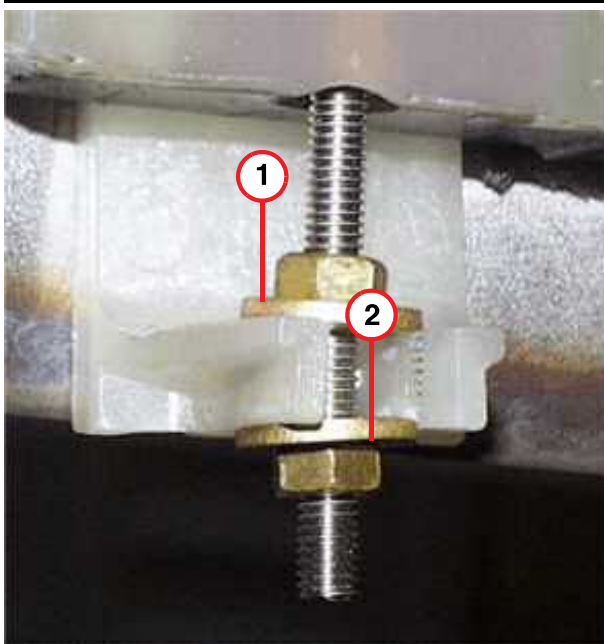
Measure the radial distance "**R**" and the Z-distance "**GR**" repeatedly.

If all distances are correct:

- Finally, hammer in all wedges cross-wise in front and back of the magnet.

- Place washers and nuts on the threaded rods for the wedges or mount the screws with washers and nut. For securing apply Loctite 221.

Fig. 132: Example of mounted wedge with threaded rod

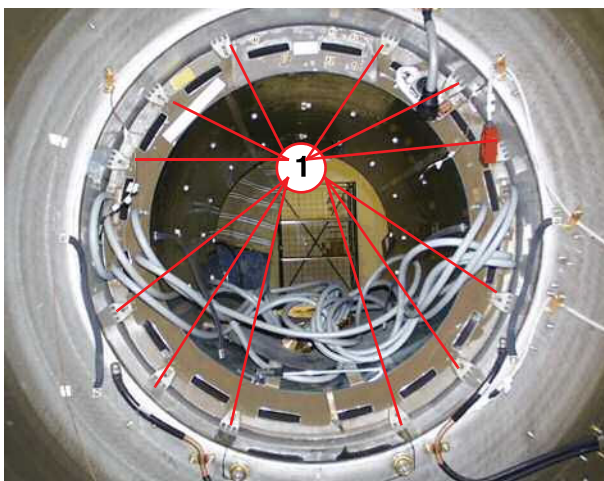


- (1) Back nut and washer
- (2) Front nut and washer

- Twist the back nuts to the front until the washers touch the wedge. Then firmly tighten the front nuts.

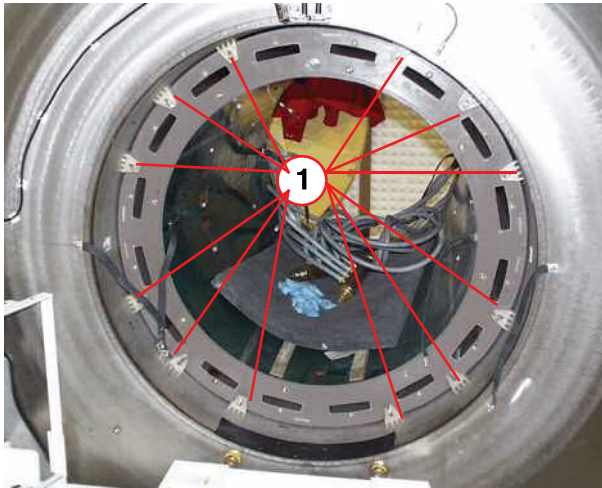
Secure the rods and nuts with Loctite 221.

Fig. 133: GC mounted in the magnet - back



- (1) Wedges

Fig. 134: GC mounted in the magnet - front



(1) Wedges

10.1.5.10 Mounting the coaxial link cables



Please follow the detailed mounting procedure for the coaxial link cables under (→ Gradient Coax Cables Avanto (material number 106 78 832) / Page 188) or (→ Gradient Coax Cables Avanto (material number 81 19 625) / Page 215).

10.1.5.11 Concluding steps of the GC replacement

- Disassemble the support beams of the lifting device and the sliding carriages.
- Push in and fasten the shim trays.
- Reassemble the water distributors and the water hoses (→ Fig. 108 Page 129), (→ Fig. 109 Page 130), (→ Fig. 110 Page 130). Secure the clamp nuts.

Fig. 135: Securing the water hose clamp nuts with "nut lock"



- Attach the foam rubber strips between the magnet bore and the GC (→ Fig. 116 Page 133).
- Mount the cover above the gradient coil connections.
- Reconnect electrical shim and temperature sensor cables.
- Reconnect the two grounding braid from the magnet to the GC in back and front of the magnet (→ Fig. 114 Page 132).
- Reassemble the BC.

10.1.5.12 Restarting the MR System

- Refill the water cooling system (for systems with KKT chiller: 35-38% ethylene glycol-water mixture); use the fluid drained from the old GC.
- Refill LHe, if necessary.
- Restart the cooling system (water pump), switch on fuses:
SEP: F3 in ACS.
IFP: F1 in ACC
- Ramp up the magnet and let stabilize.
- Switch on GPA circuit breakers 100A/F1 and GPA/GSSU/F2.



CAUTION

Hazardous voltages may be present at the connections of the gradient cables.

- » When working at the GC/BC, e.g. changing shim trays, switch off the gradient power amplifier via circuit breaker F2/SSU and F1/100A contactor in the GPA. Wait at least 20 minutes before you start working at the GC/BC.

- Switch on F11/RFPA in the ACC.
- Switch on F24/RFIS in the ACC.

10.1.6 Adjustments

- Perform TU/ Phantom Shim.
- If Phantom Shim does not come into specification, check the shim status with the array shim device and reshim the magnet if necessary.
- If the shim status is ok, reassemble the remaining covers.
- Perform a complete Tune-up.

10.1.7 Final Checks

- Perform complete General Quality Assurance.

10.2 Upgrade Espree straight bore Gradient/Body Coil to Slimline

10.2.1 Worksteps - Overview

- Ramp magnet down.
- Remove front and rear cover.
- Disconnect all attached cables and remove old straight coil with the gradient coil lifting device.
- Remove shim trays and light bars, including securing materials, from the old coil.
- Install new Slimline Gradient coil and re-connect cabling.
- Install the empty shim trays.
- Install new Body coil and re-connect cabling. Install light bars.
- Install the new divided rear cover and other covers that were removed, except for the front cover.
- Ramp magnet to field and let stabilize.

10.2.1.1 Software

Load SP1 Software patch for VB12A, if not already installed.

Go to configuration -> measurement settings -> set gradient to AS022SL

Reset gradient offsets and Super Conductive Shim Current.



Reboot the system to make the changes effective!

10.2.1.2 Worksteps - Overview continued

- Install Array Shim Device and shim magnet, starting with the bare plot.
- Re-install all covers.
- Perform system Tune-up/QA and body coil tuning

10.2.1.3 Final Work Steps

- Perform final assembly, clean covers, and return system to customer.
- Complete Installation Protocol.
- The old straight bore coil (pn 100 47 942) must be returned to Erlangen.
- Scrap old rear cover and bore tube.

10.2.2 Tools and time required

10.2.2.1 Time

Total 5 days.

2 days require 4 people for replacing the hardware.

3 days require two people (one trained Espree CSE/ISE plus one assistant).

10.2.2.2 Tools



For USA, add TT before the material number when ordering from Tool&Test

Lifting device for GC/BC with accessory parts 1225	70 14 678
Short beam 4m	11 41 725
Coil support (sliding carriages with 2 support beams)	51 37 042
Magnet power supply	83 95 613
Espree Shim Array	83 68 313
Shim plates, left over from first installation	
Helium fill kit	
8-s. socket gradient coil non-mag.	103 99 445
Torque wrench 20 - 50 Nm	83 95 803
Accessory kit for torque wrench	83 96 116
Standard tool kit	
Nonmagnetic steel tool set	77 57 706
Service aid back cover (if needed)	100 18 422
Adjustment tool for body coil decoupling	73 66 771 (non T&T item)
Tools for hammering in the wedges (→ Fig. 145 Page 162)	
Measuring tape or gauge	
Ethylene glycol and antifreeze tester	81 15 094
Double hexagon 12-point socket	100 48 130
Plumber's tape (1 to 1.5" white sand cloth (local purchase))	

Scotch Brite pad (local purchase)

Heat gun

for 220 V: material number 97 02 937

for 110 V: material number 97 02 945

10.2.3 Preliminary

- SP1 Software patch installed, for VB12A.
- All upgrade equipment & tools on site.
- Measure the pathway from the exterior to examination room; minimum clearance for 1 m (37") for the gradient coil. Allow additional room to maneuver the lifting devices around corners.
- Liquid helium and helium gas are on site if refill is necessary.

10.2.3.1 Parts required

For **all** systems to be updated: GC/BC slim complete. 100 48 100

In addition

For systems <30099: upgrade kit divided rear cover 100 18 727 (includes cable kit tunnel light)

For systems >30098: cable kit tunnel light 100 18 757

10.2.3.2 Procedure

Ramp magnet down, refer to Startup Instructions (→ Ramping the magnet / M6-010.815.01)..

For USA:

Download the MPS log file

To download the log file:

- In the ramping Software display, press the G key <Return>, this will ensure that you are at the log screen.
- Select the following in the hyper terminal: (top menu)

Capture Text/start

Type the location and name of the log file e.g. C:\Medcom\Temp \MR25123_51099RDlog.txt (Note: capture all ramp files both above and below)

- Press the L key <Return>, the data screen appears.

The data screen contains seven columns. Each line displays the data stored at a specific period of time during magnet ramping.

- After the data has stopped scrolling, select Capture Text/Stop button, transfer the file to a memory stick, and send the log to TSE Andy Wren andy.wren@siemens.com.



CAUTION

During the GC/BC replacement you have to work rather close to the non-isolated connections of the gradient cables at the GC/BC.

Hazardous voltages may be present at the connections of the gradient cables.

- » Switch off the gradient amplifier via circuit breaker F2/SSU and F1/100 A contactor in the GPA. Wait at least 20 minutes before you start working at the GC/BC.

- Move the patient table down.
 - Switch off GPA/GSSU/F2 and 100 A/F1 contactor in the GPA.
 - Switch off ACC/F11 (RFPA).
 - Switch off ACC/F24 (RFIS).
 - Switch off electrical shim F21.
 - Disassemble the front funnel and ring, the back covers of the magnet, as well as the bore cover (with the foam rubber strips) together with its supports at the magnet.
(Refer to (→ Covers / M6-020.841)).
- Remove the light bars and their securing materials from the BC.

10.2.4 Disassembling GC/BC

Prerequisites: The GC/BC is freely accessible.

10.2.4.1 Disconnecting the Water Hoses

Systems with SEP

- Switch off cutter switch F3 in the ACS (water pump).
- Disconnect the water couplings.

Systems with IFP

- Switch off ACC/F1 (water cooling circuit).
- Disconnect the water couplings.

10.2.4.2 Disconnecting Cables

- Remove the cover of the gradient cable connections at the magnet.
- Loosen the gradient cables; pull the shim plug and the plugs of the coil temperature monitoring.
- Remove the gradient coaxial link cables from the magnet.
- Loosen the ground braids from GC/BC in back of the magnet.

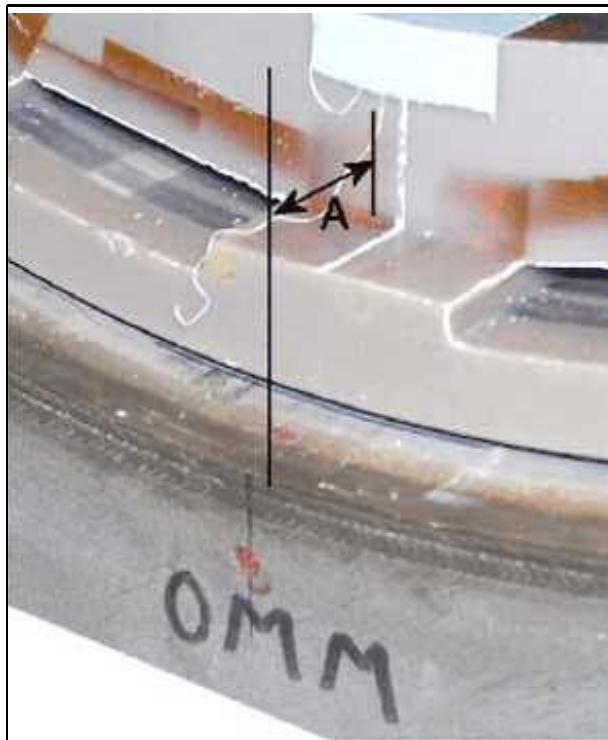
- Disconnect the two TX-cables from the BC; disconnect RFIS/ X9 and loosen the cable ties which route the cable <W5103> at the magnet.

10.2.4.3 Disassembling mechanical components

- Remove the shim trays from the GC. (Attach any missing numbering). Remove all shim plates.
- Loosen the front nuts and washers on the threaded bolts of the wedges (using a 3mm Allen key) securing the GC/BC.
- Carefully remove the foam rubber strips between the GC/BC and the magnet bore (front and back of magnet).

10.2.4.4 Measurements before removing the GC/BC

Fig. 136: Z-alignment - measure "A"



- Measure the distance **"A"** between the **inner front face** of the **BC/GC ("straight bore")** or between the **front face of the GC ("slim line")** and the front edge of the magnet at four positions.

The following table shows the specified distance A dependent on **measure M**, printed on the front of the magnet.

- Put a level on the edge at the magnet front bore that does not show welding seams and measure the distance to the inner edge of the gradient coil.

"A" (mm) "straight bore" Distance inner front face GC/BC to magnet edge	"A" (mm) "slim line" Distance front face GC to magnet edge	Measure M on magnet
70	35	-5 S
69	34	-4 S
68	33	-3 S
67	32	-2 S

"A" (mm) "straight bore" Distance inner front face GC/BC to magnet edge	"A" (mm) "slim line" Distance front face GC to magnet edge	Measure M on magnet
66	31	-1 S
65	30	0
64	29	+1 P
63	28	+2 P
62	27	+3 P
61	26	+4 P
60	25	+5 P

The new GC has to be adjusted in the Z-direction at the same distance "A" as shown in the conversion table above.

Example 1:

The old straight coil has a distance "A" of 63 mm to the front edge of the magnet. This corresponds to measure "M" of +2P.

Measure "M" is printed on front of the magnet for verification.

Install the new Slimline gradient coil at a distance of 28 mm to the front edge of the magnet, according the conversion table. This corresponds to measure "M" of +2P.

In other words:

Old straight coil "A" = 63mm

New slim coil "A" = 28 mm.

This is the correct positioning in example 1.

Example 2:

The old straight coil has a distance "A" of 66 mm to the front edge of the magnet. This corresponds to measure "M" of -1S.

Install the new Slimline gradient coil at a distance of 31 mm to the front edge of the magnet, according the conversion table. This corresponds to measure "M" of -1S.

In other words:

Old straight coil "A" = 66 mm

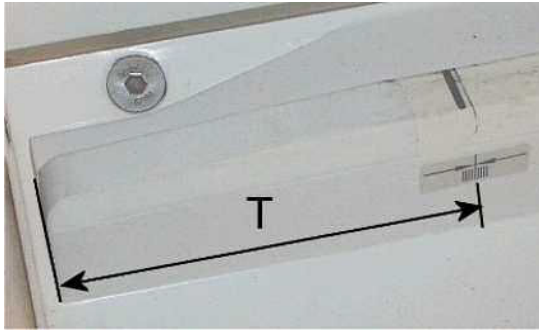
New slim coil "A" = 31 mm

This is the correct positioning in example 2.

For any other combinations please refer to the conversion table above!

- Measure and note the Z-distance (T) of the marker for the shim device inside the body coil

Fig. 137: Position of marker for shim device

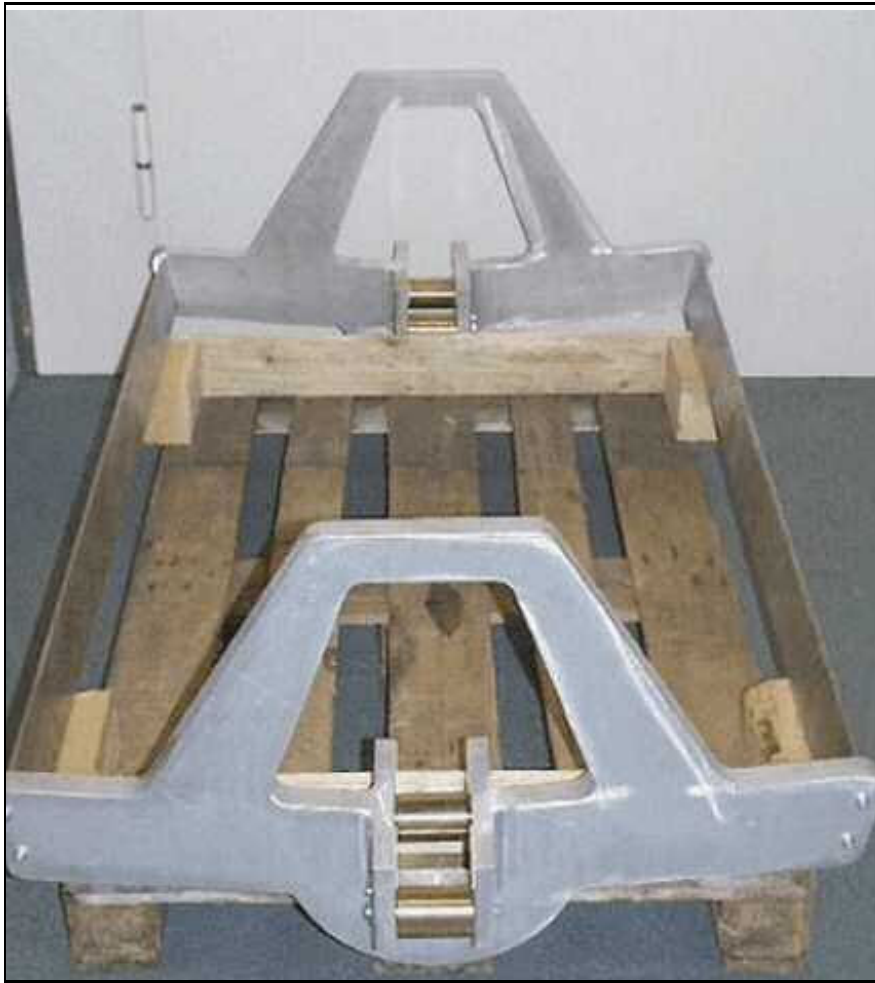


10.2.4.5 Removing the GC/BC using the Lifting Device

Lifting Device Assembly Configuration

- Mobile support 1 in front of the patient table
- Mobile support 2 under the patient table in front of magnet
- Stationary support behind the magnet (**Caution: Use foot extensions!**)
- Two connected beams
- Two sliding supports running on the connected beams.

Fig. 138: Carriages of the GC/BC lifting device



Procedure

- Disassemble the sliding carriage and put the two support beams through shim pockets 1 and 8.
- Mount the back/front sliding carriage on the support beams (on shim pocket positions 1 and 8) at the magnet back/front.
- Position the stationary beam support behind the magnet.
- To protect the RF shield, put a large piece of foam or similar inside the GC/BC.

- Put the rear (short) support beam through the GC/BC and sliding carriages and place its end on the stationary support.

Fig. 139: Beam on stationary support - GC with mounted sliding carriages



- Position the front support beam using both mobile supports 1, 2. If necessary adjust the height of the patient table.
- Put the two beams together and secure the connection with two bolts.

- Adjust the height of the beam to the position of the sliding carriages.

Fig. 140: GC/BC mounted to the carriages on the beam



- Remove the mobile support stand 2.
- Deinstall all wedges:
First remove the upper wedges; carefully loosen the wedges (by lightly hammering against both sides of the wedge with a plastic hammer).
For removing the lower wedges, raise the beam to lift the GC/BC in the magnet bore. Then remove the lower wedges.
- Put the water hoses with quick connections and the cable to the RFIS into the gradient coil. Proceed carefully. Do not damage the shim and temperature sensor cables and connectors.
- Remove the back washers, nuts and threaded bolts from the GC.

- Push out the GC/BC. Move it up to the front of mobile support 1 using the sliding carriages.

Fig. 141: GC/BC removed - mobile support 2



- Reconnect mobile support stand 2 with the foremost support beam; separate the back beam.
- Transport the GC/BC out of the magnet room using mobile support stand 1 and 2.

Fig. 142: Removal of GC/BC using carriages on the beam



10.2.4.6 Preparing the GC/BC for Shipment

- Place the GC/BC on the transport pallet, disassemble the sliding carriages and support beams. Remove the support beam of the lifting device.
- Attach the GC/BC to the transport pallet with transport safety items.



WARNING

Prior to shipping the GC/BC, any remaining water must be emptied from the gradient coil.

Otherwise, the internal water channels may freeze.

- » Open the quick connectors using the delivered adapters or by pressing in the quick connects with a piece of hose or a screwdriver. If available, use compressed air for emptying the GC/BC. The fluid should be refilled in the cooling circuit.

10.2.4.7 Return of parts

The old straight GC/BC must be returned to Germany.

Material number 100 47 942

Make sure that this material number is listed on the crate.

Remove all other labels from the crate.

Procedure for the USA:

For additional information, please contact the USC.

10.2.5 Assembly

10.2.5.1 Installing the new gradient coil

- To protect the RF shield, put a large piece of foam or similar inside the GC.
- Connect the sliding carriages to the two support beams in shim pockets 1 and 8 of the new GC.
- Mount the beam (routed through the GC and sliding carriages) on mobile support stands 1 and 2.
- Transport the GC using mobile transport stands 1 and 2 in the magnet room over the patient table.
- Reattach both support beams, secure with two bolts and remove mobile support stand 2.
- Prior to pushing the GC into the magnet, clean the magnet bore with a vacuum cleaner.
- Center the GC to the magnet bore by adjusting the height of the support beam and repositioning the mobile support. Adjust the support beam position in the Z-direction with a level.
- Push the GC into the magnet until the Z-distance "**A**" is reached. Measure "**A**" at several points between the inner front face of the GC to the front edge of the magnet (small edge without welding seams). (→ Measurements before removing the GC/ BC / Page 151)

10.2.5.2 Adjusting the GC and assembling the Wedges

- Screw in the threaded bolts for the wedge attachment and attach the back nuts and washers (up to a minimum distance of approximately 10 mm from the edge of the gradient coil).
- Verify the correct Z-adjustment and radial adjustment of the GC:
The horizontal position of the support beams is required for the correct radial position of the GC.
Check the position of the beam with a level and adjust the height of the supports, if required.

- Using the level, which is placed on the two support beams in the shim pockets, check the correct position of the GC (regarding the X-axis).

Alternatively, you can mount two screws securing the wedge in threaded holes of equal height as a support base for level measurement.

Fig. 143: X-alignment of the BC/GC using a level



- Central adjustment:

Place the GC in exactly the Center Position by either raising or lowering the support beam or by adjusting the position of mobile support stand 1:

The distance of the outer surface in the center of a shim pocket to the bore tube of the magnet "R" has to be measured in front and back of the magnet at several

locations around the coil. All measurements of "R" must be within a tolerance margin of ± 1 mm (i.e. $R \pm 1$ mm).

Fig. 144: Radial alignment - Measure "R"



- Attach all wedges on both sides by hand (lightly press them in) and test for appropriate fit:

Begin with the two lower wedges, then add two upper wedges etc. (Total: on the front side 12 wedges, in back 10 wedges).

Note:

The two lower wedges at the magnet front must be harmonized with the bore cover supports.

The lower wedge in back of the magnet as well as the wedge below the RF-connectors are secured to a threaded bolt at the magnet. (→ Fig. 149 Page 165)

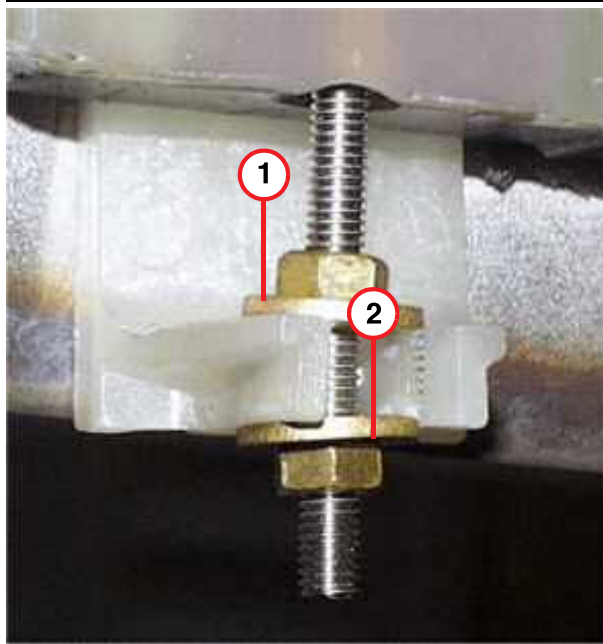
The length of the threaded bolts can be adjusted to attach the safety washers and nuts, if the wedge cannot be pushed in far enough. (Use a 3mm Allen key).

Fig. 145: Example of tools: Plastic hammer and synthetic holders



- Using a plastic hammer and synthetic or delivered brass holders, lightly hammer in all wedges.
-
- Verify the correct Z-adjustments repeatedly.
 - Recheck the horizontal position (i.e. possible shifting) of the GC by using a level across the support beams connected to the sliding rods.
 - Recheck the radial distances of the GC "R".
 - If all distances are correct, hammer in the wedges as follows:
 - Simultaneously hammer in the wedges that are located in the same position in back and front of the magnet.
Start with those in the lower back and front of the magnet, then move on to the upper back and front of the magnet.
 - Measure the radial distance "R" repeatedly.

Fig. 146: Example of mounted wedge with threaded rod



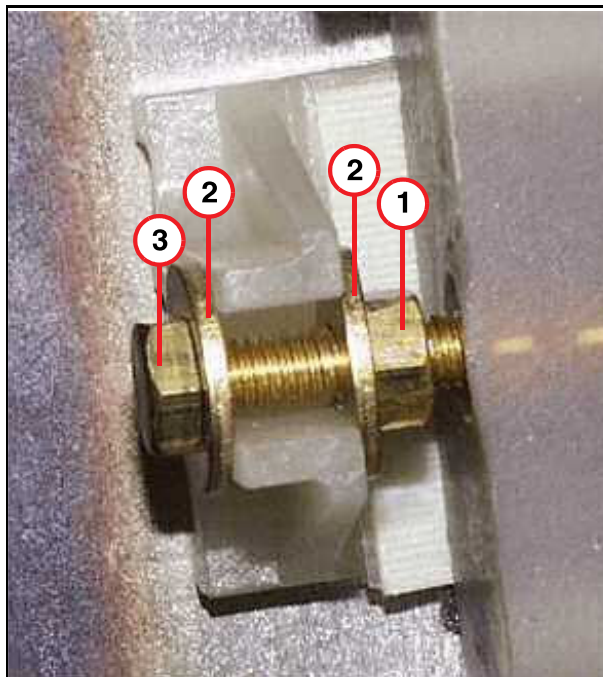
- (1) Back nut and washer
- (2) Front nut and washer

If all distances are correct:

- Finally, hammer in all wedges cross-wise in front and back of the magnet.
- Place washers and nuts on the threaded bolts for the wedges or mount the screws with washers and nut. For securing apply Loctite 221.
- If threaded bolts are used:
Twist the back nuts to the front until the washers touch the wedge. Then firmly tighten the front nuts.

Secure the rods and nuts with Loctite 221.

Fig. 147: Example of mounted wedge with screw



- (1) Back nut
- (2) Washer
- (3) Screw

- If screws are used:
Tighten the screws and then the back nut. Secure the screws and nuts with Loctite 221.
- The wedge should have a distance of 15 mm to the coil, so you can insert nut (1) and washer (2).
Tighten the screw (3) until the wedge is preloaded and the distance to the coil is approx. 10 mm.

10.2.5.3 Replacement of coaxial link Cables at Magnet

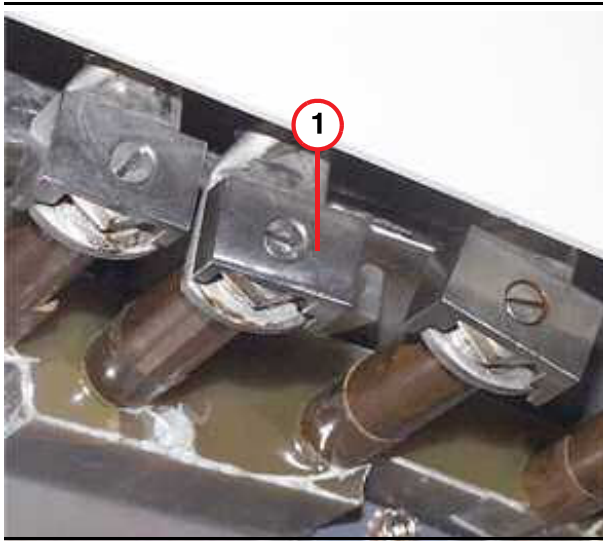
(The coaxial link cables are part of the GC/BC slim complete)



Please follow the detailed mounting procedure for the coaxial link cables under (→ Gradient Cables (Espree) / Page 236).

10.2.5.4 Concluding Steps of the GC/BC Replacement

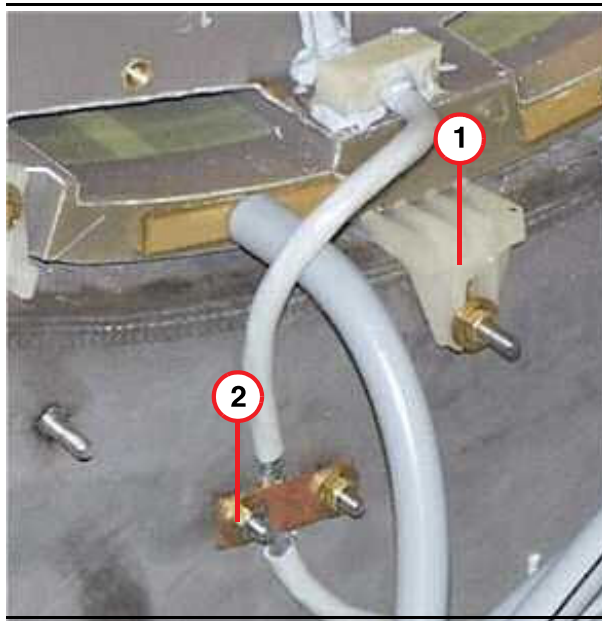
Fig. 148: Gradient cable connections at GC



(1) Securing clamp

- Disassemble the support beams of the lifting device and the sliding carriages.
- Push in and fasten the shim trays.
- **Note:**
Attach the foam rubber strips between the magnet bore and the GC.
The foam rubber strips come with the new coils.
- Reconnect the gradient cables. Use a torque of 22 ± 1 Nm for the screw connections. Mount the securing clamps on top, parallel to the cable terminal. (Minimum distance of non-insulated parts: 4 mm). Secure the screws of the securing clamps with Loctite 221.

Fig. 149: Bottom of the GC/BC, back view



- (1) Lower wedge secured at the magnet
- (2) Clamp for grounding cable shield at the magnet (cable <W 5103>)

- Mount the cover above the gradient coil connections.
- Reconnect electrical shim and temperature sensor cables.

Fig. 150: Ground braid at the GC and an example of a mounted wedge



- Reconnect the grounding braid to GC in back of the magnet.

Fig. 151: Mounted wedge with screw



- Reconnect the water quick couplings.

10.2.5.5 Installing the new body coil



CAUTION

The foamed plastics and the plastic film around the body resonator serve to provide noise protection. Do not remove this cover.

- » Never deposit the body resonator on a sharp-edged surface. The plastic cover, damping material and the capacitors could be damaged.



Place a sheet of foam on the patient table to protect the BC from damage.

- Connect the QLA cables to the RF pick-up coils at the BC:

Fig. 152: Position of QLA connector of pick-up coil on the BC; distance to back edge of BC is 180 mm.



(1) QLA connector (one of the two connectors to the right and left on the top of the BC)

- Place the new BC on the patient table and put the cabling into the BC bore.
- Slide the new BC on the two rails into the gradient coil.



Use extreme care when moving the BC into the magnet to avoid damaging the RF shield inside the gradient coil as well as the BC cabling.

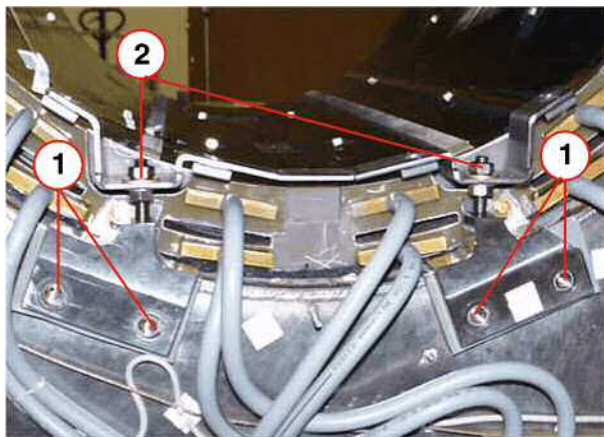
Keep the BC centered at all times on the slide rails in the gradient coil.

Be careful with the RF cables and detuning cable when pulling the BC.

- Assemble the back BC support (→ Fig. 153 Page 168) and route the detuning cable under the support. Place the BC on the coil support, insert all mounting head socket hex screws, and tighten by hand.

Assemble the front BC support as well.

Fig. 153: BC support at back of the magnet



- (1) Assemble BC support at outer vacuum chamber
- (2) Clamping nuts

10.2.5.6 BC Alignment

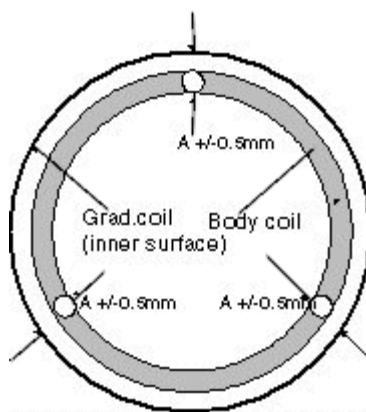
■ Check adjustment in x/y/z -direction:

The body coil has to be centered to the gradient coil in x- and y- direction. The distance between the inner surface of the BC and the inner surface of the gradient coil as well as the distance to the magnet bore must be the same all around (tolerance ± 0.5 mm).

The distance in z- direction from the front edge of the body coil to the front edge of the magnet has to be 40.25 mm ± 1 mm.

The BC is NOT centered to the GC in z- direction.

Fig. 154: Centered alignment of BC to GC



Use a spirit level to adjust the body coil horizontally in Z/ X direction at the front and back.

Fig. 155: Leveling the BC in z and x direction



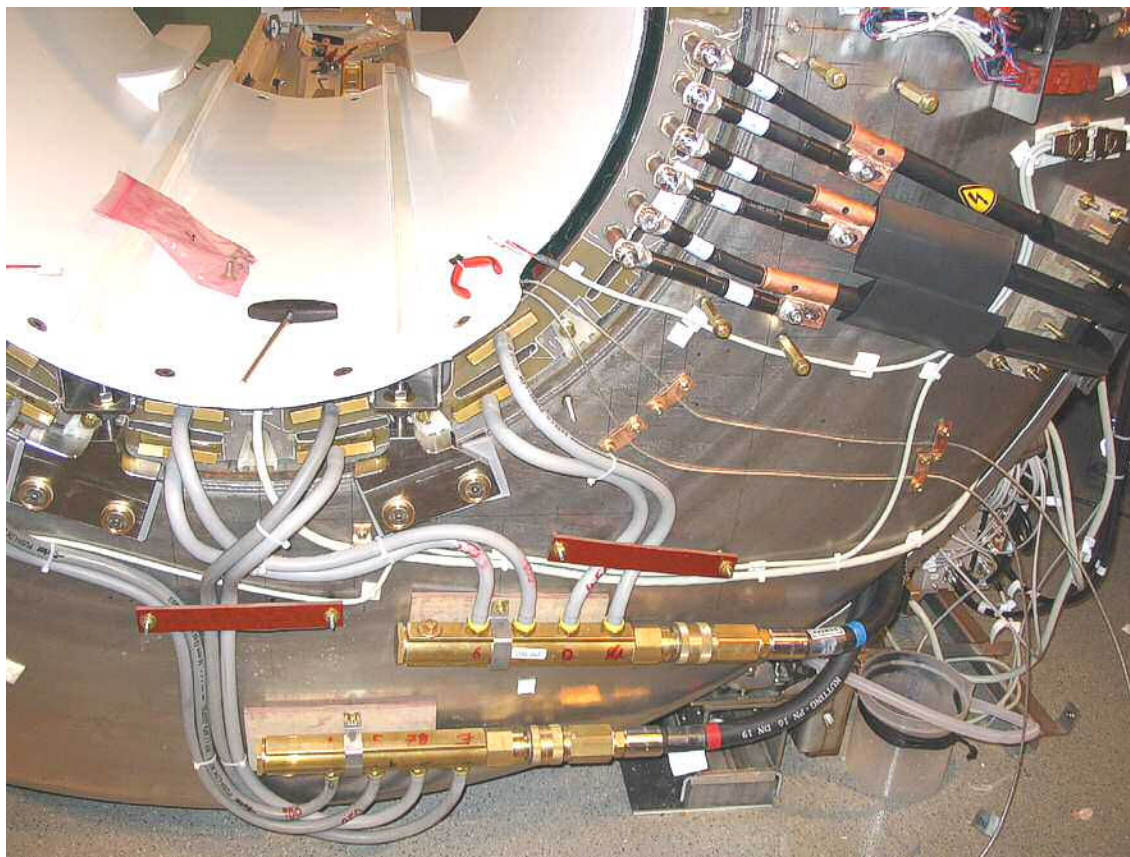
Adjust the X/Y-distances if necessary by loosening the support nuts. Measure at the back and front of the magnet at positions shifted 120 degrees.

In (→ Fig. 154 Page 168) the three measurement points "A" are shifted in intervals of 120 degrees.

- Concluding steps:
 - Tighten all screws and nuts of the coil support and check alignment again.
 - Route the RF-cables to TALES/X3/X4 and the PIN-diode cable to RFIS/ X9 and tighten the corresponding cable collars at the magnet.

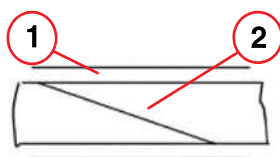
- Install the light bars. Route the bore light cables as shown in (→ Fig. 156 Page 170)
Very important: use many cable ties!

Fig. 156: Cable routing



- Attach the noise damping foam pieces between the gradient coil and the BC.
- Attach the supplied rubber sealing band (Vitolen 120) centrically onto the front and back edge of the BC. The start and end of the band should be cut at an acute angle and overlap when attached at the edge on top of the BC.

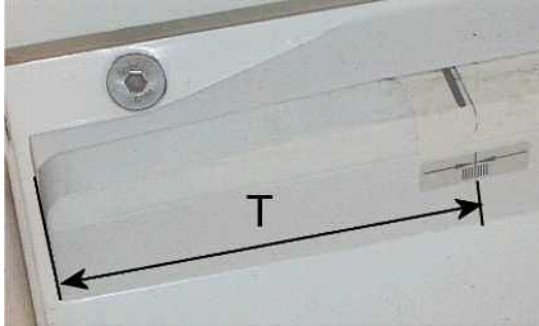
Fig. 157: Overlapping and attached start and end of sealing band on the edge of the BC



- (1) Edge of the BC
(2) Attached sealing band

- Attach (if not yet completed) the marker for the shim device at the same Z-distance **T** as measured at the old BC. The marker is delivered in a plastic bag attached at the new BC's bore.

Fig. 158: Position of marker for shim device



10.2.5.7 Restart of MR System

- Refill the water cooling system (for systems with KKT chiller: 35% - 38% ethylene glycol-water mixture); use the fluid drained from the old GC/BC.
- Refill LHe, if necessary.
- Restart the cooling system (water pump), switch on fuses:
SEP: F3 in ACS.
IFP: F1 in ACC
- Ramp up the magnet and let stabilize.
- Switch on GPA circuit breaker 100A/F1 and GPA/GSSU/F2.



CAUTION

Hazardous voltages may be present at the connections of the gradient cables.

- » If working at the GC/BC, e.g. changing shim trays, switch off the gradient power amplifier via circuit breaker F2/SSU and F1/100A contactor in the GPA. Wait at least 20 minutes before you start working at the GC/BC.

- Switch on F11/RFPA in the ACC.
- Switch on F24/RFIS in the ACC.

10.2.5.8 Software

- Install the SP1 Software patch for VB12A, if not already installed.
- Go to Configuration -> Measurement settings -> set gradient to AS022SL
- Click SAVE
- Set the gradient offsets and the Super Conductive Shim Current to zero

10.2.5.9 Procedure

- Switch to Advanced User and open a command tool.
- Type in modifymeaspara

- Set GradOffsets to zero (Refer to (→ Fig. 159 Page 172)).
- Set superConductivityShimCurrent to zero (Refer to (→ Fig. 159 Page 172)).

Fig. 159: ModifyMeasPara

```
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\meduser>modifymeaspara

-----
ModifyMeasPara V2.0
-----
  1. LM-MagnCenter           Change LightMarkerDistToMagnetCenterZ
  2. GradOffsets             Change gradient offsets
  3. SuperConductiveShimCurrent Change SuperConductiveShimCurrent
-----
Please enter number or unique text of a cmd. ('q' for quit)
-----
MMP> 2
-----
ATTENTION: The values you entered will be rounded with corresponding
           DAC steps!
-----
Enter gradient offset for x-axis [-0.0331128 mT/m]:0
Value not changed.
-----
Enter gradient offset for y-axis [-0.386078 mT/m]:0
Value not changed.
-----
Enter gradient offset for z-axis [0.783131 mT/m]:0
Value not changed.
-----
ModifyMeasPara V2.0
-----
  1. LM-MagnCenter           Change LightMarkerDistToMagnetCenterZ
  2. GradOffsets             Change gradient offsets
  3. SuperConductiveShimCurrent Change SuperConductiveShimCurrent
-----
Please enter number or unique text of a cmd. ('q' for quit)
-----
MMP> 3
-----
Enter value for superConductivityShimCurrent [-9833 mA]:0
Value not changed.
-----
ModifyMeasPara V2.0
-----
  1. LM-MagnCenter           Change LightMarkerDistToMagnetCenterZ
  2. GradOffsets             Change gradient offsets
  3. SuperConductiveShimCurrent Change SuperConductiveShimCurrent
-----
Please enter number or unique text of a cmd. ('q' for quit)q
-----
MMP>
```

- Quit the menu with q and close the command tool window.

- **Reboot the system.**

10.2.5.10 Reshim the Magnet

For shimming refer to the Installation Instructions -> Startup -> Ramping the magnet -> Shim

Start with the **bare** plot.

Make sure the shim trays are empty.

Install the Array Shim Device

10.2.5.11 Installing the front cover/funnel

- Install the covers in reverse order as described in (→ Removing the front covers/funnel / M6-020.841.14) .
- Measure the width of the gaps between BC and patient table. It should be equal on both sides and less than 8 mm; for details of the procedure, refer to UI MR036/05/SI.

10.2.6 Adjustments

- Perform a full Tune Up.
 - If Coil Power Loss fails during Tune Up, contact HSC.

10.2.6.1 BC Tuning

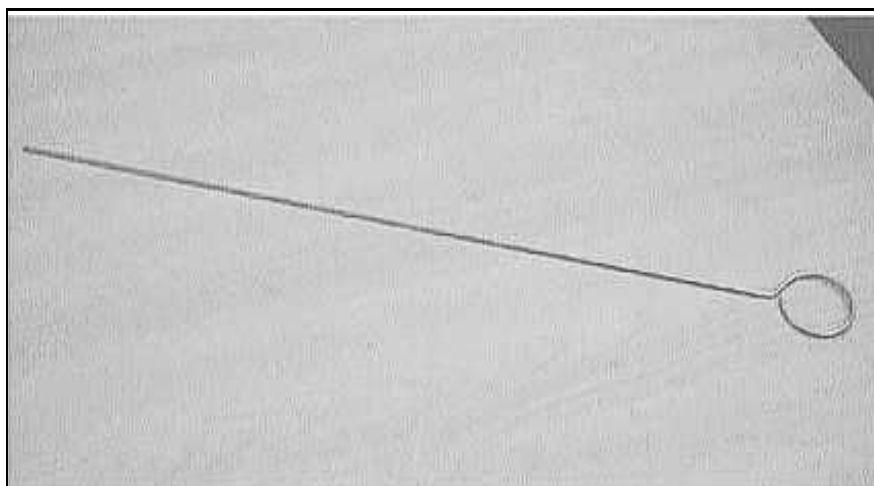
Prerequisites

Since BC Tuning uses the H and S matrices to perform calculations, the tuning calibration tune-up procedure has to be completed successfully before starting with the BC tuning.

To gain access to the trimmer capacitors at the back of the Body Coil, the back magnet cover has to be removed.

The special non-magnetic adjustment tool (i.e. long screwdriver, material number 73 66 771) is required to turn the trimmer capacitors. Refer to (→ Fig. 160 Page 174).

Fig. 160: Tool for trimmer capacitor adjustment



- Switch on ACC/ F11 (RFPA) and F24 (RFIS).
- Check under SeSo / Reports / MR Configuration: The correct gradient coil under "Measurement Settings" has to be shown: **"AS022SL"**. If not , e.g., after replacement of an old GC/BC, the coil has to be configured.



The following measurement Tune-up / BC Tuning should be done with the back funnel removed and repeated with the back funnel installed.

Frequency and Decoupling Adjustment

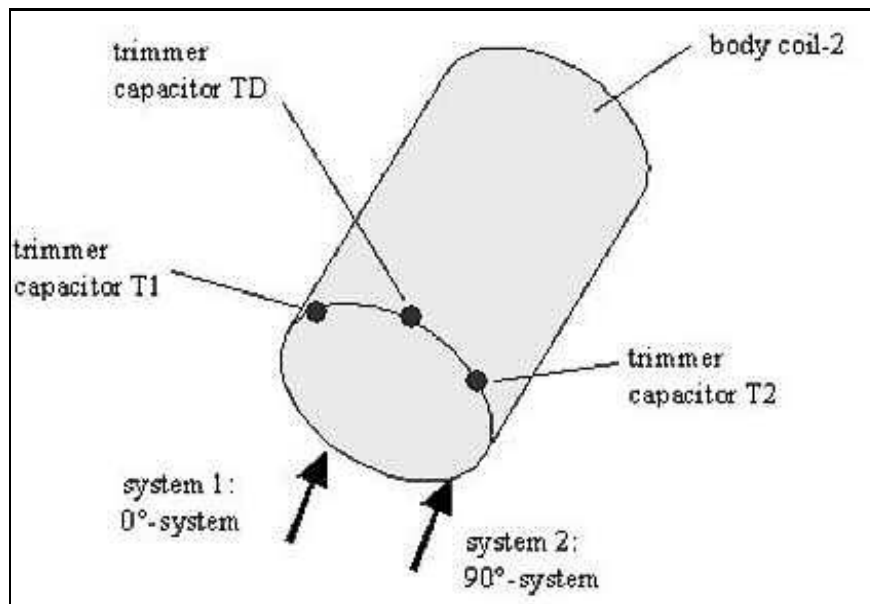
You will be guided through iterative fine adjustments. Between the adjustment steps, the resonance and decoupling of both systems (0 and 90 degree) can be varied manually using the three trim capacitors at the coil. The program stops when both resonance and decoupling are within specifications.

The trim capacitors T2 and T1 affect the resonance frequencies of systems 1 and 2. However, capacitor TD affects the decoupling between the two systems.

The decoupling has to be adjusted only if the measurement of the transmission value is not within specifications.

The trim capacitors allow 53 turns. By default, they are set in the middle position, which means you can make 26+1/2 turns clockwise or 26+1/2 turns counter-clockwise.

Fig. 161: BC Tuning - trimmer capacitors, (T1 left, TD middle, T2 right)



To prevent damage to the trimmers, use caution when approaching the end position. Do not use excessive force.

Trimmer capacitor	Main influence on	Clockwise (cw)	Counterclockwise (ccw)
T2	Resonance frequency system 1	Decrease resonance frequency 1	Increase resonance frequency 1
T1	Resonance frequency system 2	Decrease resonance frequency 2	Increase resonance frequency 2
TD	Decoupling between systems 1 and 2	Increase capacity	Decrease capacity



If the values are slightly out of spec, the system will still work. Be aware that trimmer capacitor T2 influences system 1 and vice versa.

Step 1	Reflection adjustment of delta frequency		
	delta < 100 kHz ?		
	Yes	No	
	No action required	Frequency 0 degree > frequency 90 degrees	
		Yes	No
		Turn T2 2x cw Turn T1 2x ccw	Turn T2 2x ccw Turn T1 2x cw
		Go to step 1	

Step 2	Mechanical adjustment of delta decoupling (transmission)	
	Transmission 0... -12 dB	
	Yes	No
	Adjust mechanical position of the BC in the gradient coil.	Go to Step 3
	Go to step 2	

Step 3	Electrical adjustment of decoupling (transmission)		
	Transmission T = -12 dB .. -18 dB		
	Yes		No
Step 3.1	Turn TD 3x cw, press start button at panel		No action required
	T lower		
	Yes	No	
Step 3.1.1	Turn TD 3x cw, press start button at panel	Turn TD 3x ccw, press start button at panel	
	T lower -> go to 3.1.1	T lower -> go to 3.1.1	
	No -> undo last step	No -> undo last step	
If necessary, repeat step 3.1			

10.2.6.2 Remarks about BC tuning

- T1 and T2 also influence the decoupling.
- TD also influences the frequency.
- The influence of TD relative to T1 and T2 on the decoupling is four times higher. However, it is only half as high with respect to the frequency of the two partial systems.

- To change the decoupling in the same direction, turn TD in the opposite direction as T1 and T2.

Frequency adjustment with constant decoupling:

0-degree system:

Turn T1 x turns and TD x/2 turns in the same direction.

90-degree system:

Turn T2 x turns and TD x/2 turns in the same direction.

Decoupling adjustment with constant frequency:

Turn TD x turns and T1,T2 x/2 turns in opposite direction.

10.2.7 Concluding Steps

- Install the remaining magnet covers (→ Magnet covers / M6-020.841.14)
- Perform Coil Power Loss again

10.2.8 Final Checks

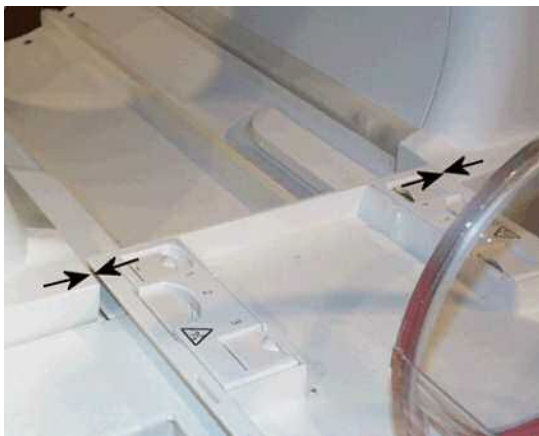
**CAUTION**

The width of the gaps between the patient table and the front cover on right and left side as well as the body coil and patient bore light bars should be of equal size and less than 8 mm. (Refer to the Start-up instructions)

Risk of injury by jamming fingers.

- » Minimize the gaps by adjusting the covers relative to the Body coil. Refer to System Start-up.

Fig. 162: Gap between patient table and cover / bore



- Perform full QA.
If Image Orientation check fails during QA, contact HSC.

- Perform Backup / Restore to save the Tune-up data.

11.1 GC Water Distributor (Avanto/Espree)

(Material number 81 19 641 for Avanto, 81 21 290 for Espree)

11.1.1 Tools and Time required

11.1.1.1 Time

120 min

11.1.1.2 Tools

- Non-magnetic tool set
- Non-magnetic scalpel



WARNING

Strong magnetic field near the gradient coil water connections exerts high attraction on magnetic objects.

Attracted objects can lead to serious physical injuries or death.

- » Use non-magnetic tools. If not available, ramp-down the magnet.

11.1.2 Preliminary



WARNING

High voltage!

Contact with voltage conducting parts leads to death or serious physical injury

- » Adhere to the described steps for your own safety

1. Switch off GPA circuit breakers F1 and F2.
2. Wait at least 20 minutes to ensure that the final stage capacitors are fully discharged.
3. **System with SEP**
 - Switch off F3 in the ACS (water pump)
4. **System with IFP**
 - Switch off ACC/F102 (water cooling circuit)

11.1.3 Disassembly

11.1.3.1 Disassembly of Magnet Covers

To get access to the water distributors, disassemble the back cover/ funnel of the magnet. For details of this procedure, refer to the "Installation Instructions / Installation of covers".

11.1.3.2 Draining the Water Circuit of the Gradient Coil

System with IFP/SEP



Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit.

If drained cooling fluid (38% ethylene glycol water mixture) requires refilling, use a hand pump (service tool) or water pump with drill.

- Disconnect the quick coupling of the water distributors (inlet and outlet flow).
- Mark and note the correct positions of the connections at the gradient coil on the replacement hoses. (The hoses are already connected to the replacement water distributor).
- Cut the water hoses to the gradient coil at a length of approximately 15 cm from the distributor. Have plastic buckets prepared to collect the cooling fluid/ water.
- Wait until cooling fluid/ water is fully drained.

11.1.3.3 Disassembly of the Water Hose from the Nozzle

Examples of hose connections at water distributors:

Fig. 163: Hose connection to water distributor



Fig. 164: Detail hose nozzle - water distributor



Examples of hose connections at gradient coils:

Fig. 165: Example 1 - water hose connection at gradient coil

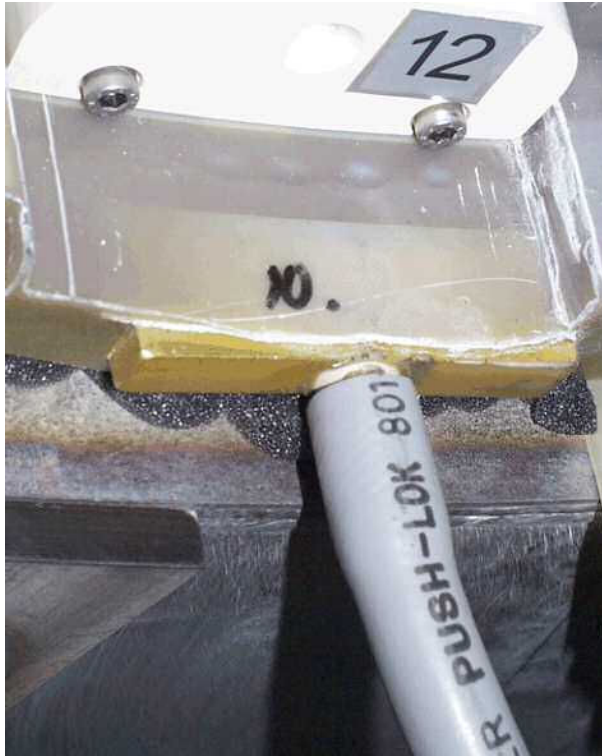


Fig. 166: Example 2 - water hose connection at gradient coil



Fig. 167: Example - hose nozzles at gradient coil



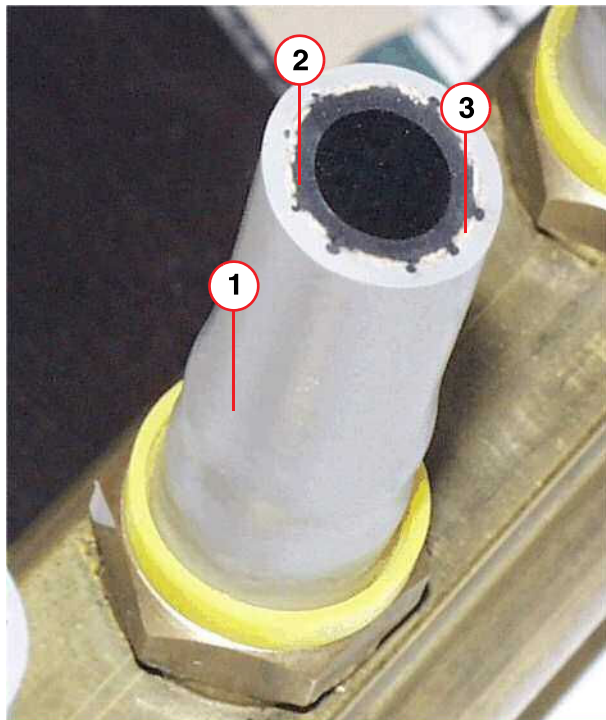
Structure and property of the water hose

The water hose is made of 3 layers (→ 1,2,3/Fig. 168 Page 182):

- the outer grey upper layer (1)
- the white web layer (3) and
- the black inner rubber layer (2).

The web (3) has the property to contract if the hose is pulled from the nozzle. Because the web is not flexible, the hose is also contracted and is fastened on the nozzle. By pulling, the hose cannot be removed from the nozzle. Otherwise, during pushing the hose onto the nozzle, the web is widened and facilitates to push the hose over the nozzle.

Fig. 168: Structure of water hose



- (1) Grey rubber layer
- (2) Black rubber layer
- (3) White web layer

Removing the hose from the nozzle

NOTICE

To remove the water hose from the nozzle, never cut perpendicularly to the surface of the hose and parallel to the brass nozzle. (→ Fig. 169 Page 183)

If the nozzle surface becomes scratched, the connection will leak, and in the worst case the gradient coil needs to be replaced.

» Scrape parallel to the hose surface. Remove the grey outer rubber layer, disrupt the white web, but don't completely remove the black inner rubber layer.

- Scrape carefully with a non-magnetic scalpel parallel to the surface of the hose.
- Cut through the web, but do not scrape too much of the underlying black rubber layer. (→ Fig. 170 Page 183)

Fig. 169: Never cut the hose perpendicular to the hose and parallel to the hose nozzle

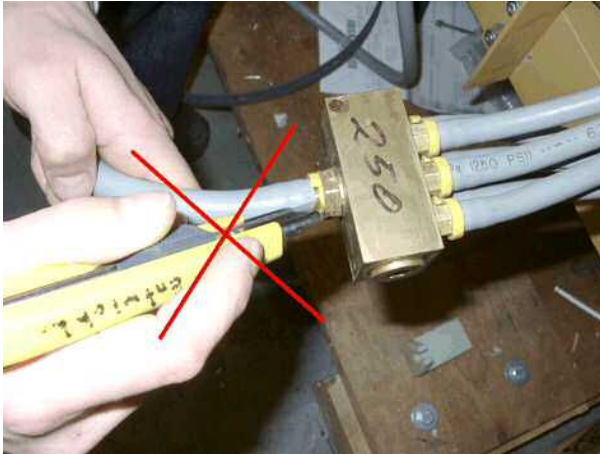


Fig. 170: Scraping the water hose at the hose nozzle



- If the web is destroyed, remove the hose from the nozzle by manually bending it strongly (→ Fig. 171 Page 183), (→ Fig. 172 Page 183).

Fig. 171: Removing the hose from the nozzle



Fig. 172: Scraping parallel to hose surface



11.1.4 Assembly

- Wet the nozzle with water.
- Push the hose applying appropriate force across the entire nozzle (→ Fig. 165 Page 181).

11.1.5 Final Steps and Checks

- **Systems with SEP:** Switch on F3 in the ACS (water pump).
- **Systems with IFP:** Switch on ACC/F102 (water cooling circuit).
- To refill the water circuit, refer to Installation Instructions.
- To check the water pressure, refer to Startup description.

- Check the water flows and temperatures in the Service Software under **Magnet & Cooling > Cooling Status**
- Check tightness of hose connections and water distributor.
- For systems with Chiller/IFP: Bleed the chiller water pump, refer to Startup description.

11.1.6 Concluding Steps

- Reassemble the magnet covers.
- Switch on GPA circuit breakers F1 and F2.

11.2 Defective NTC-Sensors of Gradient Coil

11.2.1 Tools and Time required

11.2.1.1 Time

120 min

11.2.1.2 Tools

- Non-magnetic tool set
- Non-magnetic Pin-Ejector tool for Metrimate connector,
- or if not available AMP-Ejector (material number 44 10 601 (**magnetic!**))
- DMM (with long measurement leads)



WARNING

Strong magnetic field near the gradient coil exerts high attraction on magnetic objects.

Attracted objects can lead to serious physical injuries.

- » Use non-magnetic tools. If not available, ramp-down the magnet.

11.2.2 Preliminary



WARNING

High voltage!

Contact with voltage-conducting parts leads to death or serious physical injury.

- » Adhere to the described steps for your own safety.

1. Check for PTAB error messages MRI_PXX 163 to 168 and MRI_PXX 171 to 176. At least 6 NTC sensors must be connected and working properly.
2. Switch off GPA circuit breakers F1 and F2.
3. Wait at least 20 minutes to ensure that the final stage capacitors are fully discharged.
4. Using a DMM measure the resistance of the NTCs at PTAB/ X50/ <W 5520> (see table below) to locate the defective NTC (typical resistance values of used NTCs at room temperature: $R > 15 \text{ k}\Omega$, approximately $40 \text{ k}\Omega$ at 20 deg. C).
5. Remove the magnet back cover to get access to the gradient coil backside.

6. Repeat the resistance measurement at the 18-pin Metrimate connector at the gradient coil to detect shorted cables or open connections.



Because of strong magnetic field, use long measuring leads to avoid damage to the DMM.

11.2.3 Procedure

- **Espreo:**

Interchange the pins of the defective NTC at the 18-pin connector (eject the pins from the connector) with the equivalent NTC of the 9-pin connector (see table below).

For defective Y/Z-pri NTC use a X-pri spare NTC.

- **Avanto:**

Interchange the pins of defective NTC (eject the pins from the connector) with the spare NTC at pin position 16 - 17.

NTCs at 18 pin Metrimate connector (X2) at gradient coil (Avanto/ Espreo)	Corresponding pins at connector PTAB/ A4120/ X50 (cable <W5520> from gradient coil
4 red - 5 blue X-pri (180 degree)	25 - 13
6 red - 7 blue X-pri (0 degree)	24 - 12
8 red - 9 blue Y-pri	23 - 11
10 red - 11 blue Z-pri	22 - 10
12 red - 13 blue X-sec (150 degree)	21 - 9
16 red - 17 blue (spare NTC only on Avanto)	
14 red - 15 blue X-sec (60 degree)	20 - 8
1 - 2 (Plug check)	14 - 2 (Plug check)
3 (Shield)	3,4,15,16 (Shield)
Spare NTCs at 9 pin Metrimate connector at gradient coil (Espreo only)	
2 red - 3 blue X-pri (180 degree)	
4 red - 5 blue X-pri (0 degree)	
6 red - 7 blue X sec (150 degree)	
8 red - 9 blue X sec (60 degree)	

11.2.4 Concluding Steps

- Reassemble the magnet covers.
- Switch on GPA circuit breakers F1 and F2.

11.3 Gradient Coax Cables Avanto (material number 106 78 832)

(replaces material number 81 19 625)

11.3.1 Requirements

11.3.1.1 Safety Information



WARNING

High voltage!

Contact with voltage-conducting parts leads to death or serious physical injury.

» Adhere to the described steps for your own safety.



CAUTION

During replacement you have to work close to non-isolated gradient cable connections at the GC.

Hazardous voltages may be present at gradient cable connections.

» Switch off the gradient power amplifier via circuit breaker F2 and F1 in the GPA. Wait at least 20 minutes before you start working at the gradient coil.

11.3.1.2 Estimated Completion Time and Number of Persons

Tab. 6 Completion time

Work Steps	Time	Persons
Preparations	30 min	1
Removal/ Installation	2 hr	1
Startup/ TuneUp/ QA	30 min	1
Final Steps	n.a.	n.a.
	Total = 3 hr	

11.3.1.3 Aids or Tools

- Non-magnetic standard tool set (material number 77 57 706)
- Non-magnetic torque wrench ranging from 20 - 50 Nm (material number 83 95 803)
- Non-magnetic torque wrench ranging from 5 - 20 Nm (material number 83 95 829)
- Accessory for torque wrench (material number 83 96 116)
- Jaw wrench SW17 nonmag. (material number 105 63 219)

- Set of NORD-LOCK washers GC (material number 103 58 905)

Important Information for the Proper Use of the Torque Wrench



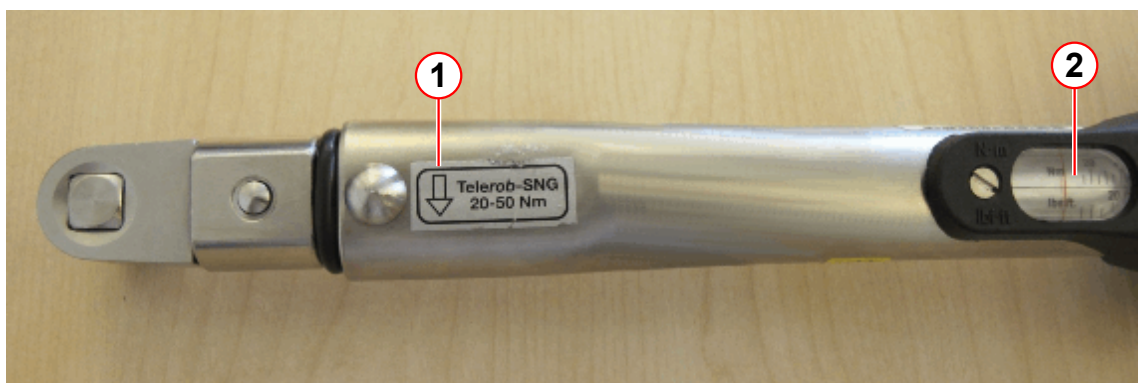
Assure the torque wrench has a valid calibration sticker before starting the cable replacement, see (→ Fig. 173 Page 189).

Fig. 173: Calibration sticker valid through November 2010



- Always use the torque wrench correctly:
 - » The scale (→ Pos.2/ Fig. 174 Page 189) must be visible during fastening.
 - » Always torque in the direction of the arrow (→ Pos.1/ Fig. 174 Page 189).

Fig. 174: Proper use of torque wrench



- (1) Hint for rotating direction
- (2) Scale



Never use the torque wrench to unscrew nuts or bolts.

- For adjusting the required torque, unlock the torque wrench first (→ Fig. 175 Page 190).
 - » After the adjustment has been performed, lock the torque wrench (→ Fig. 176 Page 190).

Fig. 175: Torque wrench - unlocked



Fig. 176: Torque wrench - locked



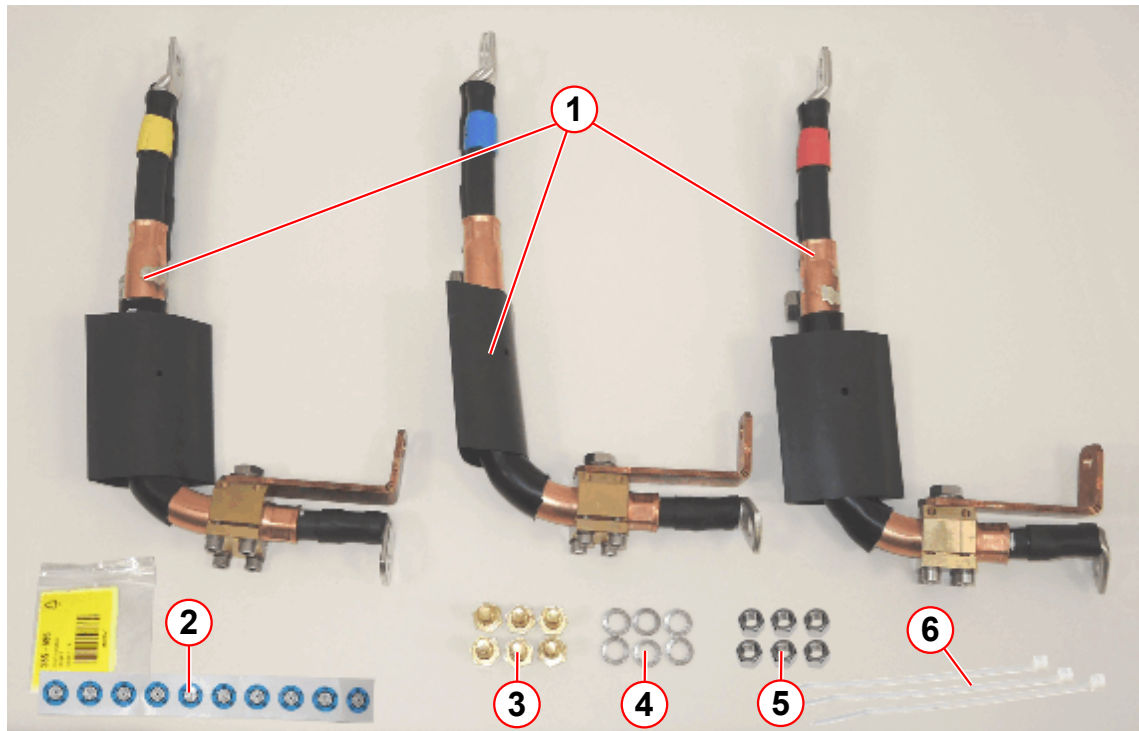
- In the next figure you can see an example of a torque adjusted to 35 Nm (→ Fig. 177 Page 190).

Fig. 177: Torque wrench set to 35 Nm



Contents of the Cable Repair Kit

Fig. 178: Set_Coax_Avanto_V05 (10678832)



The Set_Coax_Avanto_V05 (106 78 832) replaces the former material number 81 19 625.

Tab. 7 Contents of the Set_Coax_Avanto_V05 (10678832)

Item	Description	Quantity
1	Coax cable Avanto V05 (X, Y, Z)	3
2	TEMP. INDICATOR 88 TO 110 DEGREE C	12
3	SCREW M10X20 DIN 933-CU2 (CuZn37) (for GC connection)	6
4	NORD-LOCK WASHER M10-16,6MM (for GC connection)	6
5	NUT M8 ISO4032-A4-70 (for cable ground clamp at connection plate)	6
5	CABLE TIE A188X4,8 (for fastening shrink tube)	3

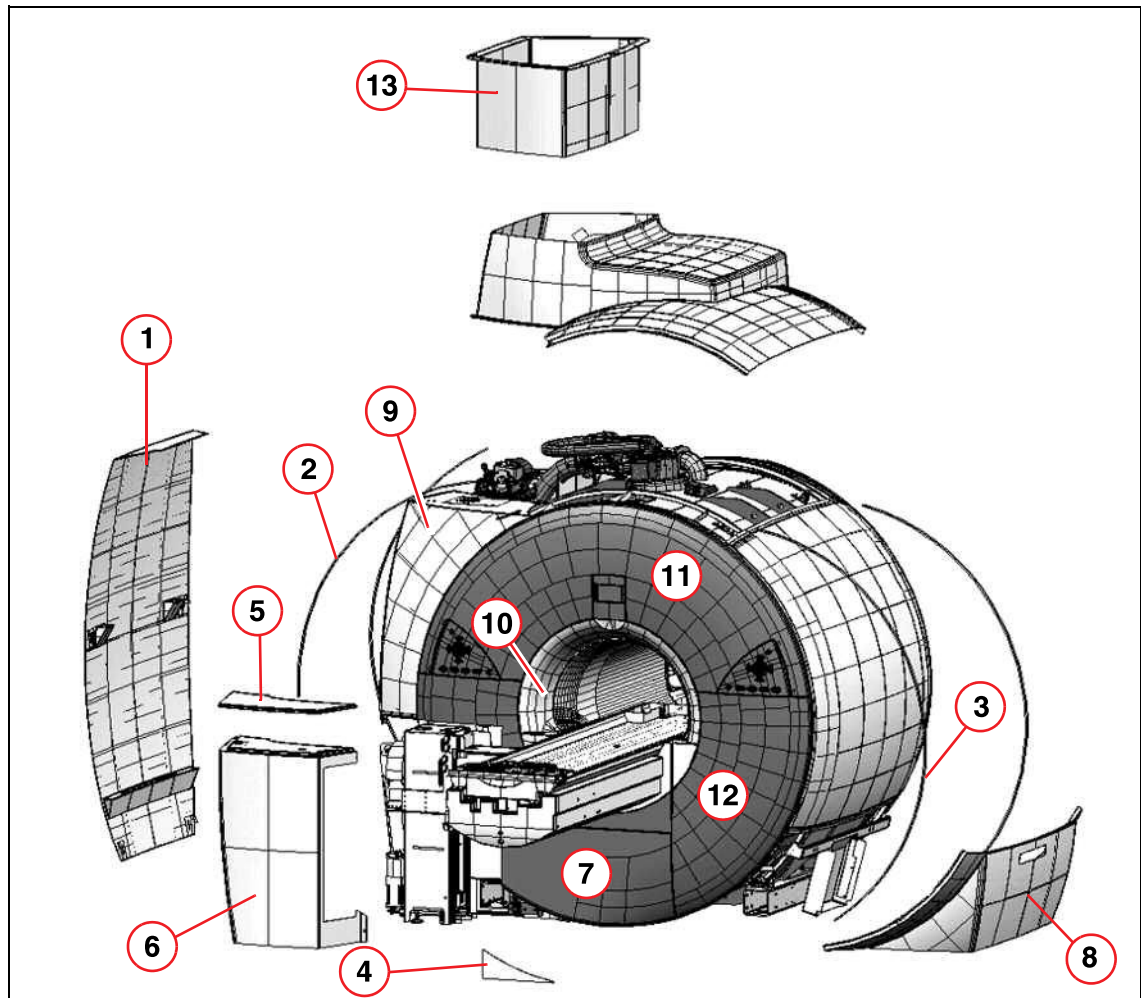
11.3.2 Preparatory Work Steps

11.3.2.1 Power switch-off

- Select **System > Control > Meas Recon > Standby**
- Switch off F2 and F1 in the GPA.
Wait at least 20 minutes before you start working at the gradient coil
- Switch off electrical shim with circuit breaker F21/ACC

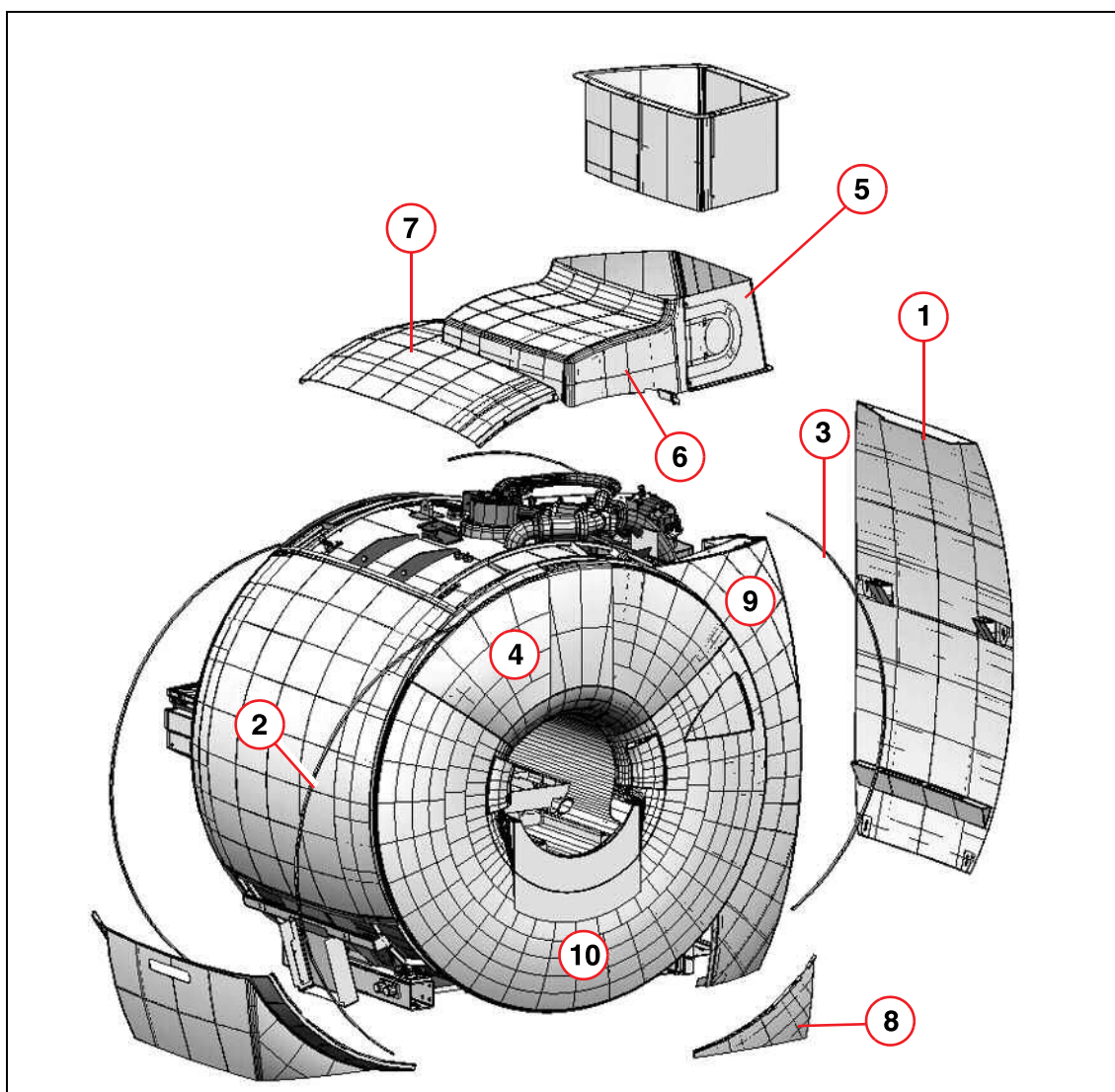
11.3.2.2 Removing Covers

Fig. 179: Overview Avanto cover, patient end



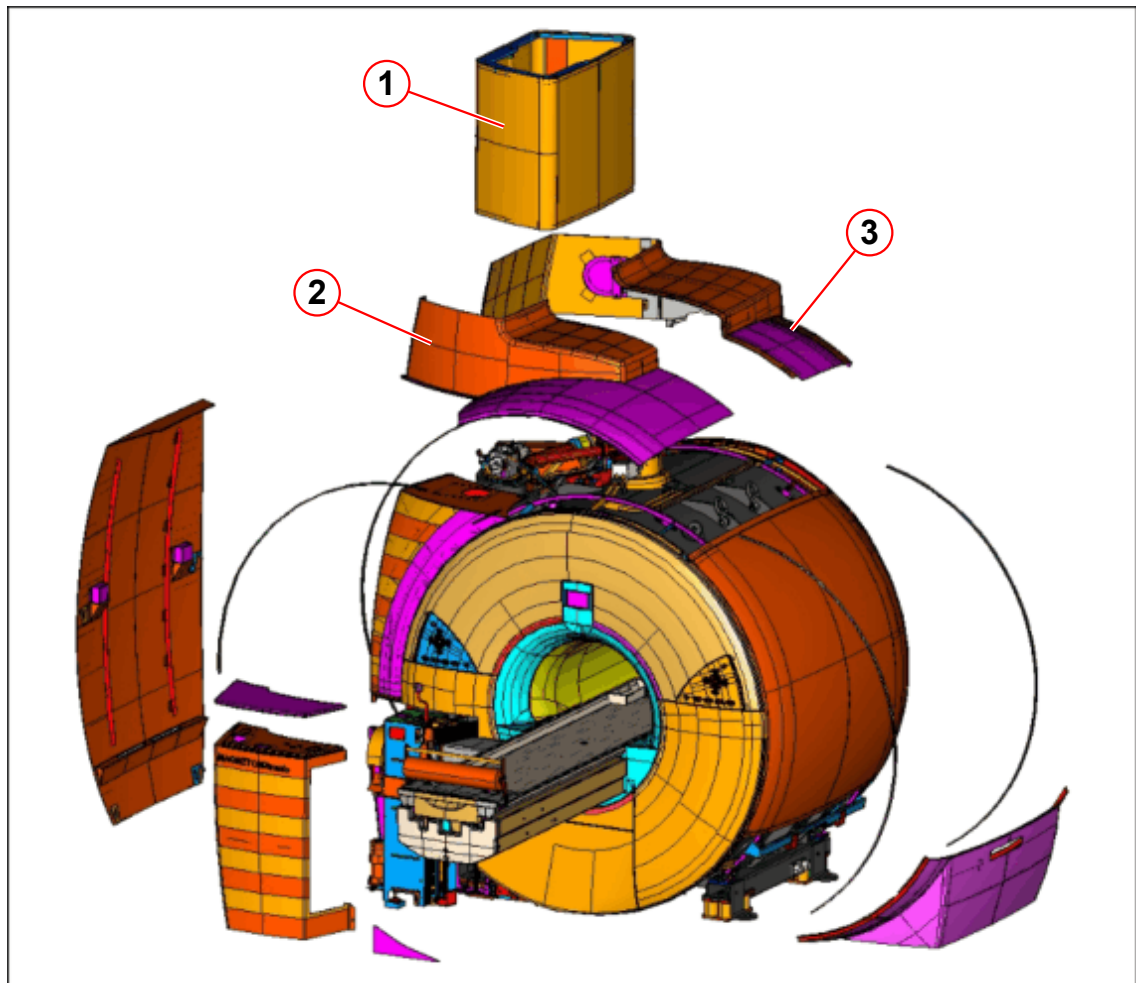
- (1) Service access cover
- (2) Left front ring segment
- (3) Right front ring segment
- (4) Lower front cover plate
- (5) Lifting column covering,
- (6) Lifting column cover
- (7) Bottom front funnel
- (8) Side panel, lower right
- (9) Left front cover
- (10) Front insert funnel
- (11) Top front cover
- (12) Middle front cover
- (13) Tower extension

Fig. 180: Overview Avanto cover, service end



- (1) Service access cover
- (2) Left rear ring segment
- (3) Right rear ring segment
- (4) Top rear cover
- (5) Left tower
- (6) Right rear tower
- (7) Right tower cover
- (8) Right lower rear cover
- (9) Right rear cover
- (10) Lower rear cover

Fig. 181: Overview, Avanto cover (second version), patient end



- (1) Tower extension (second version)
- (2) Front right tower cover
- (3) Rear right tower cover



Please handle the covers with great care.

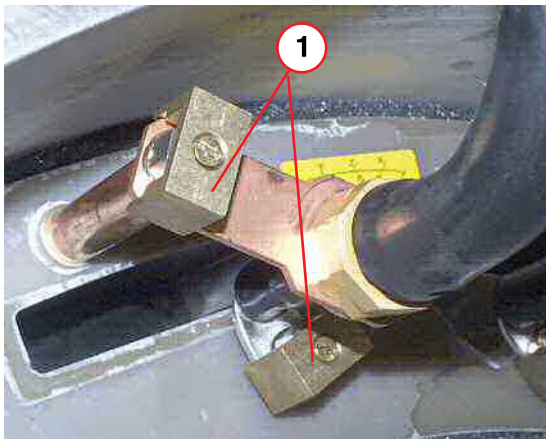
Mark the position of the covers at the base frame with a waterproof pen so that the starting position can be achieved again.

Remove the following covers:

- Remove magnet top rear cover.
- Upper service access cover
- Cover of the gradient connection plate at magnet rear-side
 - » Store the covers out of your way and line them up in the order you have removed them.

11.3.3 Removal

Fig. 182: Coax cable mounted at GC



(1) Locking clamp

Fig. 183: Removed clamp half



Old version of gradient cables: (with locking clamps)

- Remove the screw locking clamps with a flat-screwdriver.
- Remove the gradient connection screws and washers with a non-magnetic open-end wrench (SW17).

New version of gradient cables:

- Remove the gradient connection screws and washers. Use a non-magnetic wrench with extension and socket (SW17).
- If necessary, remove the coax cable clamp half by releasing 4 screws with a non-magnetic 5-mm Allen key.
 - » Now, there is more space to access the inner cable lug screw.



In case the access space for the wrench is still not sufficient, fully remove the coax cable clamp.

Fig. 184: Gradient connection plate - cover removed



- Unscrew the cable from the gradient connection plate using a 19-mm open-end wrench.
- Remove the cable

11.3.4 Installation

NOTICE

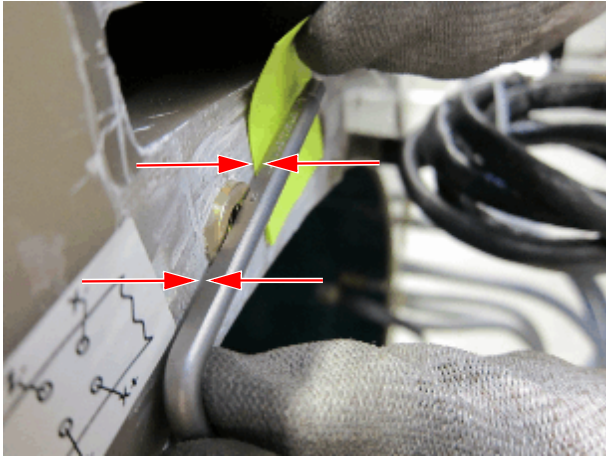
Incorrect mounting of gradient cables under mechanical strain.

Risk of gradient coax cable defects due to fatigue cracks!

- » Carefully adjust the bolts at the gradient Connection Plate within the long holes to achieve a strain-free mounting of gradient cables.
- » The coaxial cables and gradient Connection Plate must not touch vibrating parts like magnet housing, covers etc.

11.3.4.1 Check and Cleaning of Contact Surfaces

Fig. 185: Checking the space between the contact surface and the coil front



- Check the space between the contact surface and the coil front (→ Fig. 185 Page 198).
Use a straight non-magnetic tool (e.g. Allen key or a plastic ruler).
 - » There must be some space around the contact surface to enable full contact with the coax cable lugs.
- If you find a contact without space and full contact with the cable lug is at risk, contact HSC for further assistance.

Fig. 186: Cleaning the gradient coil insert surface



- Clean the contact surfaces at the gradient coil, using Scotch-Brite.

Fig. 187: Cleaning the coax cable copper contact

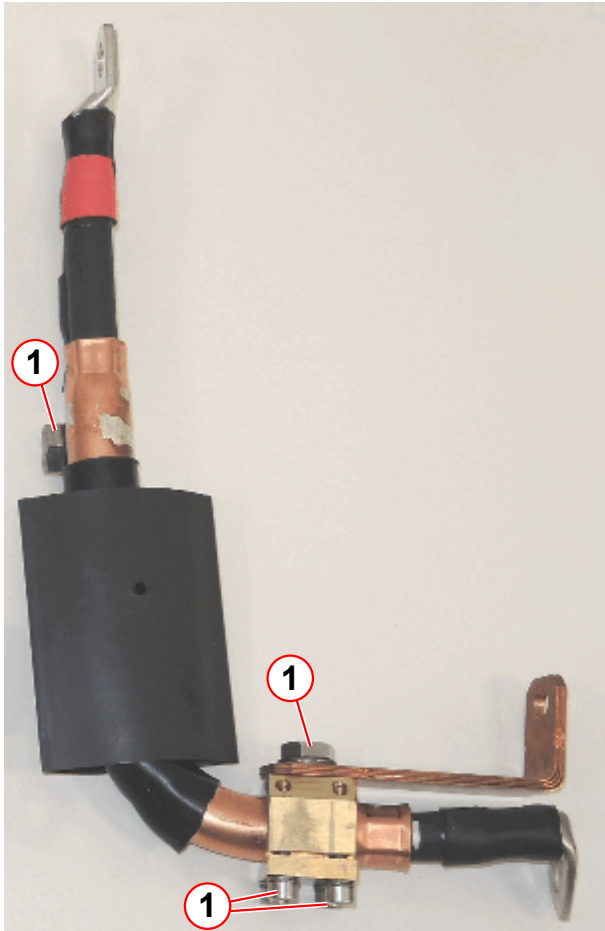


- Clean the copper contact surfaces at the gradient cables, using Scotch-Brite.

Note:

Do not clean galvanized cable lugs with Scotch-Brite.

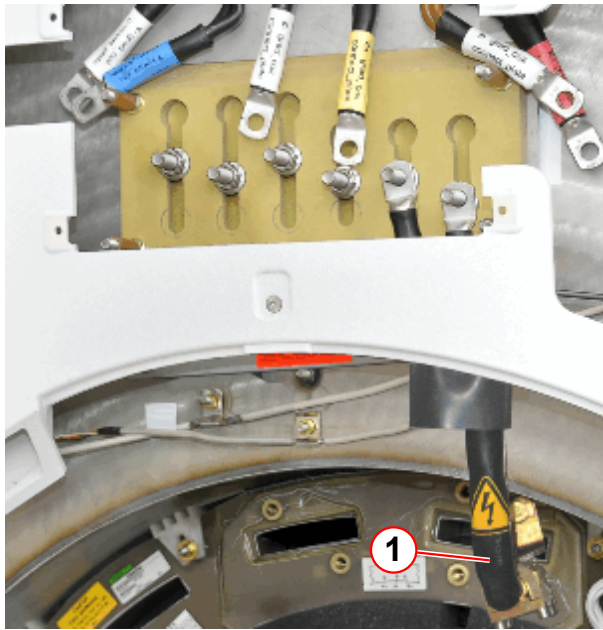
11.3.4.2 Coax Cable Alignment

Fig. 188: Single coax cable

(1) Adjustment screws

- Assure, all coax cable adjustment screws (1) are fastened just hand-tight.
- » It must be possible to align all coax cable components easily by hand.

Fig. 189: Loosely mount the gradient cable



(1) Shrink tube attached

- Pull a shrink tube (1) over the coax cable.
- Provisionally attach the gradient cable to the mounting bolts of the gradient connection plate.



Access space to the screws is limited. Always install the gradient cables from right to left, i.e., start with the cable for the x-axis.

Inner cable lug alignment:

- At the GC, check if the inner cable lug is correctly aligned to the GC connector.
- The right figure below shows a correct alignment without offset between inner cable lug and GC connector (→ Fig. 191 Page 202).
 - » If the coax cable alignment is NOT OK (shown in the left figure below), the mounting bolts at the gradient connection plate have to be adjusted !
 - » **Installation of the cable must be possible without bending or mechanical stress !**

Fig. 190: Inner cable lug alignment is NOT OK !

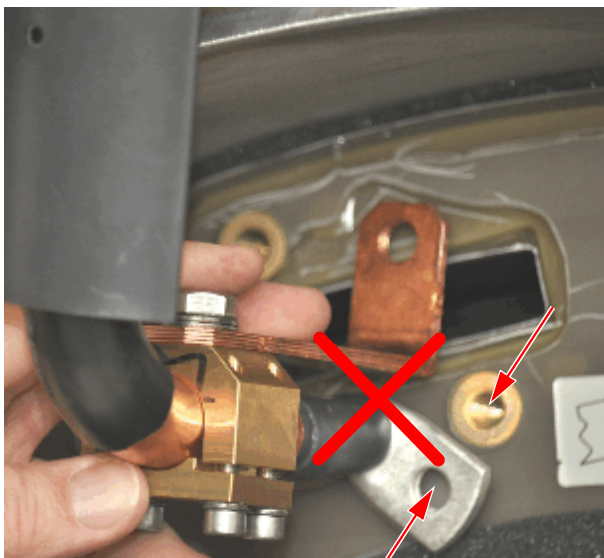


Fig. 191: Inner cable lug alignment is OK

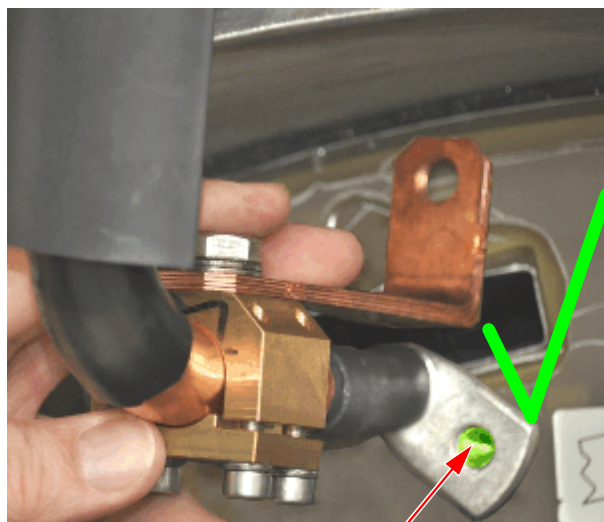
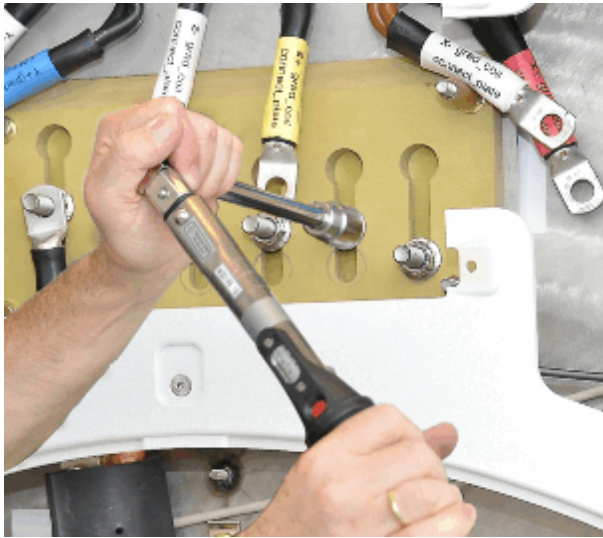


Fig. 192: Adjustment of mounting bolt position

**Cable position adjustment:**

- If necessary, release the securing nut (flange nut) of the mounting bolts with an 18-mm open-end wrench.
- Adjust the mounting bolt position within the long holes until the cables fully match to the GC connectors without any strain, see (→ Fig. 191 Page 202).
- Fasten the securing nut in the optimum position.

Fig. 193: Fastening securing nuts

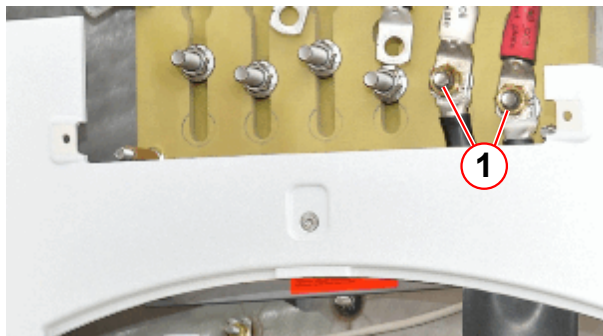


- Remove the coax cable again.
- Fasten the securing nuts with the non-magnetic torque wrench.



- Apply a torque of 21 ± 2 Nm.

Fig. 194: Installed coax and gradient cable



(1) NORD-LOCK washer and nut

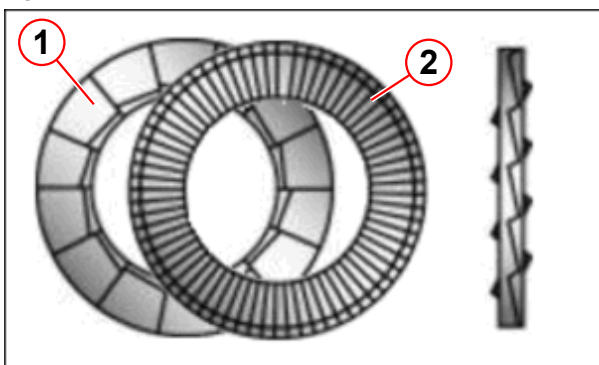
- Re-attach the coax cable to the bolts.
- Mount the gradient cables which come from the RF-filter plate on top of the coax cable lugs.

Note:

The correct assembly is shown in figure:

(→ Fig. 206 Page 209)

Fig. 195: NORD-LOCK washer



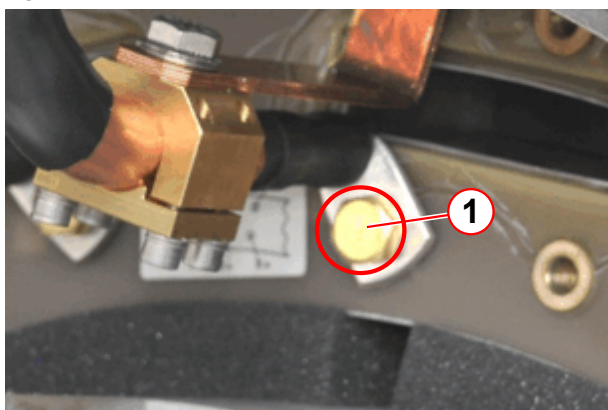
(1) Cam side
(2) Ridged side

- Use a NORD-LOCK washer and fasten the connection provisionally with the nut. Use a 19 mm open-end wrench.
- A NORD-LOCK washer **always** consists of two halves.
 - » Both ridged sides must point towards outside.



Original NORD-LOCK washers can be reused. In case of missing or defective ones, use new NORD-LOCK washers (material number 103 58 905).

Fig. 196: Cable attached to GC connectors



(1) Brass hex screw M10x20 + NORD-LOCK washer

- Fasten the inner cable lug hand-tight with a screw (M10x20) and NORD-LOCK washer.

Outer cable bracket alignment:

- Align the outer cable bracket and clamp to achieve a connection without mechanical stress !
- The right figure below shows a correct alignment without offset between inner cable lug and GC connector (→ Fig. 198 Page 204)
 - » **Installation of the cable must be done without bending or mechanical stress !**

Fig. 197: Outer cable bracket alignment is NOT OK !

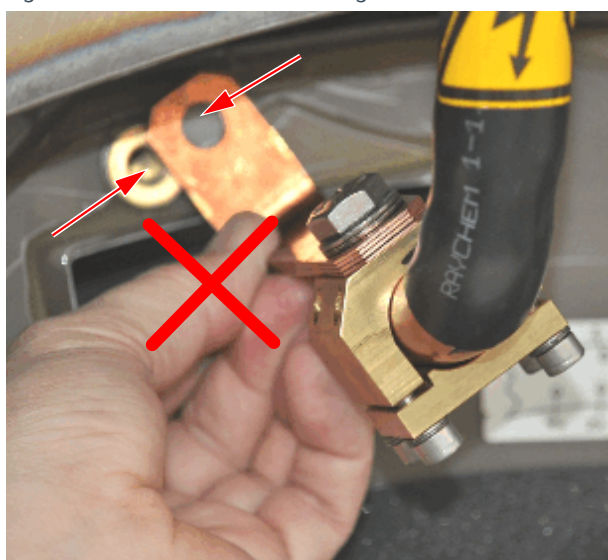


Fig. 198: Outer cable bracket alignment is OK

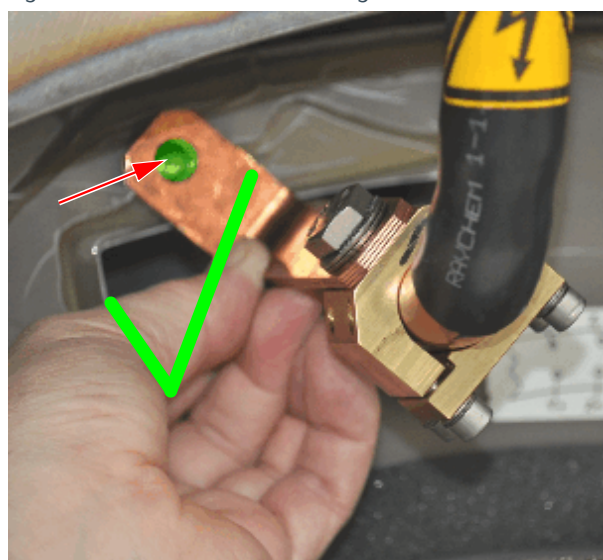
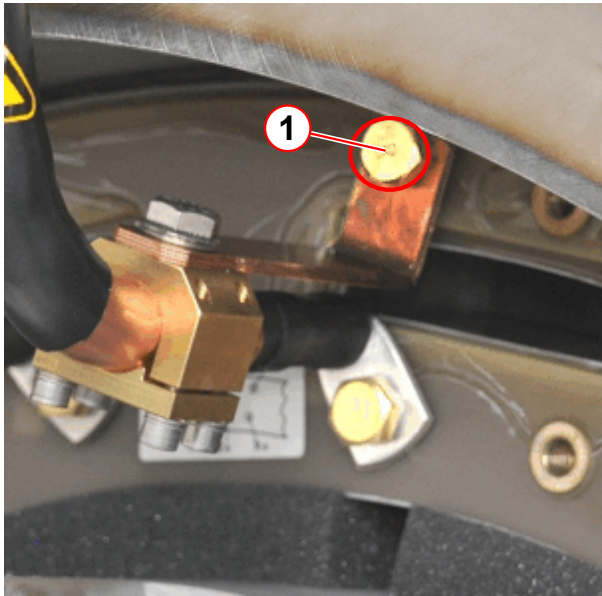


Fig. 199: Cable attached to GC connectors



(1) Brass hex screw M10x20 + NORD-LOCK washer

- Fasten the outer cable bracket hand-tight with a screw (M10x20) and NORD-LOCK washer.

11.3.4.3 Coax Cable Fastening

Fig. 200: Removed clamp half



- Remove the coax cable clamp half by releasing 4 screws with a non-magnetic 5-mm Allen key.
 - » Now, there is enough space to fasten the inner cable lug screw with a torque wrench.

Fig. 201: Fasten GC connectors with 30 Nm

**Inner cable lug fastening:**

- Fasten the inner cable lug screw with a non-magnetic torque wrench (SW17).



- Apply a torque of 30 ± 3 Nm.



In case the access space for the wrench is still not sufficient, fully remove the coax cable clamp.

Fig. 202: Installed clamp half

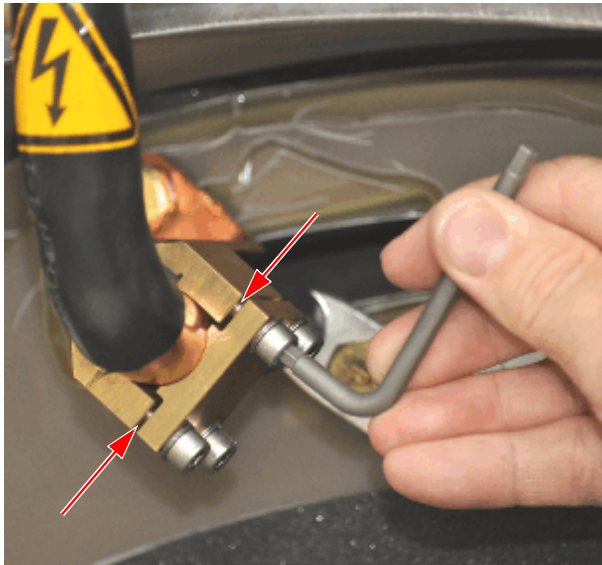


(1) Tighten Allen screws crosswise

Outer cable bracket fastening:

- Re-attach the clamp with the 4 Allen screws and NORD-LOCK washers.
- Fasten the clamp crosswise with a non-magnetic 5-mm Allen key to achieve an equal contact pressure.

Fig. 203: Gap between clamp halves



(1) Check for equal gaps

- Check the gap on both sides of the clamp
 - » The gap must be equal on both sides.
- Tighten the clamp firmly (apply a torque of approx. 6 - 8 Nm).

Fig. 204: Fasten cable bracket hex screw with 30 Nm



- Fasten the hex cable bracket screw with the non-magnetic wrench and the jaw wrench insert SW17 (material number 105 63 219).



- Apply a torque of 30 ± 3 Nm.

Fig. 205: Fasten hex cable bracket screw with 30 Nm



- Fasten the bracket hex brass screw at the GC with a non-magnetic torque wrench (SW17).



- Apply a torque of 30 ± 3 Nm.

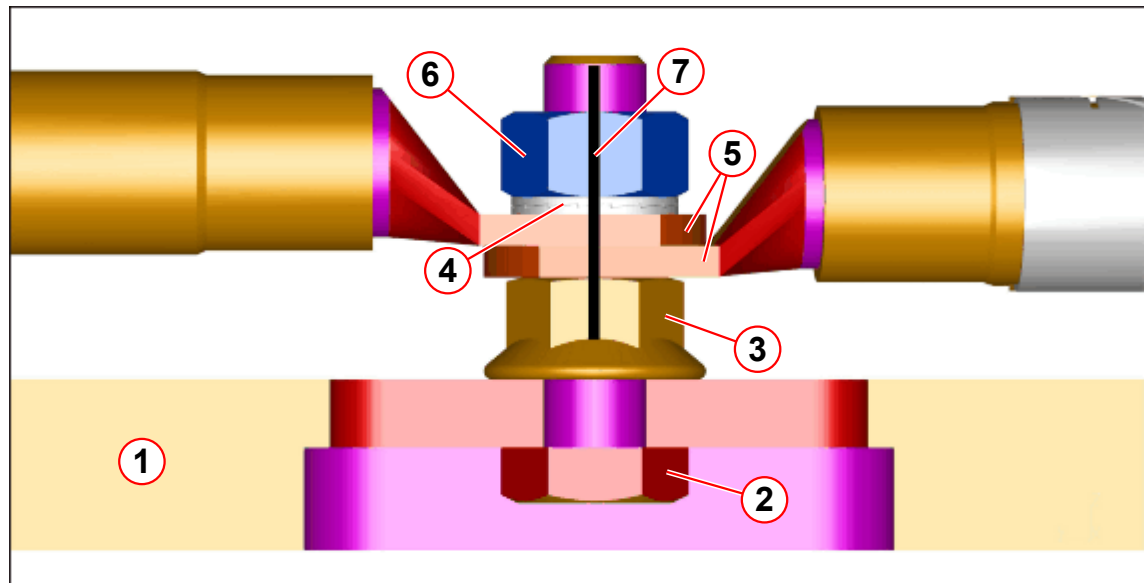
At the Gradient Connection Plate:

- Check that all cable lugs are fitted to the bolts without any strain. If necessary, reposition the bolts (→ Fig. 192 Page 202).
- Fasten the cables with NORD-LOCK washer and nut (6) using a non-magnetic torque wrench (SW19).



- Apply a torque of 40 ± 4 Nm.
- With a waterproof pen, mark the positions of all nuts with a line over the connections.

Fig. 206: Correct set-up of the gradient connection plate at the magnet



- (1) Connection plate
- (2) 12 mm bolt (SW 19)
- (3) Flange nut (SW 18) (torque 21 ± 2 Nm)
- (4) NORD-LOCK washers
- (5) Gradient cable lugs
- (6) Nut (SW 19) (torque 40 ± 4 Nm)
- (7) Mark with waterproof pen

Fig. 207: Fastening of gradient cable joint screw with 40 Nm - Example



At the coax cable joints:

- Fasten the coax cable joint screws with a non-magnetic torque wrench (SW17).



- Apply a torque of 40 ± 4 Nm.
- With a waterproof pen, mark the positions of the nuts.

11.3.4.4 Shrink Tube Fastening

Fig. 208: Fastening of shrink tube with a cable tie



- Feed a cable tie through the hole in the shrink tube.
- Fasten the shrink tube at the coax cable joint.
 - » The cable tie is mounted above the cable joint screw.
 - » Shrinking the tube with a heat gun is not necessary here.

11.3.4.5 Ground Clamp Fastening

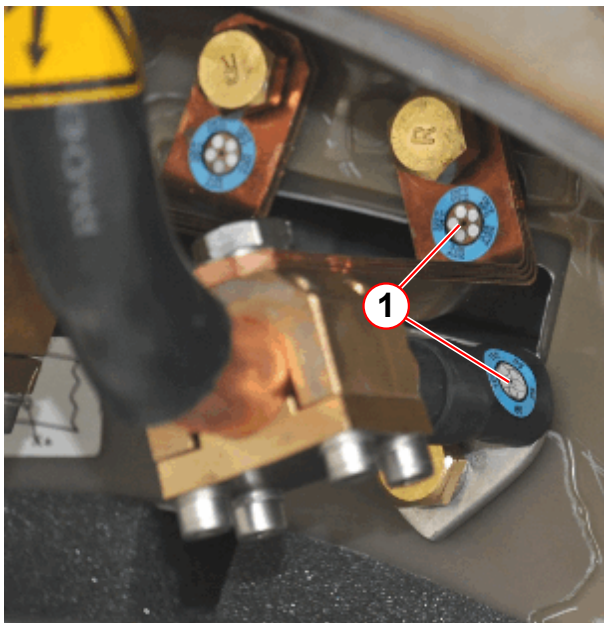
Fig. 209: Fastening of ground clamps



- Fasten the nuts of the cable ground clamps with a non-magnetic wrench (SW13).
- Attach a second lock nut on top (contained in spare part kit (→ Pos.5/Fig. 178 Page 191)).
- Tighten the lock nut with the wrench while holding the nut with an open-end wrench (SW13).

11.3.4.6 Temperature Indicator Attachment (Clock Stickers)

Fig. 210: GC cable connection

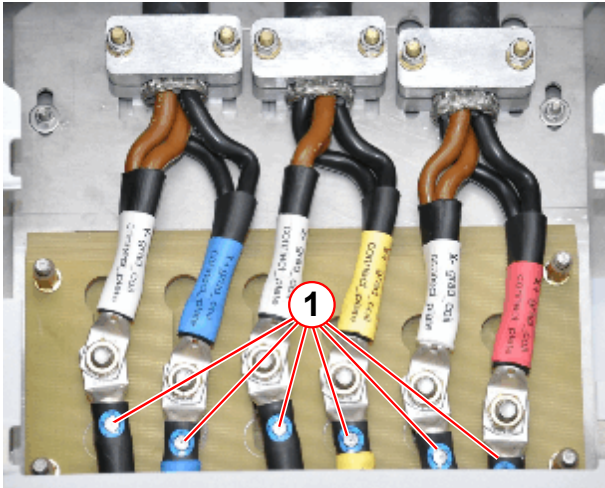


(1) Temperature Indicators (Clock Stickers)

Attach Temperature Indicators close to the screwed connections:

- at each GC connection (inner and outer lug).

Fig. 211: Gradient Connection Plate

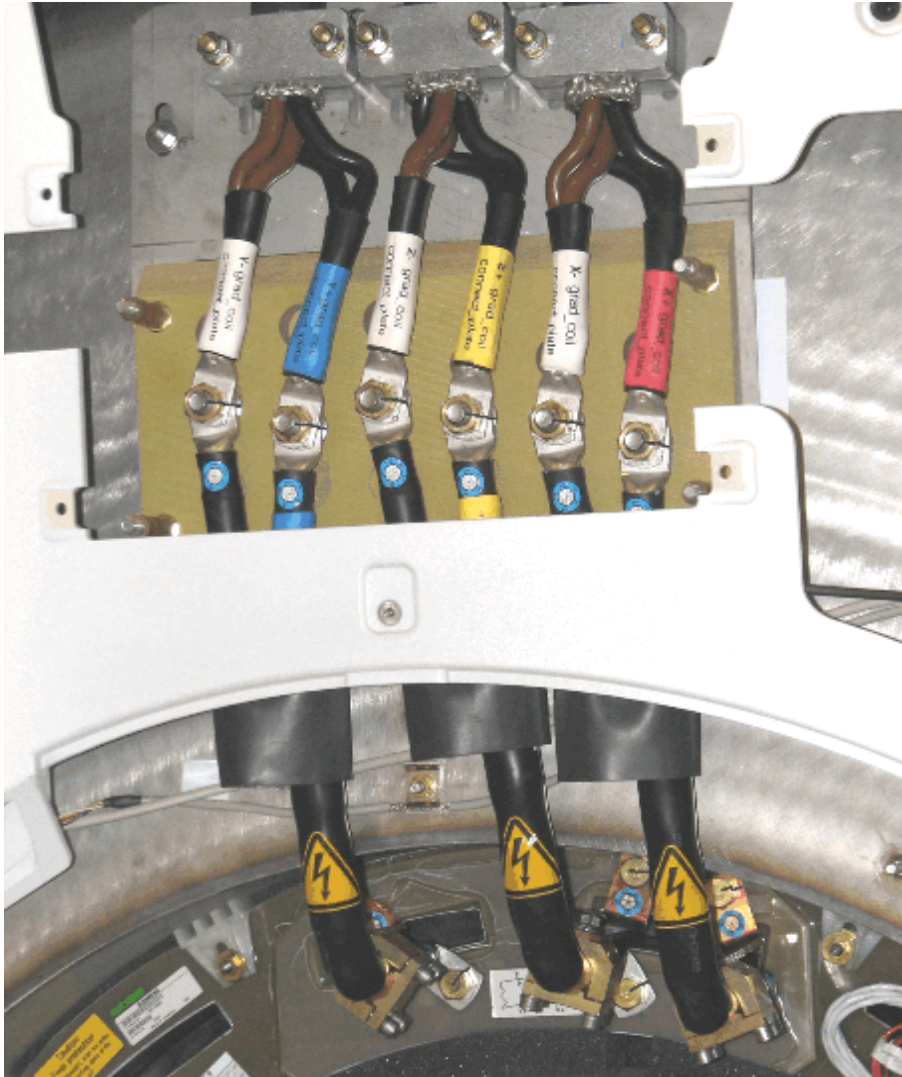


(1) Temperature Indicators (Clock Stickers)

- attach temperature indicators at each connection of the gradient connection plate
- in total, 12 stickers are attached.

- The figure below shows all gradient cables completely assembled.

Fig. 212: Gradient cables completely assembled



11.3.5 Startup

11.3.5.1 Mechanical Work Steps

- Re-install the covers
- Switch on GPA with circuit breaker F1 and F2/ GPA.
- Switch on electrical shim with circuit breaker F20/ ACC.
- Switch the MR system "ON" at the alarm box.

11.3.5.2 TuneUp/ Adjustments

n.a.

11.3.5.3 Function Tests/ Q Tests

- Run QA/ Expert mode/ Spike Check (repeat 5 times)
- Run QA/ Image Orientation Check.

11.3.6 Final Steps

n.a.

11.4 Gradient Coax Cables Avanto (material number 81 19 625)

(original cable version)



This procedure is valid for Avanto systems only.

11.4.1 Requirements

11.4.1.1 Safety Information



WARNING

High voltage!

Contact with voltage-conducting parts leads to death or serious physical injury.

- » Adhere to the described steps for your own safety.



CAUTION

During BC/GC disassembly, you have to work close to non-isolated gradient cable connections at the GC.

Hazardous voltages may be present at gradient cable connections.

- » Switch off the gradient power amplifier via circuit breaker F1/GPA and F2.1/GPA/GSSU.
- » Wait at least 20 minutes to allow capacitor discharge before you start working at gradient connections.

11.4.1.2 Estimated Completion Time and Number of Persons

Tab. 8 Completion time

Work Steps	Time	Persons
Preparations	30 min	1
Removal/ Installation	2 hr	1
Startup/ TuneUp/ QA	30 min	1
Final Steps	n.a.	n.a.
	Total = 3 hr	

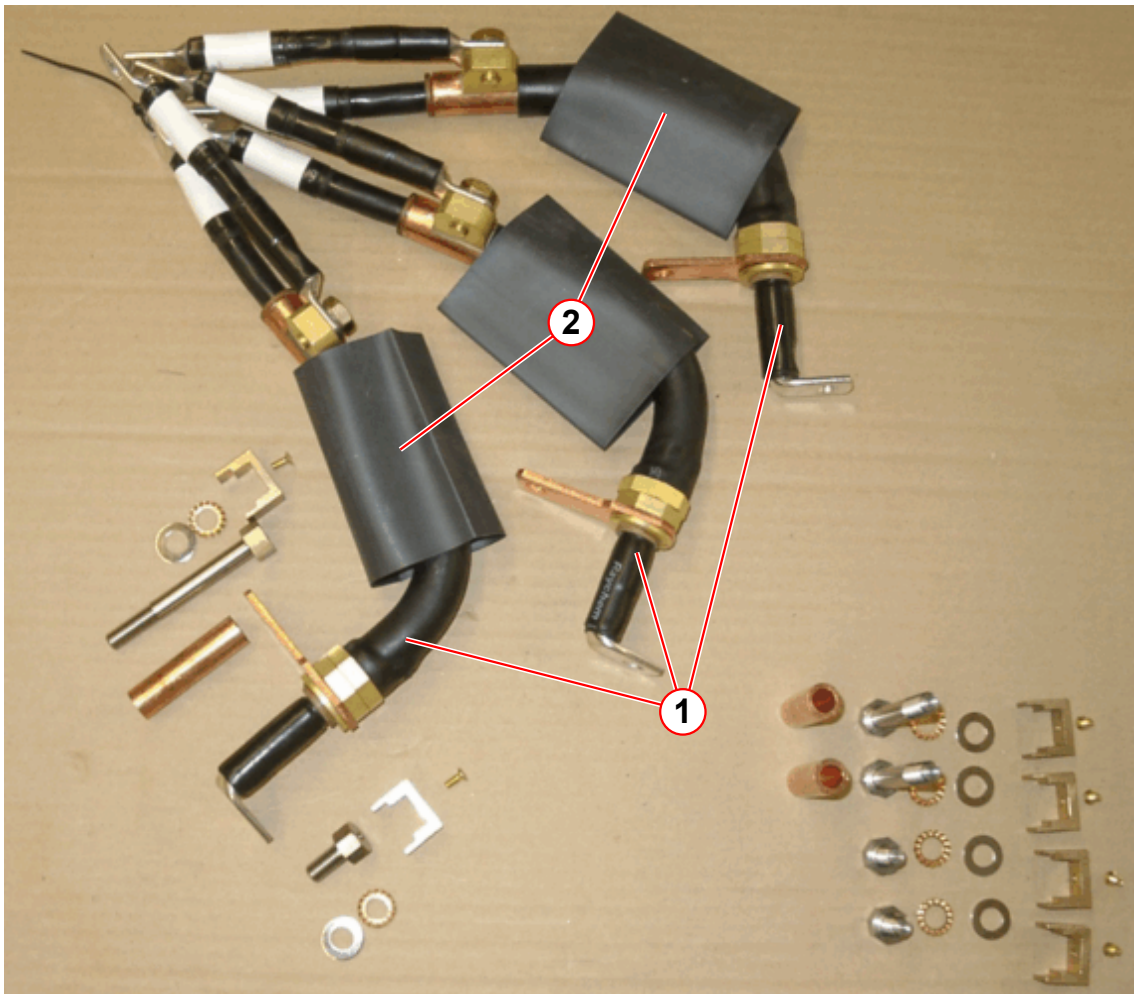
11.4.1.3 Aids or Tools

- Non-magnetic standard tool set (77 57 706) including a 32-mm open-ended wrench

- Non-magnetic torque wrench ranging from 20 - 50 Nm (83 95 803)
- Accessory for torque wrench (83 96 116)
- 8-s. socket gradient coil, non-magnetic (103 99 445)
- Heat gun (for 220 V: material number 97 02 937/ for 110 V: material number 97 02 945)
- Set of NORD-LOCK washers GC (103 58 905)
- Loctite 221 (soft-locking, 20 48 874)

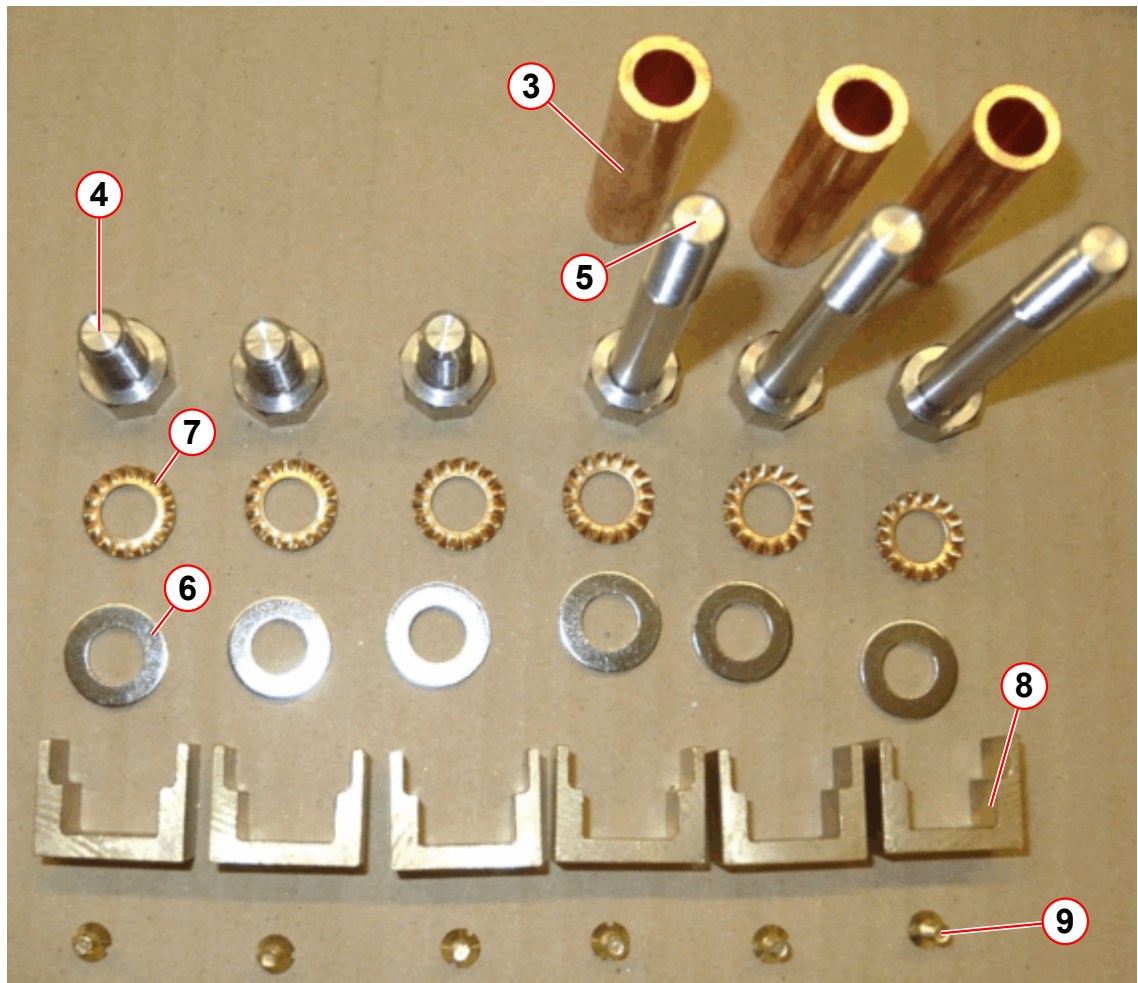
11.4.1.4 Contents of the Cable Repair Kit

Fig. 213: Coax cable kit Avanto - 8119625



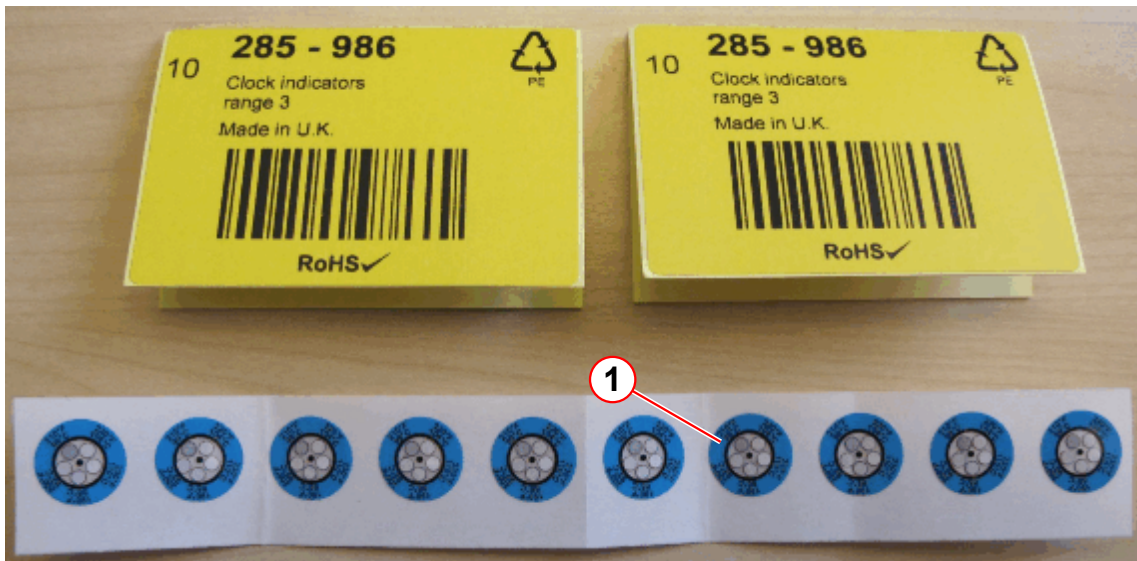
- (1) Coax cable kit Avanto
(2) Heat shrink

Fig. 214: Small parts coax cable kit Avanto



- (3) Copper bush
- (4) Screw M10x20
- (5) Screw M10x80
- (6) Washer A10.5
- (7) Star washer A10.5
- (8) Retaining clip
- (9) Flat head screw M4x10

Fig. 215: Temperature indicator



(1) Temperature indicator

11.4.2 Preparatory Work Steps

11.4.2.1 Power switch-off



WARNING

High voltage!

Contact with voltage-conducting parts leads to death or serious physical injury.

» Adhere to the described steps for your own safety.

1. Switch the MR system to "Standby" at the alarm box.
2. Switch off GPA circuit breakers F1 and F2.1.
3. Wait at least 20 minutes to ensure the power stage capacitors are fully discharged before starting to work on the gradient cables.

11.4.2.2 Removing Covers

- Remove magnet top rear cover.
- Unscrew and remove the old/defect cable from the gradient coil and from the connection plate.

11.4.3 Removal

- Unscrew the old/defect cable from the connection plate.
- Unscrew the old/defect cable from the Gradient Coil.

- Remove the old/defect coax cable.

11.4.4 Installation

NOTICE

Incorrect mounting of gradient cables under mechanical strain.

Risk of Gradient coax cable defects due to fatigue cracks!

- » Carefully adjust the bolts at the Gradient Connection Plate within the long holes to achieve a strain-free mounting of coax cables.
- » The coaxial cables and gradient connection plate must not touch vibrating parts like magnet housing, covers etc.

11.4.4.1 Mounting the Coax Cables

Fig. 216: Cleaning the copper bush



- Clean all contact surfaces including the copper bush (-es) metallic bright, using Scotch Brite.

Fig. 217: Cleaning the gradient coil insert surface





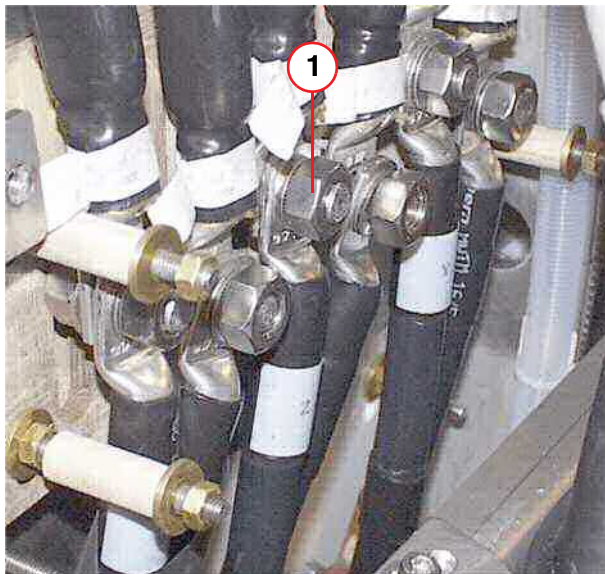
Do not clean galvanized parts (e.g. cable lugs) with Scotch Brite!

Fig. 218: Cleaning the copper lug



- Clean the copper lug with Scotch Brite but leave out the galvanized lug.

Fig. 219: Upper GC cable connection at magnet



(1) Connection between GC cable and coaxial link cable
(tightened with 40 +/- 4 Nm)

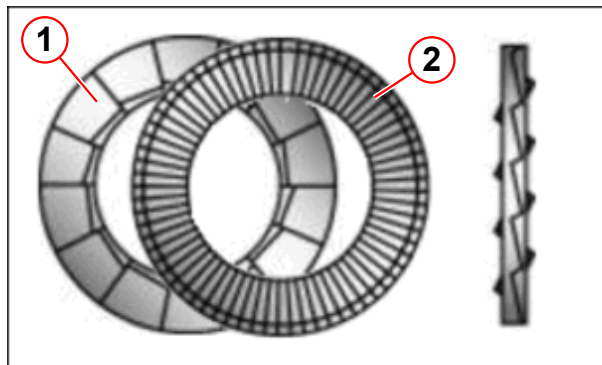
At the Gradient Connection Plate:

- Loosely hang in the gradient cables to the bolts of the connection plate.
- Adjust the mounting bolt position within the long holes to achieve a strain-free installation of the coax cable. .
 - » The coax cable must be routed without mechanic stress.
- Fasten the nuts - at this point just hand-tight.

Note:

Ensure, the delivered shrink tube is loosely attached to the joint of the gradient cable.

Fig. 220: NORD-LOCK washer



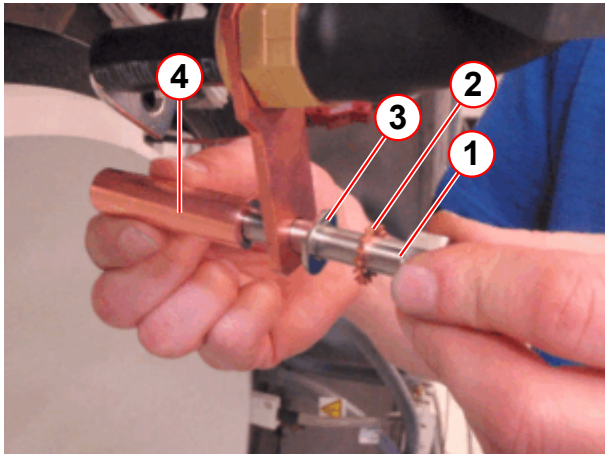
- (1) Cam side
- (2) Ridged side

A NORD-LOCK washer (consisting of 2 parts) must be used between the cable lugs and the nut. The correct assembly of the plates must be maintained: refer to the figure on the left. Please refer to (→ Fig. 242 Page 231) for the correct order of parts.



Usually, the originally fitted NORD-LOCK washers can be reused. Just in case of missing or defective ones, use new parts (set of NORD-LOCK washers material number 103 58 905).

Fig. 221: Correct arrangement of fixing elements



- (1) Screw M10x80
- (2) Star washer A10.5
- (3) Washer A10.5
- (4) Copper bush

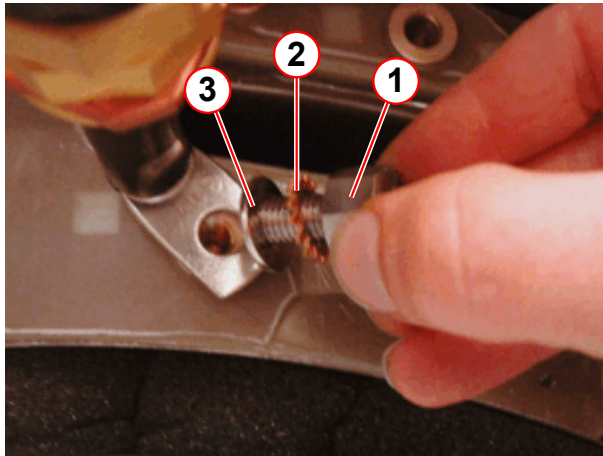
- Prepare the gradient cable and bush using the fixing elements (→ Fig. 221 Page 222).

Fig. 222: Mounting the gradient cable to the coil with the bush



- Connect the gradient cable bush screw to the coil insert - just hand-tight at this point.

Fig. 223: Mounting the 90° cable shoe to the gradient coil



- (1) Screw M10x20
- (2) Star washer A10.5
- (3) Washer A10.5

- Screw the M10x20 bolt into the 90° gradient cable shoe - just hand-tight at this point.

Fig. 224: Tightening the rectangular gradient coil cable shoe (torque: 24 +/- 3 Nm)



- Tighten the gradient cable lug with the non-magnetic torque wrench and 8-s. socket gradient coil non-mag (103 99 445)



- » Apply a torque of 24 ± 3 Nm



For tightening the gradient cables at the gradient coil, use a non-magnetic torque wrench with 8-s. socket gradient coil non-mag (material number 103 99 445).

Fig. 225: Tightening the copper bush to the gradient coil (torque: 24 ± 3 Nm)

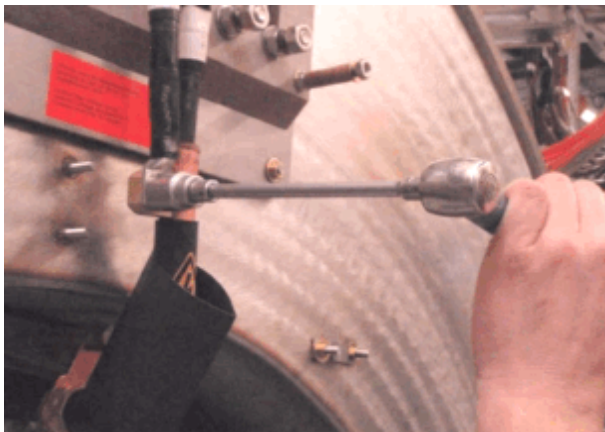


- Tighten the copper bush connection with the non-magnetic torque wrench and 8-s. socket gradient coil non-mag (103 99 445)



» Apply a torque of 24 ± 3 Nm

Fig. 226: Tightening the cable joint (torque: 30 ± 3 Nm)



- Fasten the coax cable joint screws with a non-magnetic torque wrench (SW17) .



» Apply a torque of 30 ± 3 Nm.

At the Gradient Connection Plate:

Fasten the GC mounting bolts with the correct torque.

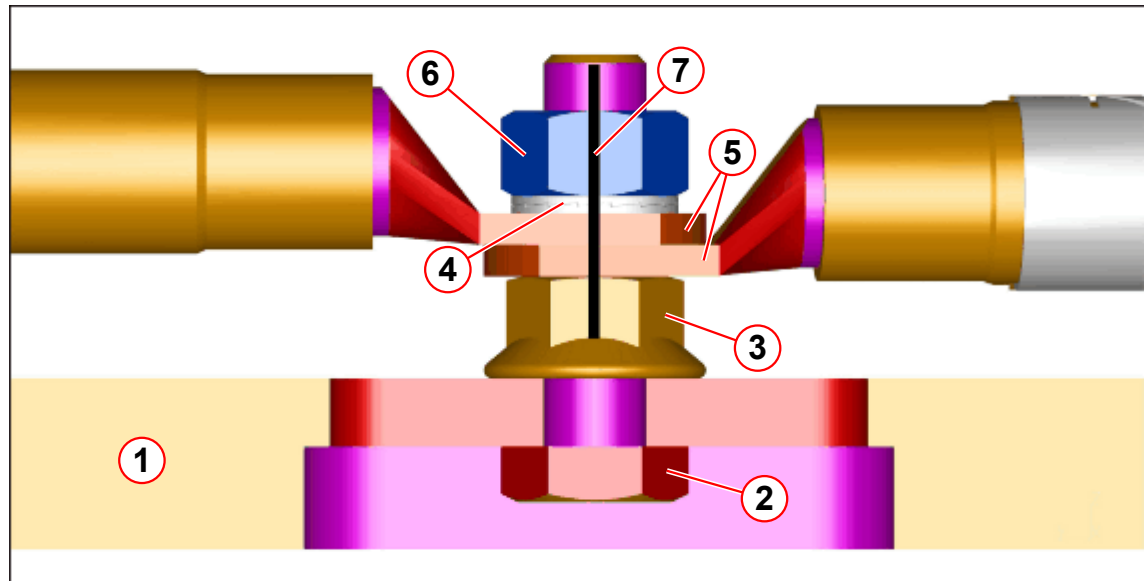
A: New version (with flange nut)

- Fasten the flange nut (3) with a non-magnetic torque wrench (SW18).



- » Apply a torque of 21 ± 2 Nm.

Fig. 227: Correct set-up of the gradient connection plate at the magnet



- (1) Connection plate
- (2) 12 mm bolt (SW 19)
- (3) Flange nut (SW 18) (torque 21 ± 2 Nm)
- (4) NORD-LOCK washers
- (5) Gradient cable lugs
- (6) Nut (SW 19) (torque 40 ± 4 Nm)
- (7) Mark with waterproof pen

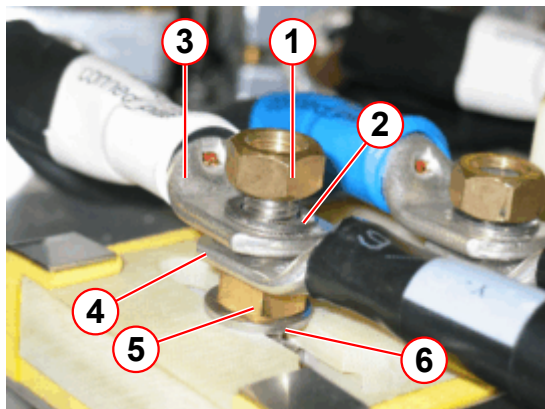
B: Old version (with washer and nut)

- Fasten securing nut (5) with a non-magnetic torque wrench.



- » Apply a torque of 21 ± 2 Nm.

Fig. 228: Cable connection (tightened with 40 ± 4 Nm)



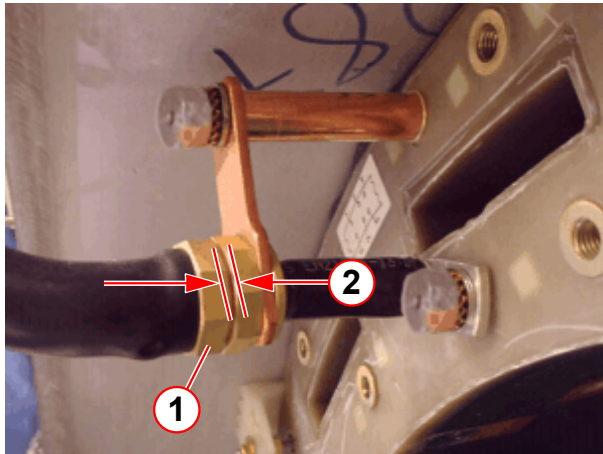
- (1) 19 mm nut (torque 40 ± 4 Nm)
- (2) NORD-LOCK washer
- (3) Cable lug gradient cable from filter plate
- (4) Cable lug coax cable
- (5) Nut (torque 21 ± 2 Nm)
- (6) Washer

Fig. 229: Checking the minimum distance of the connection to the bore

**At the GC:**

- Check the minimum distance between the copper connection parts and the magnet bore.
 - » The **gap must be at least 4 mm** (e.g. check with a 4 mm Allen key).
 - » In case the gap is too small (less than 4 mm) check again the mounting bolt position adjustment at the Gradient Connection Plate.

Fig. 230: Releasing the rear nut SW32



- (1) Rear nut released
- (2) Gap

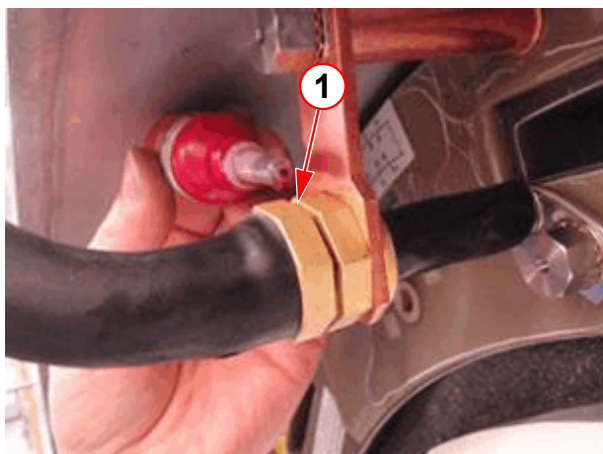
- Release the rear nut (SW32) a few turns to obtain a gap between both nuts.

Fig. 231: Tightening the front nut (SW32)



- Tighten the front nut using a non-magnetic open-ended wrench (SW32).

Fig. 232: Securing the nuts with Loctite 221



- (1) Apply a few drops Loctite 221 into the gap

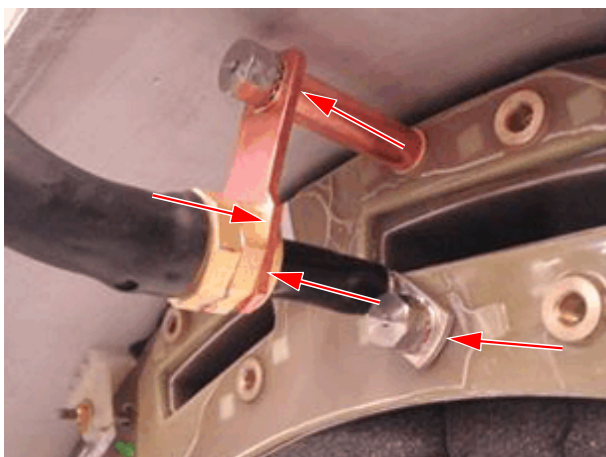
- Apply a few drops of Loctite 221 into the gap between both nuts.

Fig. 233: Tightening the rear nut (SW32)



- Tighten the rear nut (SW32), using a non-magnetic open-ended SW32 wrench .
 - » There should be **NO gap** between the nuts!

Fig. 234: Checking for proper alignment



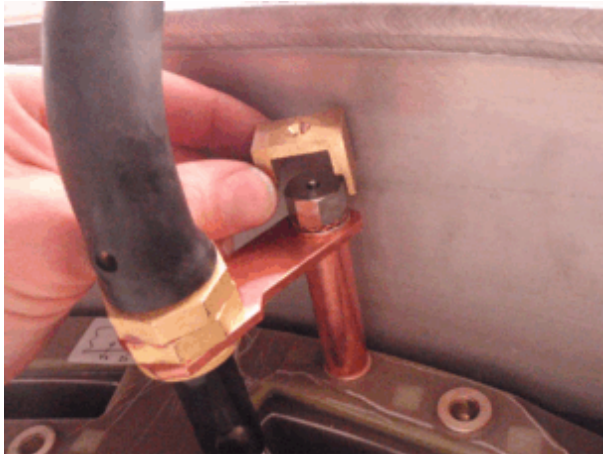
- Check all contact surfaces for proper alignment and mounting.
 - » The current conducting contact surfaces should align fully flat.
 - » There should be **NO gap** between the contacts.



Check the torque of the gradient coil connection screws again.

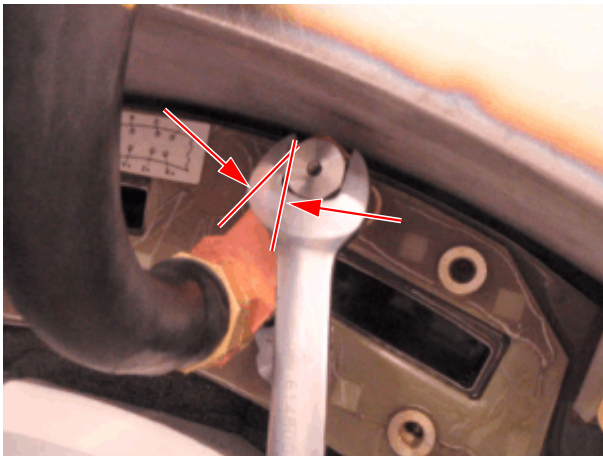
Apply a torque of 24 ± 3 Nm.

Fig. 235: Inserting the retaining circlip



- Insert the retaining circlip over the hex head bolt (→ Fig. 235 Page 229). Ensure, that the outer contour of the hex screw is parallel to the copper tab (CU tab).

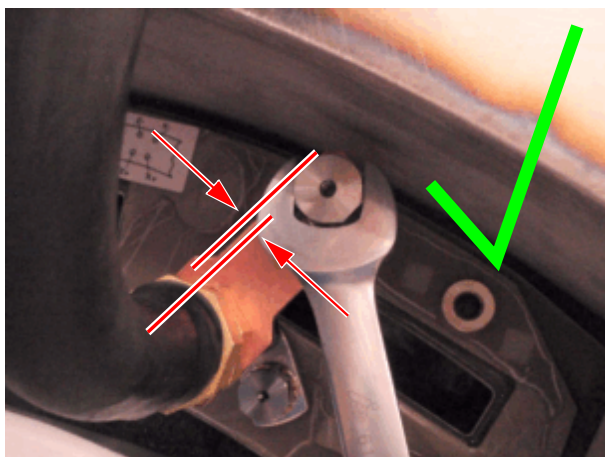
Fig. 236: Incorrectly oriented octagonal screw (NOT parallel to CU tab)



- If necessary (→ Fig. 236 Page 229), tighten the screw further (clock-wise) until it is parallel and the retaining circlip fits snugly (→ Fig. 237 Page 229).

» If the circlip does not fit properly, tighten the gradient connection screw a little bit more (clockwise) until the circlip fits snugly. Never loosen the connection to fit a circlip!

Fig. 237: Correctly oriented octagonal screw (parallel to CU tab)



- » The left figure shows the correct alignment of the octagonal screw.

Fig. 238: Installing the retaining circlip on the coaxial bushing

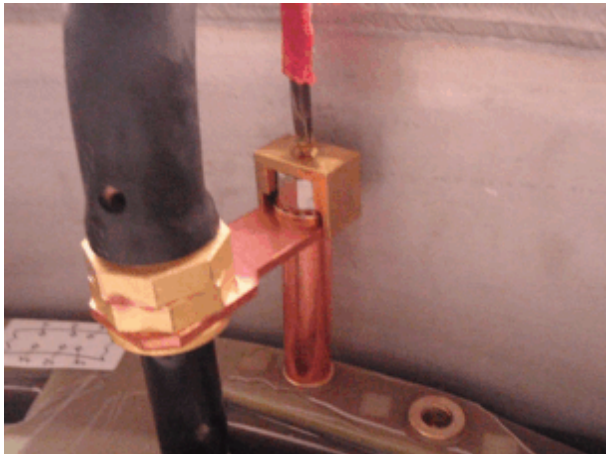


Fig. 239: Installing the retaining circlip at the gradient coil cable socket

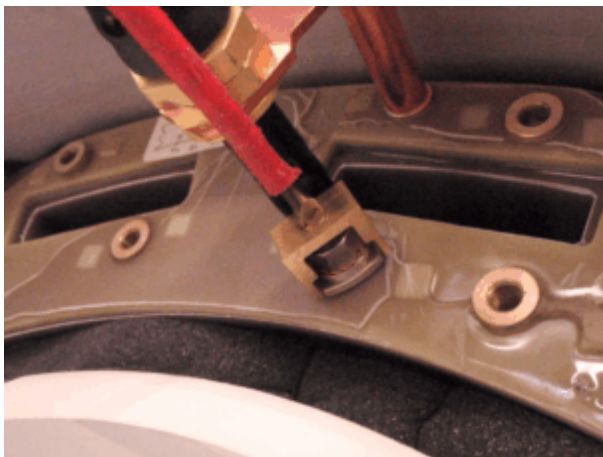
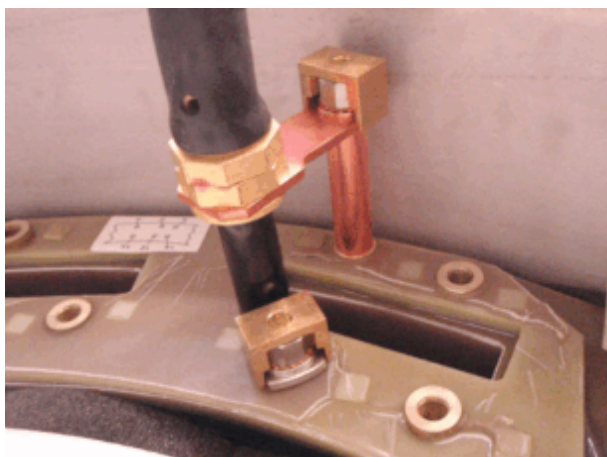


Fig. 240: Fully installed and secured gradient connection

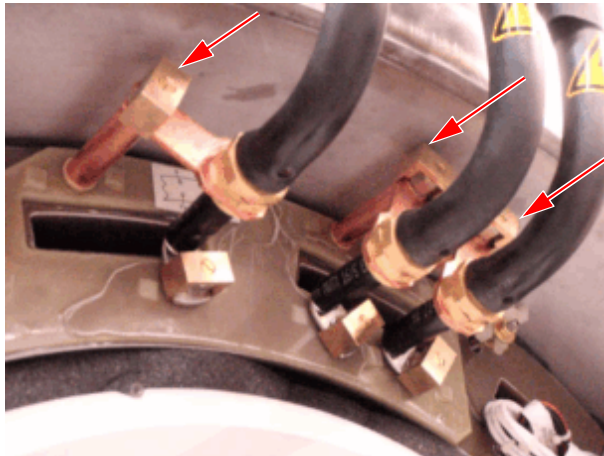


At the gradient coil coaxial bushing:

- Insert the retaining circlip.
- Apply a drop of Loctite 221 to the thread of the M4x10 locking screw and fasten it with a non-magnetic screwdriver (→ Fig. 238 Page 230).

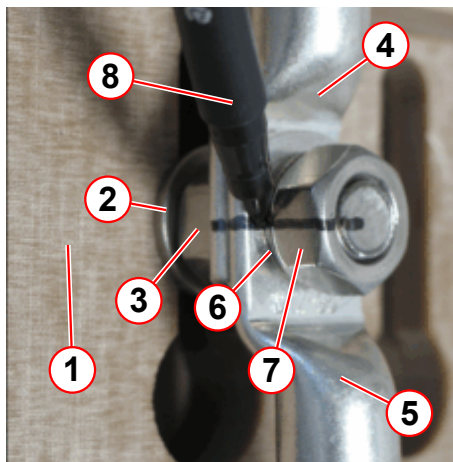
- Do the same at the gradient coil cable socket (→ Fig. 239 Page 230).

Fig. 241: Fully installed coaxial cable kit

**At the GC:**

- Check the minimum distance between the brass circlip and the magnet bore.
 - » The gap must be at least 4 mm (e.g. check with a 4-mm Allen key).

Fig. 242: Gradient connection plate

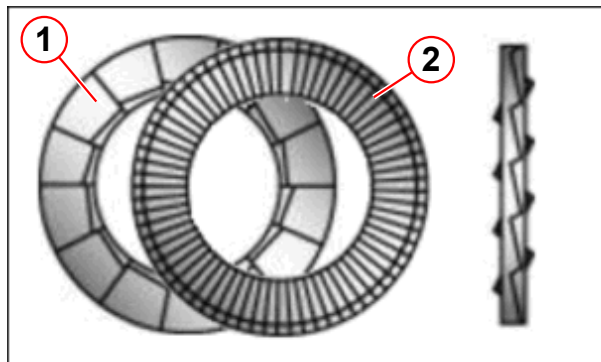


- (1) Connection plate
- (2) Washer A13
- (3) Nut SW19
- (4) Upper cable lug (gradient cable to filter plate)
- (5) Lower cable lug (gradient cable to gradient coil)
- (6) Nordlock washer
- (7) Nut SW19
- (8) Waterproof pen

**At the Gradient Connection Plate:**

- Tighten the nut (→ Pos.7/Fig. 242 Page 231) with the non-magnetic torque wrench (SW19).
 - » Apply a torque of 40 ± 4 Nm.
- Mark the connections with a waterproof pen (→ Pos.8/Fig. 242 Page 231).

Fig. 243: NORD-LOCK washer



- (1) Cam side
- (2) Ridged side

- A NORD-LOCK washer always consists of 2 halves.
 - » Both ridged sides must point towards outside.



Usually, the originally fitted NORD-LOCK washers can be reused. Just in case of missing or defective ones, use new parts (set of NORD-LOCK washers material number 103 58 905).

11.4.4.2 Fastening of Shrink Tubes



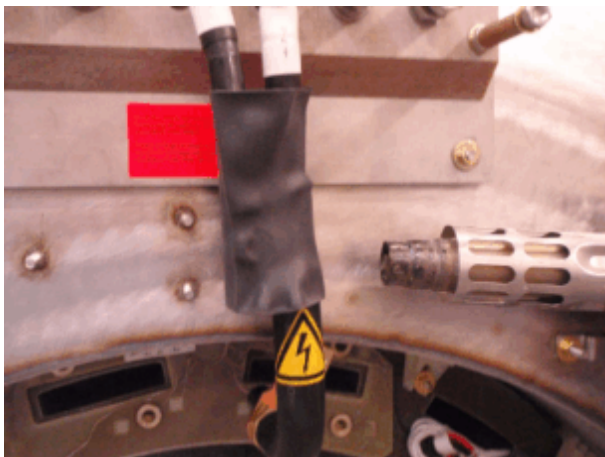
WARNING

Strong magnetic field near the gradient coil exerts high attraction on magnetic objects.

Attracted objects can lead to serious physical injuries.

- » Only use a heat gun with extension hose (for 220 V: material number 97 02 937/ for 110 V: material number 97 02 945).
- » The fan motor must be safely placed and fastened in the maximum possible distance away from the magnetic field!

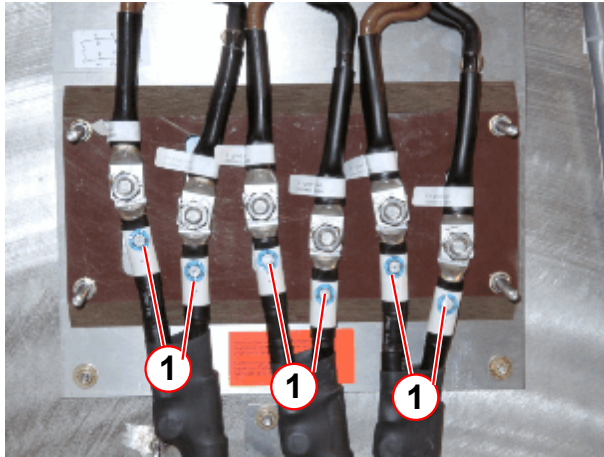
Fig. 244: Heat shrinking the gradient cable joint



- Push the shrink tube over the joint and shrink it using the heat gun (→ Fig. 244 Page 232).

11.4.4.3 Temperature Indicator Attachment

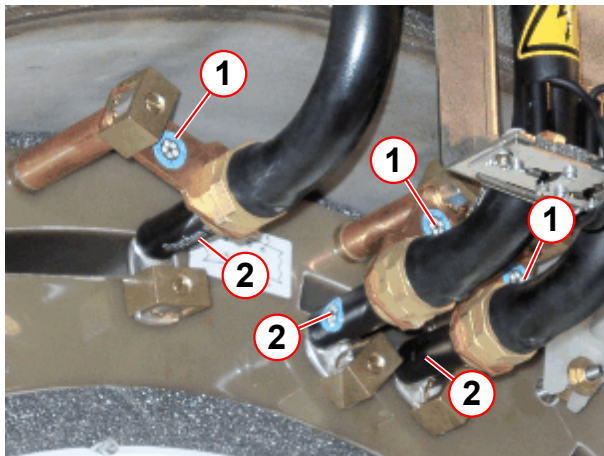
Fig. 245: Position of the thermo sensitive labels - Avan-
to connection plate



(1) Labels on lower coax cables (6x)

- Attach the clock indicators (thermo-sensitive labels) at the positions (1) shown in the left figure.

Fig. 246: Position of the thermo-sensitive labels - Avan-
to gradient coil



(1) Labels on copper lugs (3x)
(2) Labels on inner tube (3x)

- Attach the clock indicators (thermo-sensitive labels) at the positions (1) and (2) shown in the left figure.

11.4.5 Startup

11.4.5.1 Mechanical Work Steps

- The next figure shows the cables finally installed at the connection plate.

Fig. 247: Overview - all labels attached

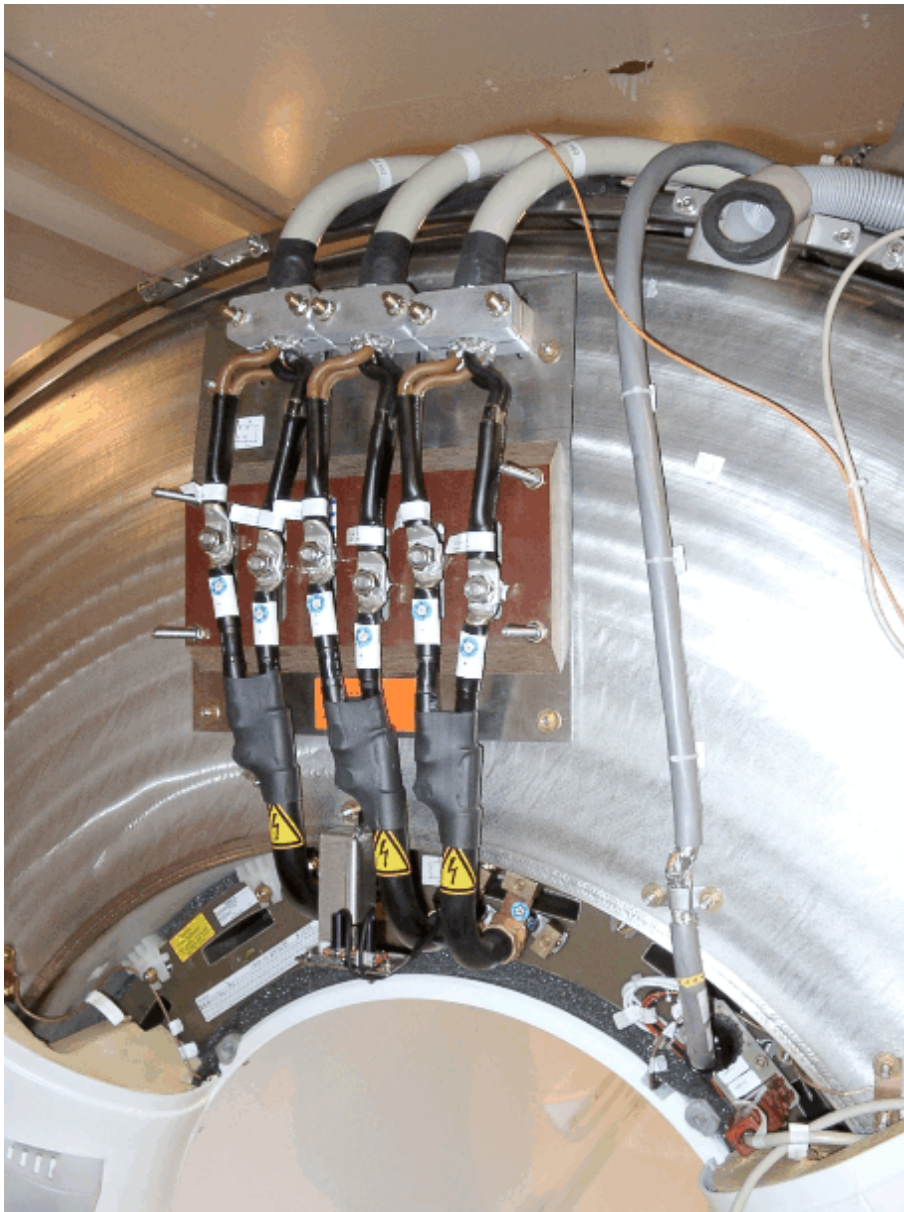
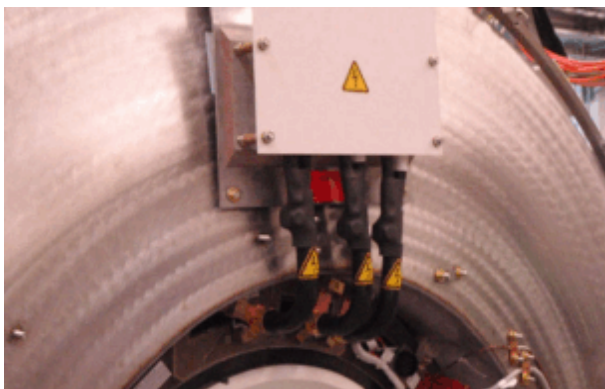


Fig. 248: Cover installed



- Install the cover at the Gradient Connection Plate.
- Reinstall the magnet top rear cover.
- Switch-on circuit breakers F1/GPA and F2.1/GPA/GSSU.
- Switch the MR system "ON" at the alarm box.

11.4.5.2 TuneUp/ Adjustments

n.a.

11.4.5.3 Function Tests/Q Tests

- Run QA/ Expert mode/ Spike Check (repeat 5 times)
- Run QA/ Image Orientation Check.

11.4.6 Final Steps

n.a.

11.5 Gradient Cables (Espree)

(Material number 81 23 684 for straight coil) or (Material number 81 21 860 for slim coil)



This procedure is valid for Espree systems only.

11.5.1 Requirements

11.5.1.1 Safety Information



WARNING

High voltage!

Contact with voltage-conducting parts leads to death or serious physical injury.

- » Adhere to the described steps for your own safety.



CAUTION

During BC/GC disassembly, you have to work close to non-isolated gradient cable connections at the GC.

Hazardous voltages may be present at gradient cable connections.

- » Switch off the gradient power amplifier via circuit breaker F1/GPA and F2.1/GPA/GSSU.
- » Wait at least 20 minutes to allow capacitor discharge before you start working at gradient connections.

11.5.1.2 Estimated Completion Time and Number of Persons

Tab. 9 Completion time

Work Steps	Time	Persons
Preparations	30 min	1
Removal/ Installation	2 hr	1
Startup/ TuneUp/ QA	30 min	1
Final Steps	n.a.	n.a.
	Total = 3 hr	

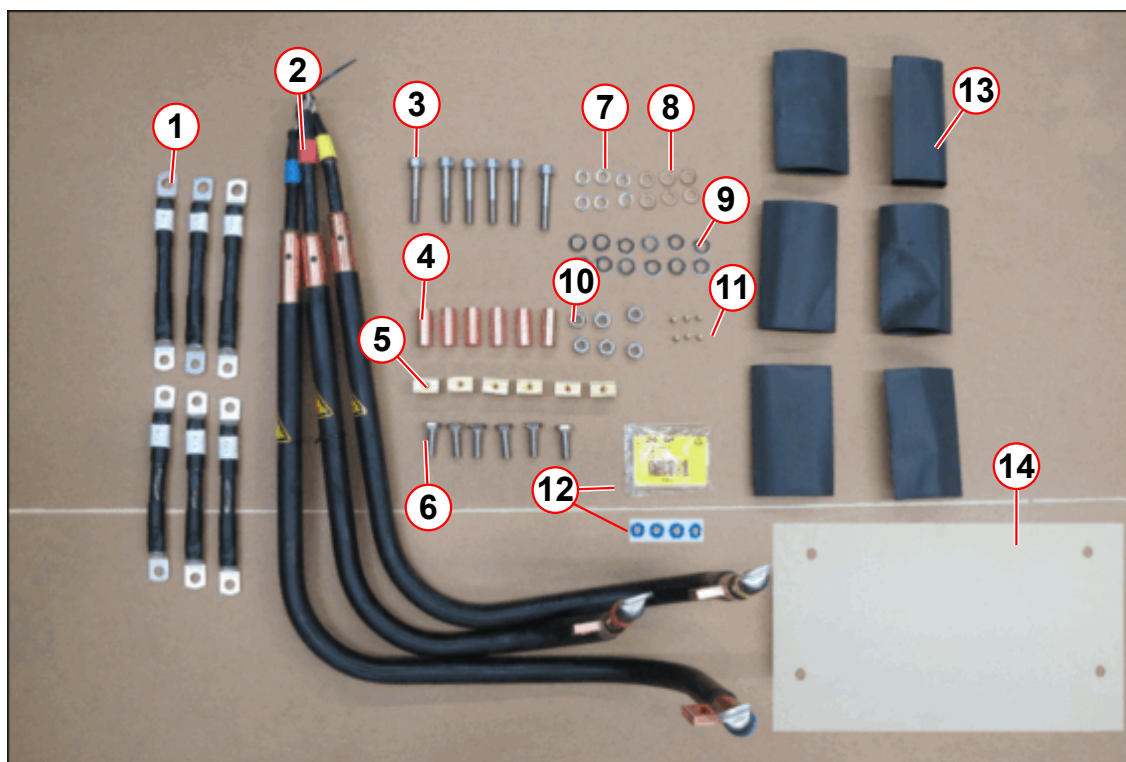
11.5.1.3 Aids or Tools

- Non-magnetic standard tool set (7757706)

- Non-magnetic torque wrench ranging from 20 - 50 Nm (8395803)
- Accessory for torque wrench (8396116)
- 8-s. socket gradient coil, non-magnetic (10399445)
- Heat gun (for 220 V: material number 97 02 937/ for 110 V: material number 97 02 945)
- Set of NORD-LOCK washers GC (10358905)
- Loctite 221 (soft-locking, 2048874)

11.5.1.4 Contents of the Cable Repair Kit

Fig. 249: Gradient cable kit - 8121860



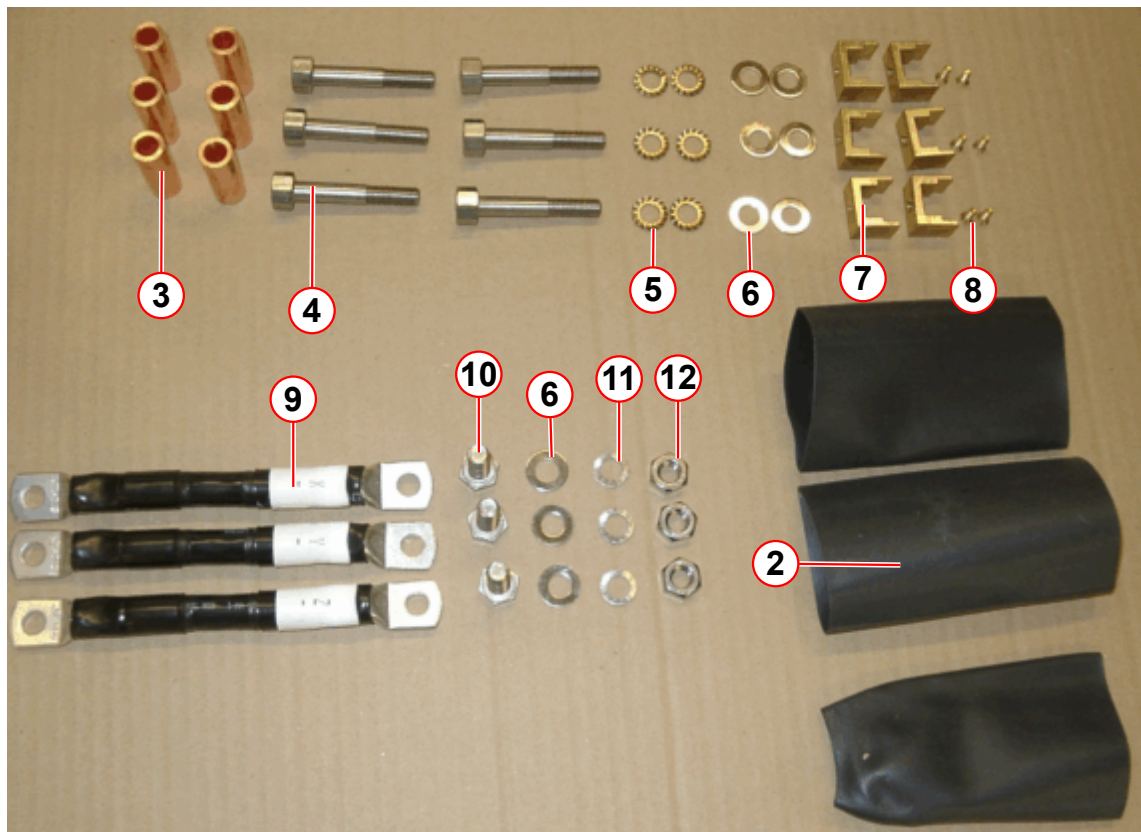
- (1) Cable joint
- (2) GC-coax pipe
- (3) Screws M10 straight
- (4) Copper brush
- (5) Retaining circlip
- (6) Screws M10x30
- (7) Nordlock washer
- (8) Star washer A 10.5
- (9) Washer A 10.5
- (10) Nut M10
- (11) Flat head screw M4x10
- (12) TEMP.INDICATOR 88 TO 110 DEGREES C
- (13) Heat shrink
- (14) Insulating plate Espree

Fig. 250: Complete coax cable kit Espree - 8123684



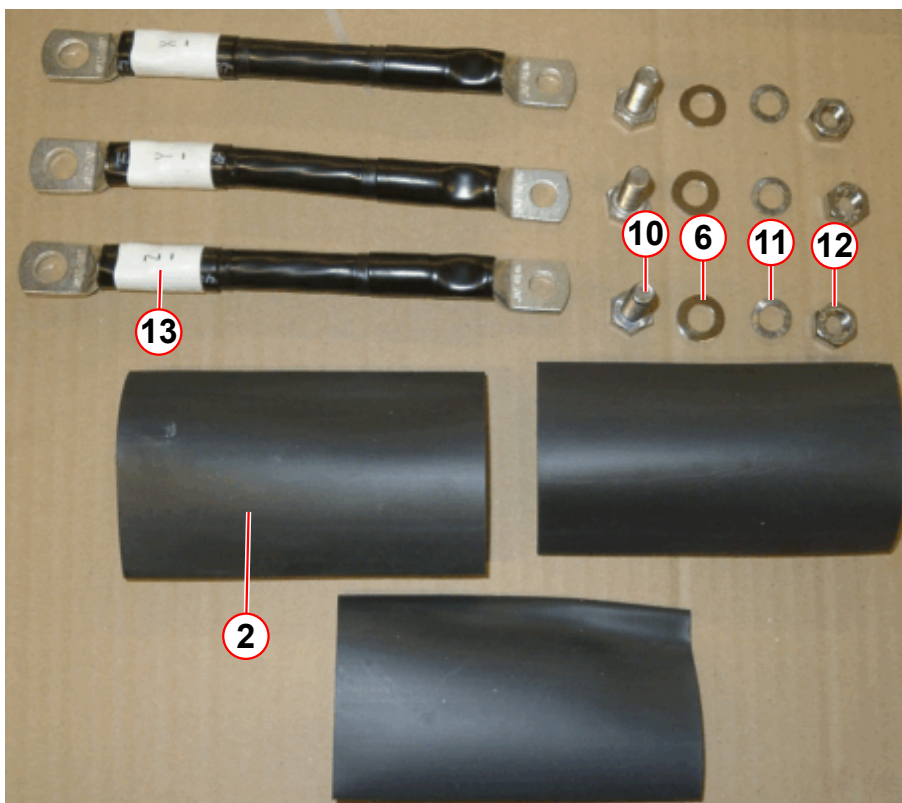
(1) Gradient cable kit Espree

Fig. 251: Fixing elements for gradient coil side



- (2) Heat shrink
- (3) Copper bush
- (4) Screw M10_straight
- (5) Star washer A10.5
- (6) Washer A10.5
- (7) Retaining circlip
- (8) Flat head screw M4x10
- (9) Short cable joints (10mm cable lugs on both ends)
- (10) Screw M10x30
- (11) Nordlock washers
- (12) Nuts M10

Fig. 252: Fixing elements for connection plate side



- (2) Heat shrink
- (6) Washer A10.5
- (10) Screw M10x30
- (11) Nordlock washers
- (12) Nuts M10
- (13) Long cable joints (10mm / 12mm cable lugs)

11.5.2 Preparatory Work Steps

11.5.2.1 Power switch-off



WARNING

High voltage!

Contact with voltage-conducting parts will lead to death or serious physical injury.

» Adhere to the described steps for your own safety.

1. Switch the MR system to "Standby" at the alarm box.
2. Switch off GPA circuit breakers F1 and F2.1.

3. Wait at least 20 minutes to ensure the final stage capacitors are fully discharged before starting to work on the gradient cables.

11.5.2.2 Removing Covers

Fig. 253: Magnet gradient cover



(1) Protective cover

- Remove magnet covers.
- Remove the protective cover of the Gradient Connection Plate (4x10 mm nuts) (→ Pos.1/Fig. 253 Page 241).

11.5.3 Removal

- Unscrew the old/defect cable from the connection plate.
- Unscrew the old/defect cable from the Gradient Coil.
- Remove all gradient cable clamps (→ Pos.2/Fig. 267 Page 251).
- Remove the old/defect coax cable.

11.5.4 Installation

NOTICE

Incorrect mounting of gradient cables under mechanical strain.

Risk of Gradient coax cable defects due to fatigue cracks!

- » Carefully adjust the bolts at the Gradient Connection Plate within the long holes to achieve a strain-free mounting of coax cables.
- » The coaxial cables and gradient Connection Plate must not touch vibrating parts like magnet housing, covers etc.

11.5.4.1 Mounting the Coax Cables

Fig. 254: Cleaning the copper bush

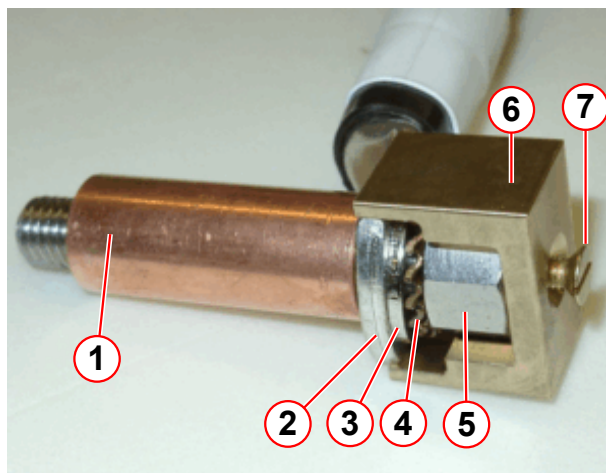


- Clean all copper contact surfaces including the copper bushes metallic bright, using Scotch Brite.

Note:

Do not clean galvanized parts (e.g. cable lugs) with Scotch Brite!

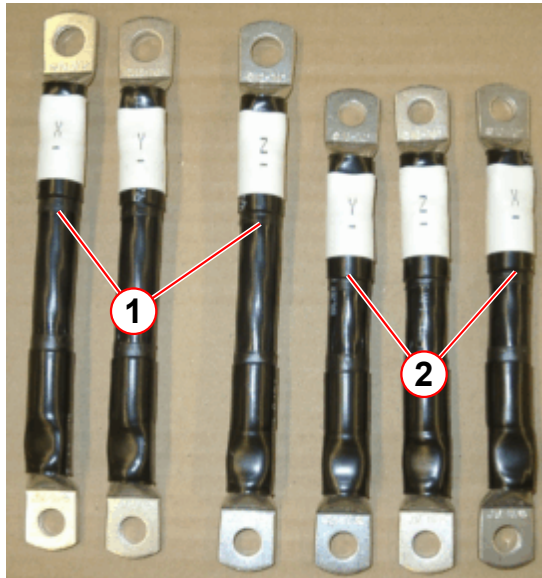
Fig. 255: Correct order of fixing elements (at gradient coil)



- Prepare the gradient cables and bushes using the fixing elements (→ Fig. 255 Page 242).

- (1) Copper bushing
- (2) Cable lug
- (3) Washer A10.4
- (4) Star washer A10.5
- (5) Screw M10 straight
- (6) Retaining circlip
- (7) Flat head screw M4x10

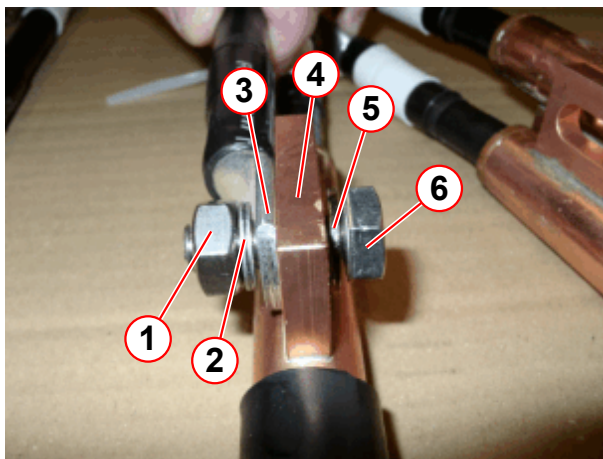
Fig. 256: Different types of cable joints



- (1) Longer cable joint (10mm / 12 mm)
 (3) Shorter cable joint (10 mm / 10mm)

- Sort the delivered cable joints according to the different length.
 - » The 12-mm cable lugs at the slightly longer cable joints (→ Pos.1/ Fig. 256 Page 243) must be fitted at the gradient connection plate, the shorter cable joints must be fitted at the gradient coil.

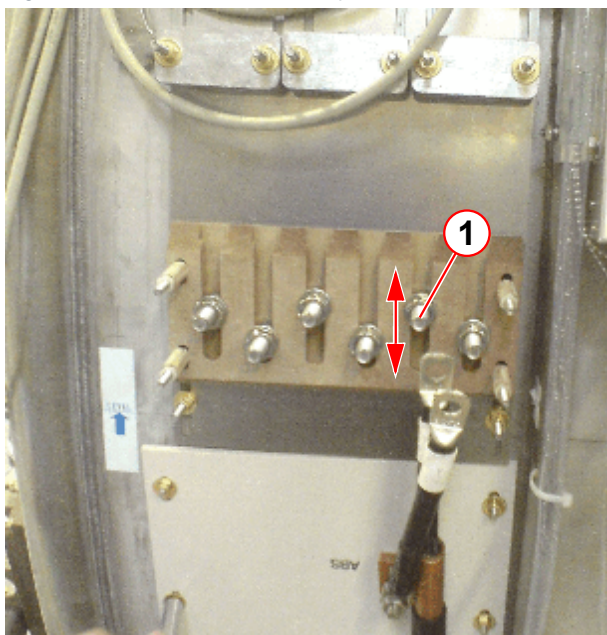
Fig. 257: Correct arrangement of fixing elements on Y-cable joints



- (1) Nut M10
 (2) Nordlock washers
 (3) Cable joint
 (4) Copper bar of coax cable
 (5) Washer A10.5
 (6) Screw M10x30

- Prepare the 6 cable joints as shown in the left figure. Just hand-tighten the M10 nuts at this point.
 - » Check the correct use and position of the cable joints. There are 2 different types included in the kit (→ Fig. 256 Page 243).

Fig. 258: Gradient connection plate



(1) Bolt adjustment range

- Provisionally attach the gradient cables to both ends, i.e. gradient coil and gradient connection plate.
- Adjust the mounting bolt position within the long holes to achieve a strain-free installation of the coax cable. (→ Pos.1/Fig. 258 Page 244).
 - » The coax cable must be routed without mechanic stress.

Note:

Ensure, there is a delivered shrink tube loosely attached to both ends of the gradient cable.

Fasten the GC mounting bolts with the correct torque.

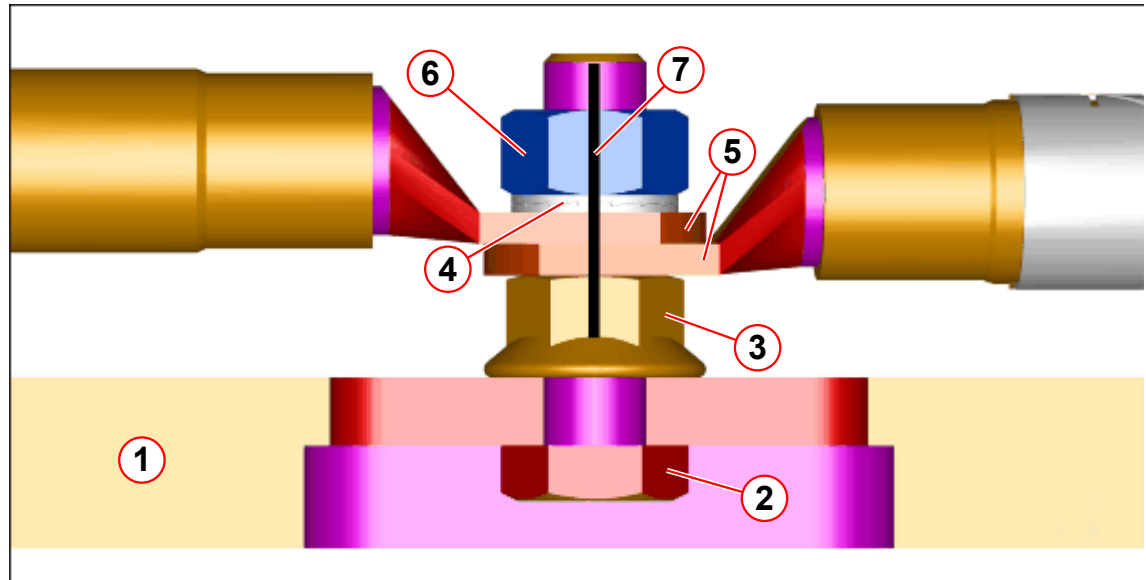
A: New version (with flange nut)

- Fasten the flange nut (3) with a non-magnetic torque wrench (SW18).



» Apply a torque of 21 ± 2 Nm.

Fig. 259: Correct set-up of the gradient connection plate at the magnet



- (1) Connection plate
- (2) 12 mm bolt (SW 19)
- (3) Flange nut (SW 18) (torque 21 ± 2 Nm)
- (4) NORD-LOCK washers
- (5) Gradient cable lugs
- (6) Nut (SW 19) (torque 40 ± 4 Nm)
- (7) Mark with waterproof pen

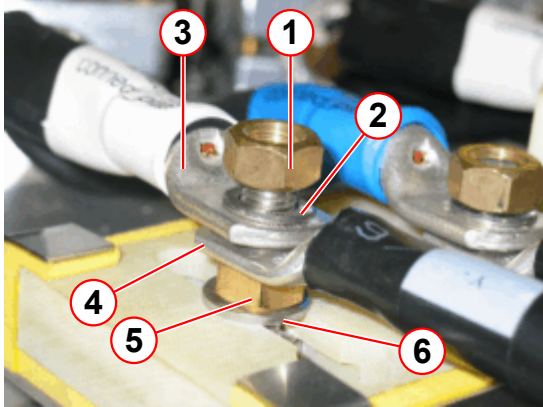
B: Old version (with washer and nut)

- Fasten securing nut (5) with a non-magnetic torque wrench.



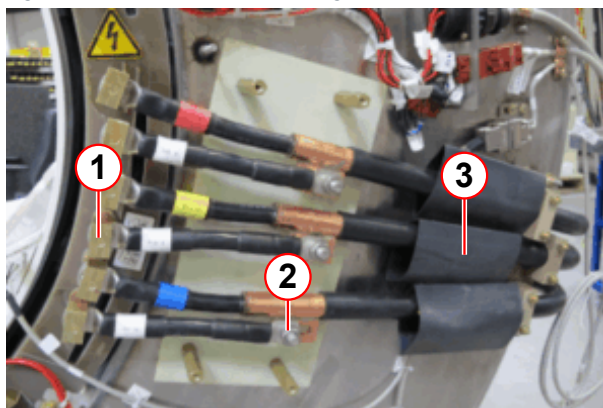
» Apply a torque of 21 ± 2 Nm.

Fig. 260: Cable connection (tightened with 40 ± 4 Nm)



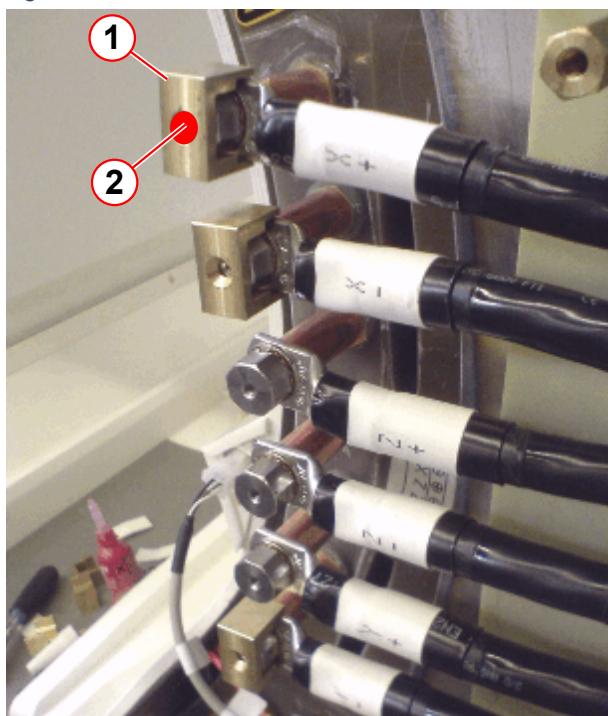
- (1) 19 mm nut (torque 40 ± 4 Nm)
- (2) NORD-LOCK washer
- (3) Cable lug gradient cable from filter plate
- (4) Cable lug coax cable
- (5) Nut (torque 21 ± 2 Nm)
- (6) Washer

Fig. 261: Cable connection at gradient coil



- (1) Screw connection (tightened with 24 Nm) with mounted securing clamps
- (2) Connection tightened with 30 Nm
- (3) Heat shrinkable tube

Fig. 262: Gradient Coil connections

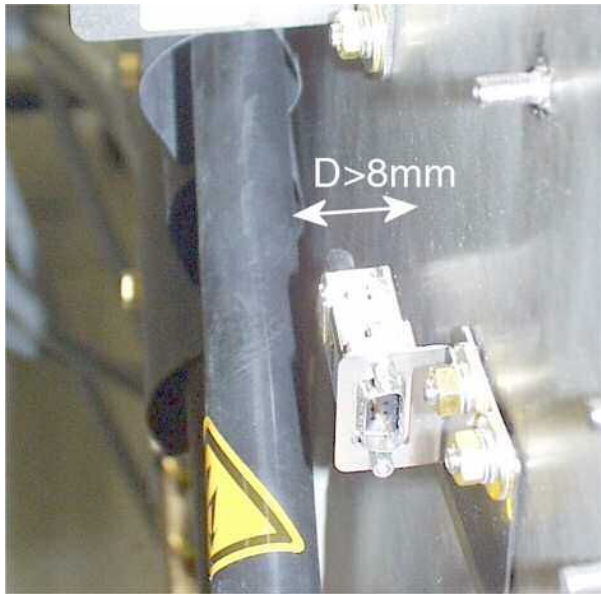


- (1) Securing clamp
- (2) M4 locking screw

**At the GC:**

- Fasten the gradient coil connection screw (1) using a non-magnetic torque wrench and 8-s. socket gradient coil non-mag.(103 99 445) .
 - » Apply a torque of 24 ± 3 Nm.
- Tighten the cable joint connections (2).
 - » Apply a torque of 30 ± 3 Nm.
- Install the securing clamps (1) parallel to the cable terminal.
 - » Minimum distance of non-insulated parts: 4 mm.
 - » **If the securing clamp does not fit properly, tighten the gradient connection screw a little bit more (clockwise) until the clamp fits snugly. Never loosen the connection to fit a clamp!**
- Apply a drop of Loctite 221 to the thread of the M4 locking screw (2) and fasten it with a non-magnetic screwdriver.

Fig. 263: Distance D: Coaxial cable to OVC at the back of the magnet



- The distance D of the coaxial cables to the OVC should be greater than 8 mm.

Fig. 264: Coaxial link cable connection at magnet

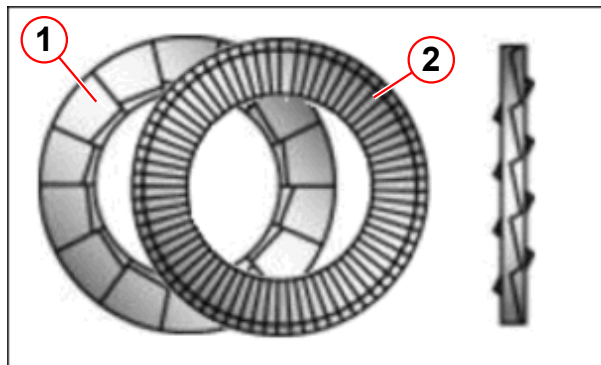


- (1) Upper connections tightened with 40 Nm
- (2) Lower connection tightened with 30 Nm
- (3) Shrink tube

**At the Gradient Connection Plate:**

- Install the lugs of the gradient cables which come from the RF-Filter panel on top of the coax cable lugs.
- Fasten the cables with NORD-LOCK washer and nut using a non-magnetic torque wrench (SW19) (→ 1/Fig. 264 Page 249)
 - » Apply a torque of 40 ± 4 Nm.
- Fasten the coax cable joint screws with a non-magnetic torque wrench (SW17) (→ 2/Fig. 264 Page 249)
 - » Apply a torque of 30 ± 3 Nm.

Fig. 265: NORD-LOCK washer



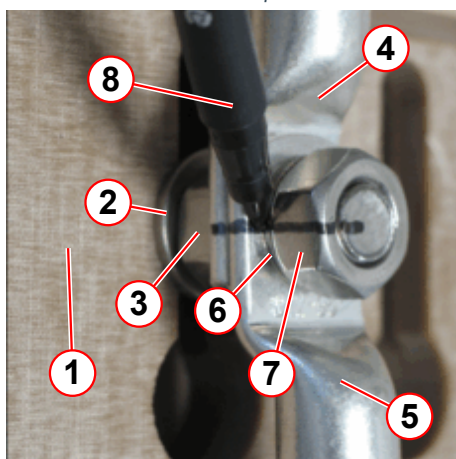
- (1) Cam side
- (2) Ridged side

- A NORD-LOCK washer always consists of 2 halves.
 - » Both ridged sides must point towards outside.



Usually, the originally fitted NORD-LOCK washers can be reused. Just in case of missing or defective ones, use new parts (set of NORD-LOCK washers material number 103 58 905).

Fig. 266: Gradient connection plate



- (1) Connection plate
- (2) Washer A13
- (3) Nut SW19
- (4) Upper cable lug (gradient cable to filter plate)
- (5) Lower cable lug (gradient cable to gradient coil)
- (6) Nordlock washer
- (7) Nut SW19
- (8) Waterproof pen

- Mark the connections with a waterproof pen
(→ Pos.8/Fig. 266 Page 250).

11.5.4.2 Fastening of Shrink Tubes



WARNING

Strong magnetic field near the gradient coil exerts high attraction on magnetic objects.

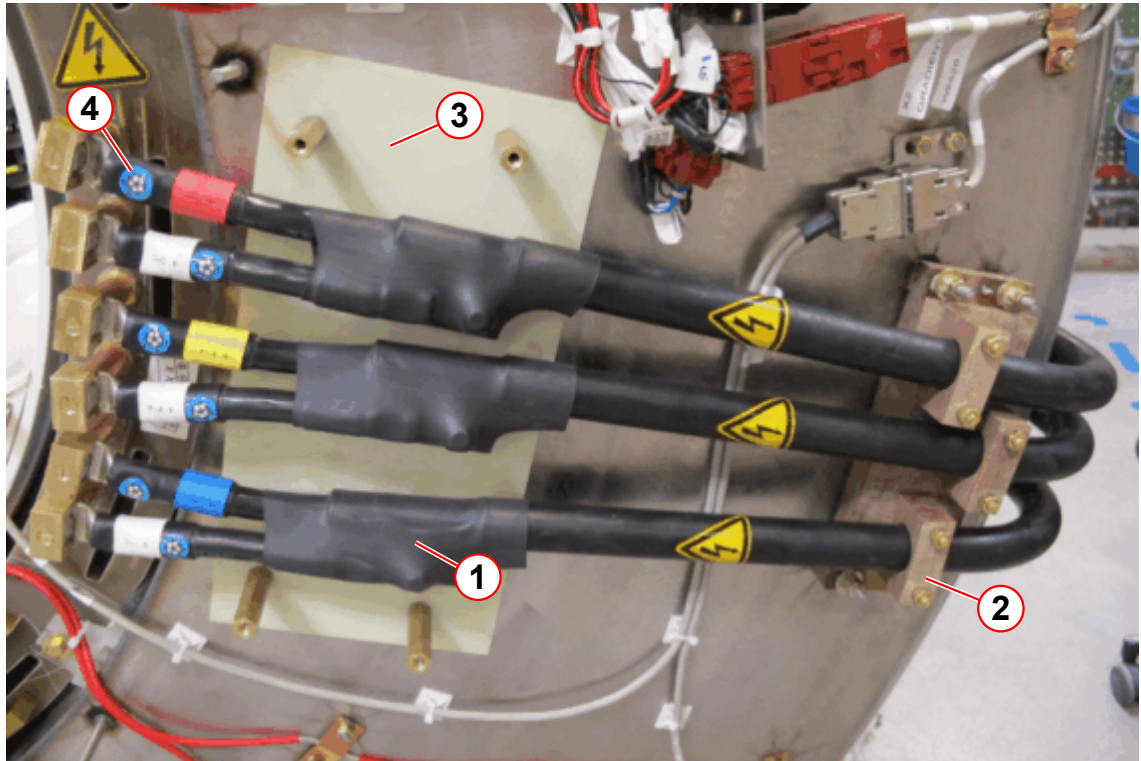
Attracted objects can lead to serious physical injuries.

- » Only use a heat gun with extension hose (for 220 V: material number 97 02 937/ for 110 V: material number 97 02 945).
- » The fan motor must be safely placed and fastened in the maximum possible distance away from the magnetic field!

At the GC:

- Shrink the tube (→ Pos.1/Fig. 267 Page 251) over the cable joints using the heat gun.

Fig. 267: Cable kit fully installed (coil side for slim)



- (1) Heat shrink
- (2) Cable holder
- (3) Insulating plate
- (4) TEMP.INDICATOR 88 TO 110 DEGREES C



The insulation plate is only used for the Gradient Cable kit 8121860 (slim coil).

Fig. 268: Coaxial link cable connection at magnet



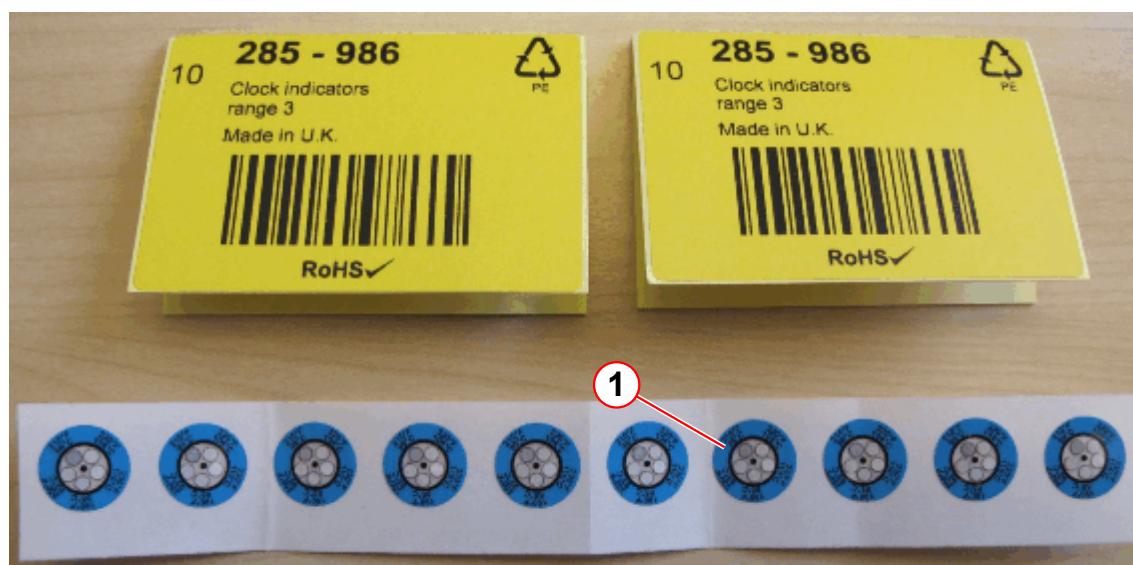
- (1) Upper connections tightened with 40 Nm
- (2) Lower connection tightened with 30 Nm
- (3) Shrink tube

At the Gradient Connection Plate:

- Shrink the tube over the cable joint using the heat gun
(→ 3/Fig. 264 Page 249)

11.5.4.3 Temperature Indicator Attachment

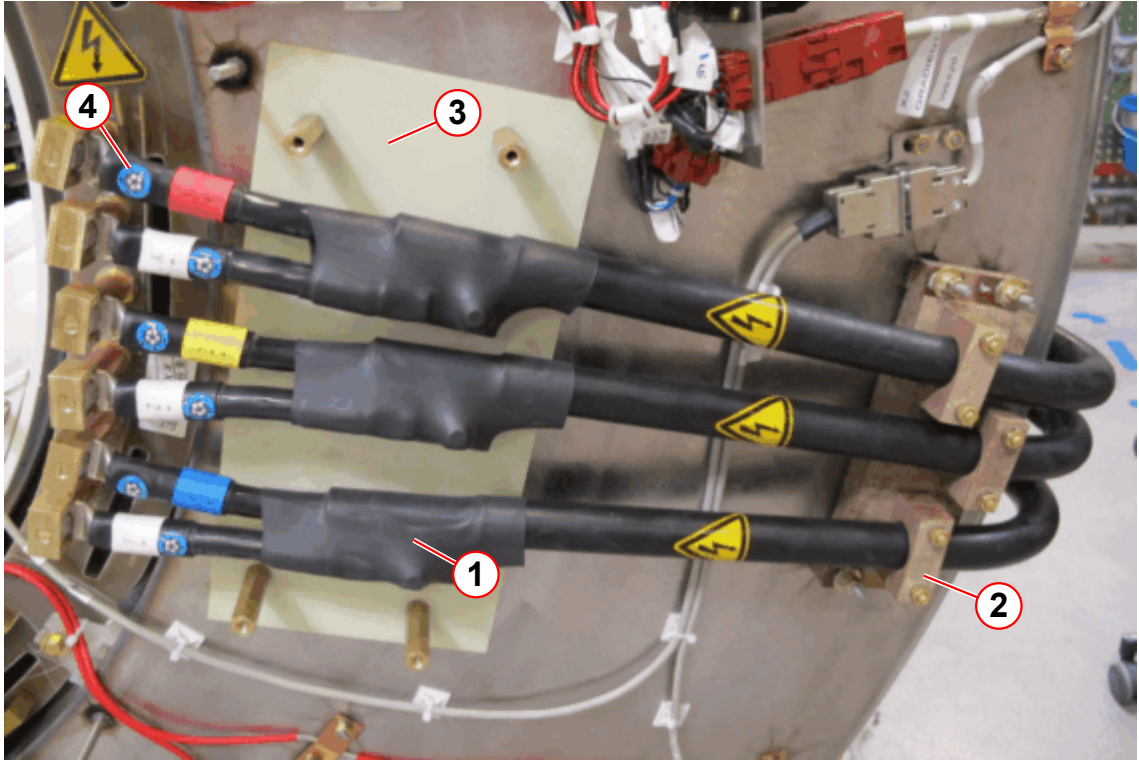
Fig. 269: Temperature indicator



- (1) Temperature indicator

- Fasten all gradient cable clamps (→ Pos.2/Fig. 270 Page 253).
- Attach temperature indicators close to the screwed connections at the GC (→ Pos.4/Fig. 270 Page 253).

Fig. 270: Cable kit fully installed (coil side for slim)



- (1) Heat shrink
- (2) Cable holder
- (3) Insulating plate
- (4) TEMP.INDICATOR 88 TO 110 DEGREES C

- Attach temperature indicators close to the screwed connections at Gradient Connection Plate (1)..

Fig. 271: Coax cable kit Espree - fully installed



(1) TEMP.INDICATOR 88 TO 110 DEGREES C

11.5.5 Startup

11.5.5.1 Mechanical Work Steps

- Reinstall the cover to the gradient connection plate (→ Fig. 253 Page 241).
- Reinstall the magnet covers.
- Switch-on circuit breakers F1/GPA and F2.1/GPA/GSSU
- Switch the MR system "ON" at the alarm box.

11.5.5.2 TuneUp/ Adjustments

n.a.

11.5.5.3 Function Tests/Q Tests

- Run QA/ Expert mode/ Spike Check (repeat 5 times)
- Run QA/ Image Orientation Check

11.5.6 Final Steps

n.a.

Version 10:

- The FRU overview table has been reworked:
Adjustments and final checks, column time changed in total time.
- Components of the gradient coil:
Add information for the torque wrench 20 Nm on the connection plate.
- All procedures reworked:
Time, adjustment and final checks.

Version 11

- Chapter (Power Stage Supply) - D100 description updated CHARM_MR_00457473.
- Chapter (Component of the Gradient Coil) - new version of Gradient Coax Cables Avanto (material number 106 78 832) added (CHARM_MR_00457661).
- Chapter (General Information), section FRU Overview - Manual pump RP50 (material number 81 15 136) added for refilling of cooling water.
- All chapters - minor editorial changes.

Version 12

- Chapter (Component of the Gradient Coil) fastening torque tolerance for coax connectors adjusted 35 ± 2 Nm changed to 40 ± 4 Nm,
 20 ± 1 Nm to 21 ± 2 Nm , 28 ± 1 Nm to 30 ± 3 Nm, 22 ± 1 Nm to 24 ± 3 Nm and 22 ± 1 Nm to 30 ± 3 Nm (CHARM_MR_00459191).

Version 13

- Chapter (FRU Overview) - original Gradient Coax Cables Avanto (material number 106 78 832) added to FRU list.
- Chapter (Components of Gradient Coil) - section (→ Gradient Coax Cables Avanto (material number 106 78 832) / Page 188) revoked from Version 10 of this document (CHARM_TD_00016737).
- All chapters - minor editorial changes.

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Print No.: M6-020.841.16.13.02 | Replaces: M6-020.841.16.12.02
Doc. Gen. Date: 05.16 | Language: English
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